An IoT-Enabled System for Real-Time Fuel Monitoring, GPS Tracking, and Anomaly Detection in Trucks Using Non-Parametric Unsupervised Machine Learning

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ABSTRACT

The logistics and transportation industry continues to face challenges in fuel management, anomaly detection, and real-time fleet monitoring. Existing solutions often operate in silos, resulting in fragmented data, undetected fuel irregularities, and delayed maintenance responses. This study developed and evaluated an IoT-based, context-aware fleet monitoring system that first acquires standard OBD-II vehicle parameters and retrieves fuel level through a vehicle-specific PID adapter, implemented in the tested vehicle through the Toyota meter Mode 22 PID 21/29 channel. The system tracks GPS location and detects anomalous fuel consumption in trucks using non-parametric unsupervised machine learning models, including Matrix Profile and Isolation Forest. The system uses an ESP32-based in-truck HMI with a fuel-filter cascade, a prioritized LoRa, LTE, GSM 2G, WiFi, and SPIFFS-offline communication stack, and embedded alert hardware for driver rest and maintenance notifications. Sensor and vehicle data are synchronized and transmitted to a cloud backend where anomalies are automatically flagged and actionable alerts are generated. The prototype was evaluated using controlled-injection trip data and field-test telemetry data to measure detection accuracy, false positive rates, and response latency. Three user-facing interfaces were delivered: an embedded in-truck HMI for drivers, a mobile driver web application for companion GPS and trip-state synchronisation, and an administrative dashboard for SGSA fleet monitoring.

Index Terms—Fuel Monitoring, GPS Tracking, Anomaly Detection, Internet of Things, Unsupervised Machine Learning, Isolation Forest, Matrix Profile

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ABBREVIATIONS

ACID	Atomicity, Consistency, Isolation, And Durability	71
ADC	Analog-to-Digital Converter	76
AI	Artificial Intelligence	4
API	Application Programming Interface	15
AUC	Area Under The Curve	48
BT	Bluetooth	74
BW	Bandwidth	102
CAN	Controller Area Network	23
CE	Conformité Européenne (European Conformity)	70
CPS	Cyber-Physical System	59
CSS	Chirp Spread Spectrum	65
CSQ	Cell Signal Quality	106
DB	Database	75
DTC	Diagnostic Trouble Code	76
EBM	Explainable Boosting Machine	46
ECU	Engine Control Unit	13
EMA	Exponential Moving Average	29
ERD	Entity-Relationship Diagram	108
ESP32	Espressif Systems Microcontroller With Wi-Fi And Bluetooth	28
FAIR	Findable, Accessible, Interoperable, And Reusable	71
FCC	Federal Communications Commission	70
FPR	False Positive Rate	15
GDPR	General Data Protection Regulation	71
GNSS	Global Navigation Satellite System	65
GO	General Objective	12
GPIO	General-Purpose Input/Output	70
GPS	Global Positioning System	3
GSM	Global System For Mobile Communications	5
HDV	Heavy-Duty Vehicle	43
HMAC	Hash-based Message Authentication Code	105
HMI	Human-Machine Interface	10
HTML	Hypertext Markup Language	77
HTTP	Hypertext Transfer Protocol	15
HTTPS	Hypertext Transfer Protocol Secure	105
HSPI	Hardware Serial Peripheral Interface	93

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I2C	Inter-Integrated Circuit	70
IEC	International Electrotechnical Commission	70
IEEE	Institute Of Electrical And Electronics Engineers	31
IETF	Internet Engineering Task Force	71
IF	Isolation Forest	54
IIR	Infinite Impulse Response	62
IOT	Internet Of Things	4
IQR	Interquartile Range	44
ISO	International Organization For Standardization	61
ITU	International Telecommunication Union	38
J1939	SAE J1939 Heavy-Duty Vehicle Network Standard	61
JSON	JavaScript Object Notation	77
KMPL	Kilometers Per Liter	121
LOF	Local Outlier Factor	43
LORA	Long Range (Low-power Wide-area Radio)	15
LOS	Line Of Sight	102
LOTO	Leave-One-Trip-Out	54
LPWAN	Low-Power Wide-Area Network	37
LSTM	Long Short-Term Memory	78
LTE	Long-Term Evolution	15
M2M	Machine-to-Machine	34
MAF	Mass Air Flow	76
MAPE	Mean Absolute Percentage Error	90
ML	Machine Learning	7
MP	Matrix Profile	80
MQTT	Message Queuing Telemetry Transport	7
NLOS	Non-Line Of Sight	102
NTC	National Telecommunications Commission	42
OBD-II	On-Board Diagnostics, Second Generation	9
OSRM	Open Source Routing Machine	126
PGN	Parameter Group Number	61
PHP	Philippine Peso	3
PRISMA	Preferred Reporting Items For Systematic Reviews And Meta-Analyses	31
PWA	Progressive Web Application	7
PWM	Pulse-Width Modulation	94
QOS	Quality Of Service	79
RAM	Random Access Memory	74
REST	Representational State Transfer	77
RFID	Radio Frequency Identification	4

NOTATION

x_t	current raw sensor value at time t	60
x_t^{smooth}	smoothed sensor value at time t after applying moving average	60
N	window size for the moving average filter	60
x_t^{med}	median-filtered sensor value at time t	62
W	sliding window length of the median filter	62
y_t	exponential moving average output at time t	62
α	smoothing factor of the exponential moving average	62
τ	time constant of the exponential moving average	62
δ	dead-band publishing threshold	62
y_{pub}	last published dead-band fuel value	62
$V_{\text{sensor},i}$	fuel volume estimated by the sensor at sample i	126
$V_{\text{manual},i}$	actual (manual) fuel volume measurement at sample i	126
T_{sent}	timestamp when a data packet is sent	65
T_{received}	timestamp when a data packet is received	65
L	data transmission latency	65
T_i	GPS timestamp at index i	65
t_i	time difference between successive GPS logs	65
T	time series of fuel usage	66
m	window length for subsequence in time series analysis	66
$P[i]$	matrix profile value at index i	66
$\mathbf{x}$	input feature vector for anomaly detection	66
$s(\mathbf{x})$	anomaly score for vector $\mathbf{x}$ from Isolation Forest	66
$E(h(\mathbf{x}))$	expected path length for $\mathbf{x}$ in an Isolation Forest	66
$c(n)$	average path length for unsuccessful search in a BST with n points	66
TP	true positive	63
FP	false positive	63
TN	true negative	63
FN	false negative	63
T_{detected}	timestamp when an anomaly is detected	63
T_{actual}	timestamp of actual anomaly occurrence	63
$\hat{p}$	estimated binomial proportion (sample success rate)	64
z	standard normal quantile used in the Wilson score interval	64
n	number of samples or validation events	64
C_{tank}	nominal tank capacity used to convert fuel volume to percentage	126
$V_{\text{pump},i}$	fuel volume dispensed by the pump during validation event i	126

493	$V_{\text{HMI},i}$	fuel volume displayed on the HMI during validation event i	126
494	e_i	fuel-volume error for validation event i	126
495	f_i	filtered fuel-level percentage at telemetry row i	119
496	Δf_i	change in filtered fuel level between successive telemetry rows	119
497	d_i	distance travelled between telemetry rows $i-1$ and i	119
498	v_i	vehicle speed at telemetry row i	119
499	kmpl_i	fuel-economy estimate for telemetry row i	119
500	Precision	precision of the anomaly detector, computed as $TP/(TP+FP)$	63
501	FPR	false-positive rate of the anomaly detector, computed as $FP/(FP+TN)$	63
502	$q_{0.985}$	0.985 quantile of clean training anomaly scores used as the decision threshold	125
503	$\mathcal{D}_{\text{train}}$	training partition used within a leave-one-trip-out fold	125
504	$\mathcal{D}_{\text{test}}$	held-out trip partition used for evaluation within a leave-one-trip-out fold	125

505 # GLOSSARY

506 **anomaly detection** — A technique used to identify data points, events, or observations that deviate significantly from expected patterns. In fuel monitoring, this helps detect irregularities such as theft, leakage, or abnormal consumption.

507 **Artificial Intelligence** — A branch of computer science focused on building systems capable of performing tasks that typically require human intelligence, such as decision-making, pattern recognition, and problem-solving.

508 **cloud analytics** — Data analysis performed on cloud-based platforms, enabling real-time access to fleet data (e.g., GPS and fuel metrics) and supporting informed decision-making.

509 **deep learning** — A subset of machine learning involving deep neural networks with multiple layers. Used in predictive analytics and anomaly detection in real-time fleet systems.

510 **fleet management** — The administration and coordination of commercial vehicle operations, including monitoring of fuel use, driver behavior, vehicle maintenance, and routing.

511 **float sensor** — A sensor that uses a floating mechanism to measure liquid levels, such as fuel in a tank. Commonly used in IoT-based real-time monitoring systems.

512 **Global System for Mobile Communications** — A standard for mobile communication that enables real-time alerts and remote data transmission for vehicle monitoring systems.

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513	Global Positioning System	A satellite-based system that provides real-time geolocation data, used in fleet operations for tracking vehicle movement and optimizing routes.
514	Internet of Things	A network of interconnected physical devices that collect and share data via the internet. In logistics, this includes in-vehicle sensors for real-time tracking and monitoring.
515	Isolation Forest	An unsupervised machine learning algorithm for anomaly detection that isolates outliers based on random partitioning of data points.
516	machine learning	A subset of AI where systems learn from data and improve performance over time without being explicitly programmed.
517	Matrix Profile	A time-series data mining method used for pattern recognition and anomaly detection in sequential datasets like fuel usage over time.
518	unsupervised machine learning	A type of machine learning where the model learns patterns from unlabeled data. Useful for anomaly detection where normal vs. abnormal behavior is not predefined.
519	On-Board Diagnostics	A standardized vehicle diagnostic interface that exposes engine and sensor data, such as fuel level, speed, and engine parameters, through the on-board diagnostic port without requiring physical modification of the vehicle.
520	Controller Area Network	A robust serial communication bus used within vehicles to allow electronic control units to exchange data without a host computer.

521	edge computing	A computing approach in which data is processed locally on the device that generates it, such as the embedded microcontroller in the fleet vehicle, reducing latency and bandwidth use before transmission to the cloud.
522	Human-Machine Interface	The in-vehicle interface that displays fuel, route, and alert information to the driver and captures driver acknowledgments such as rest and snooze actions.
523	telemetry	The automated measurement and wireless transmission of data, such as fuel level, location, and vehicle status, from the fleet vehicle to a remote backend for monitoring and analysis.
524	Leave-One-Trip-Out	A cross-validation protocol in which each trip is held out once as the test set while the model is trained and calibrated on the remaining trips, preventing information leakage from within-trip temporal correlation.
525	Cyber-Physical System	A system that tightly integrates physical processes with embedded computation and networked communication, where sensing, processing, and actuation operate together in real time.
526	confusion matrix	A table that summarizes the performance of a classifier by tabulating true positives, false positives, true negatives, and false negatives.
527	point anomaly	A single observation that deviates significantly from the rest of the data, such as a sudden fuel drop while the vehicle is stationary.
528	contextual anomaly	An observation that is anomalous only with respect to its surrounding context, such as high fuel consumption recorded at zero speed.

529	collective anomaly	A sequence of individually unremarkable observations that together form an anomalous pattern, such as gradual fuel siphoning over time.
530	discord	The subsequence of a time series that is least similar to all other subsequences, identified by the Matrix Profile as the most likely anomalous segment.
531	median filter	A non-linear order-statistic filter that replaces each sample with the median of a sliding window, robust to impulse noise and sensor spikes.
532	exponential moving average	A first-order recursive low-pass filter that smooths a signal by weighting recent samples more heavily than older ones according to a smoothing factor.
533	dead-band	A hysteresis mechanism that publishes a new value only when the smoothed signal departs from the last published value by more than a set threshold, preventing display jitter.
534	fuel slosh	The apparent fluctuation in a fuel-tank reading caused by liquid movement inside the tank during acceleration, braking, and turning.
535	Receiver Operating Characteristic	A curve that plots the true-positive rate against the false-positive rate of a classifier across thresholds; its area summarizes ranking performance.
536	Wilson score interval	A confidence interval for a binomial proportion that remains accurate for small samples and few positive cases, preferred over the normal-approximation interval.
537	Explainable Artificial Intelligence	Machine learning techniques that make a model's predictions interpretable, providing human-understandable reasons for why an input was flagged as anomalous.

538	Self-Organizing Map	An unsupervised neural network that maps high-dimensional data onto a low-dimensional grid, used to cluster and detect abnormal fuel-consumption patterns.
539	Chirp Spread Spectrum	A modulation technique used by LoRa in which data is encoded onto frequency sweeps (chirps), providing long range, low power use, and resistance to interference.
540	administrator dashboard	The fleet-manager interface that presents fleet-vehicle status, trips, maps, telemetry logs, analytics, alerts, and configuration controls for operational supervision.
541	alert acknowledgement	The user action or system record showing that an alert has been reviewed, acted upon, snoozed, or resolved by the driver or administrator.
542	backend	The cloud-side service layer that receives telemetry, stores records, synchronizes client interfaces, runs alert rules, and coordinates anomaly-detection results.
543	telemetry row	A single database record produced by the prototype telemetry pipeline, containing synchronized fuel, speed, location, trip-state, and communication fields for one sampling instant.
544	trip	A completed operating interval of a fleet vehicle, bounded by start and end events, used as the unit for route review, fuel analysis, and leave-one-trip-out evaluation.
545	fuel sender	The vehicle's built-in fuel-level sensing mechanism whose reported value is read through the diagnostic interface and filtered before being displayed or transmitted.

546	HMI reading	A measured fuel-volume or fuel-percentage reading taken from the vehicle's dashboard or embedded interface and compared with pump and receipt evidence during validation.
547	pump receipt	The receipt generated by the fuel pump during refuelling, used as supporting ground-truth evidence for dispensed fuel volume during validation.
548	odometer reading	A vehicle distance reading used to verify trip progress, fuel economy, and consistency between field observations and recorded telemetry.
549	project-native partition	The collection of telemetry records captured by the implemented prototype and used as the primary labelled evaluation partition for the thesis results.
550	public partition	A publicly derived clean-route telemetry corpus used as an out-of-distribution stress test rather than being pooled with the labelled project-native evaluation data.
551	out-of-distribution test	A validation condition in which clean data come from a different route, vehicle, or operating domain from the project-native data, used to test whether the detector creates unnecessary false alarms.
552	fuel-economy deviation feature	A derived feature that compares observed fuel use with the expected fuel economy for the fleet vehicle, driver, route, or operating state.
553	contextual gate	A deterministic rule layer that checks model flags against operational context, such as refuelling, vehicle motion, and fuel-change direction, before an alert is surfaced.

554	precision	The proportion of flagged anomaly rows that are truly anomalous, used in this study as the primary quality measure for avoiding misleading fuel alerts.
555	false-positive rate	The proportion of normal rows incorrectly flagged by the detector, used in this study to measure false-alarm behavior against the SO4 ceiling.
556	communication stack	The prioritized combination of LoRa, cellular, WiFi, and offline buffering used by the prototype to preserve telemetry delivery across changing network conditions.
557	offline buffer	A local storage mechanism on the embedded device that temporarily preserves unsent telemetry rows when no communication channel is available.
558	field-test telemetry	Telemetry collected during operational trials of the prototype, including fuel, speed, location, trip-state, and alert records used to evaluate real-world system behavior.
559	fuel anomaly	An event in which fuel behavior differs from the expected pattern for the vehicle context, such as a drop while stationary, unusually high fuel use, or a leak-like downward trend.
560	fuel-economy baseline	A reference profile of expected fuel economy for a fleet vehicle, driver, or route, used to judge whether current consumption is unusually high for that context.
561	HMI state machine	The embedded logic that controls the driver's display states, button actions, buzzer behavior, fleet-vehicle selection, trip status, rest status, and alert visibility.

562	offline replay	A recovery procedure in which locally buffered telemetry is uploaded after connectivity returns, allowing delayed records to be preserved instead of discarded.
563	route history	The stored sequence of GPS positions and trip metadata used to review where a fleet vehicle travelled and how fuel behavior changed along the route.
564	trip session	A bounded period of vehicle operation recorded by the system from trip start to trip end, including fuel, route, distance, driver, fleet vehicle, and alert data.
565	vehicle-context feature	Derived model inputs that describe the operating situation of the vehicle, such as motion state, speed, engine load, RPM, distance, route type, and refuel state.
566	dashboard usability survey	The structured questionnaire used to evaluate whether administrators found the dashboard understandable, usable, and useful for monitoring and resolving fleet events.
567	partner fleet	The participating fleet environment used for prototype consultation, vehicle inspection, demonstration, and controlled operational feedback.
568	prototype demonstration	A supervised presentation and hands-on review of the working prototype, including the embedded HMI, dashboard, telemetry flow, and alert behavior.

Chapter 1

INTRODUCTION

1.1 Background of the Study

The transportation and logistics industry is a critical driver of economic development but is increasingly facing operational inefficiencies and security challenges, especially in fuel management and vehicle tracking. Fuel theft, inaccurate fuel monitoring, inefficient routing, and lack of real-time data remain persistent issues for fleet operators, particularly in developing countries. These challenges are exacerbated by the absence of scalable, integrated technological systems that provide reliable data-driven solutions [Ahmed et al., 2017, Alquhali et al., 2019, Crisgar et al., 2021].

Fuel is one of the most substantial and recurring expenses in logistics operations, with its accurate monitoring being essential to maintaining profitability and operational efficiency. Even minor inconsistencies in fuel tracking—such as errors in manual readings, unreported consumption, or unnoticed leakage—can accumulate into significant losses over time. Komal Shoukat Ali et al. (2018) [Komal Shoukat Ali et al., 2018] report that fuel-related fraud and inefficiencies can contribute up to 30% of the total operational overhead in large fleet operations. These losses are often compounded by issues such as unauthorized vehicle use, fuel siphoning, and unoptimized driving routes, which not only drive up fuel costs but also disrupt delivery schedules, vehicle maintenance cycles, and driver accountability. Despite efforts to manage these concerns, many logistics providers continue to rely on fragmented or outdated systems that offer limited visibility into fuel-related metrics and driver behavior [Dhivyasri and Mariappan, 2015, Obikoya, 2014].

These pressures are especially acute in the Philippine setting, where fleet operations contend with some of the most severe road and logistics conditions in the region. The Japan International Cooperation Agency estimates that traffic congestion costs Metro Manila

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roughly PHP 3.5 billion per day, a figure projected to reach PHP 5.4 billion per day by 2035 without intervention [Japan International Cooperation Agency, 2022, Japan International Cooperation Agency, 2025]. Logistics operations are further constrained by congestion at the Port of Manila, which has operated at roughly 94% utilization against an ideal of about 80%, and by regulatory measures such as the 2014 Manila truck ban, which the Philippine Institute for Development Studies estimated caused around PHP 43.85 billion in economic losses and doubled container trucking costs [Patalinghug et al., 2015b, Llanto, 2017, Patalinghug et al., 2015a]. Compounding these conditions, fuel fraud and adulteration in the national supply chain have been estimated to cost up to USD 750 million annually [Asian Development Bank, 2016]. For trucks operating under such conditions, prolonged idling, stop-and-go consumption, and route variability are routine rather than exceptional, which makes fixed-threshold fuel-monitoring rules, calibrated for free-flowing conditions, especially unreliable.

In addition to financial implications, traditional fuel monitoring systems struggle to detect abnormal fuel consumption patterns in real time. These anomalies may be caused by aggressive or inefficient driving habits, extended engine idling, mechanical malfunctions (such as injector faults or fuel line leaks), or variable environmental conditions like terrain and traffic. However, without intelligent tools that correlate fuel data with contextual factors—such as vehicle speed, route topography, and time of day—fleet managers are often unable to distinguish between acceptable fuel use and problematic deviations. As highlighted by Mumcuoglu et al. (2023) [Mumcuoglu et al., 2023], conventional systems lack the sophistication to perform this level of analysis, leading to delayed detection of inefficiencies and reduced responsiveness. This underscores the urgent need for smart, data-driven solutions that combine fuel monitoring with GPS tracking and anomaly detection models to proactively manage consumption and enhance overall fleet performance [Barbado and Corcho, 2022, Beteto et al., 2022, Fortela et al., 2023].

To address these problems, researchers and engineers are increasingly exploring the convergence of IoT, GPS, and AI—particularly machine learning—to develop smart, real-time monitoring systems [Alquhali et al., 2019, Joshi et al., 2024, Prabhu et al., 2022]. These systems aim to optimize fuel usage, enhance fleet security, and improve operational efficiency through automation and predictive analytics.

The transportation and logistics sector has attracted significant scholarly attention, particularly in the development of smart systems for operational efficiency and security. The academic literature reflects a growing interest in leveraging IoT, GPS, and AI technologies to enhance fleet management, particularly in areas such as fuel monitoring, vehicle tracking, and anomaly detection. Several studies in recent years have investigated IoT-enabled fuel monitoring, GPS tracking, and anomaly detection solutions [G et al., 2022, G, 2023, Patil and Patil, 2024, Prabhu et al., 2022, RS et al., 2024, S et al., 2024].

Because partner HINO in the Philippines do not permit tank-cap or sender-loom modifications on production trucks, any field-deployable monitoring approach must be non-invasive. This constraint was the dominant driver of the acquisition-layer design adopted in this study.

- **Intelligent IoT-Enabled Real-Time Monitoring System for Logistics Management:** Joshi et al. [Joshi et al., 2024] developed a hybrid logistics management system that integrates IoT sensors, GPS, Radio Frequency Identification (RFID), and cloud analytics. Their solution achieved real-time tracking of assets and vehicles, enhancing transparency and decision-making in logistics operations. However, the

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study was limited by the type of hardware used and did not include advanced AI components for anomaly detection.

- **Fuel Consumption Classification for Heavy-Duty Vehicles:** Mumcuoglu et al. [Mumcuoglu et al., 2023] employed machine learning to classify fuel consumption anomalies based on naturalistic driving data from 57 trucks. With an accuracy of 92.2%, their approach effectively identified abnormal fuel consumption linked to driver behavior. Nevertheless, the dataset was relatively small (606 records), and integration with real-time vehicle diagnostics was not tested.

- **Development of an Intelligent Fuel Monitoring and Theft Detection System Equipped with GPS & GSM Integration:** Mathi et al. [Mathi et al., 2024] introduced an IoT-based solution combining resistance-based fuel level sensors, GPS, Global System for Mobile Communications (GSM), and real-time alarm mechanisms. Their system successfully monitored fuel and battery anomalies in hybrid vehicles but lacked higher-precision sensors and deep learning integration for smarter detection.

Despite these advancements, several limitations and research gaps remain:

- **Scalability Constraints:** Many existing solutions are limited to small-scale deployments or specific vehicle types (e.g., motorcycles, generators) and have not been validated for large truck fleets in diverse environments [RS et al., 2024, Tatababu et al., 2024].

- **Data and Sensor Limitations:** Several studies rely on basic sensors (resistance, ultrasonic) that may lack the accuracy and robustness needed for high-volume commercial operations. Furthermore, datasets used are often small or simulated, limiting generalizability [Nada, 2017, aut, 2019].

- **Fragmented Solutions:** Many systems only address a single issue — fuel monitoring, GPS tracking, or theft detection — without offering a comprehensive, unified solution integrating these components using AI and IoT [G et al., 2022, Crisgar et al., 2021].

These gaps underline the need for a robust, scalable, and integrated IoT-enabled system capable of monitoring fuel levels in real time, tracking vehicle location, and detecting operational anomalies using machine learning algorithms. Such a system holds the potential to significantly reduce fuel fraud, enhance operational efficiency, and support more informed decision-making in fleet management. This study focuses on designing and implementing such a system within an active fleet operation, specifically addressing the technical and logistical challenges encountered by commercial truck fleets operating across varied conditions. By leveraging IoT sensors, GPS modules, and unsupervised machine learning-based analytics, the proposed solution aims to deliver a comprehensive, automated, and intelligent monitoring platform that advances the current landscape of smart logistics technologies.

1.2 Prior Studies

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TABLE 1.1 LIST OF PRIOR STUDIES

Literature Title (APA Citation)	Research Problem	Method	Results	Limitations
Intelligent IoT-Enabled Real-Time Monitoring System for Logistics Management [Joshi et al., 2024]	Logistics systems lack integration and real-time monitoring capabilities for secure and efficient operations.	Hybrid IoT-based logistics management combining sensors, GPS, RFID, and cloud analytics for real-time monitoring and decision-making.	Provides real-time tracking of assets, vehicles, and equipment.	Constrained by hardware type and limited test cases; lacks advanced AI for anomaly detection.
Real Time Automatic Vehicle Monitoring System Using IoT [G et al., 2022]	Lack of real-time vehicle tracking and passenger/driver identity verification in institutional transport.	Used RFID, GPS, and NodeMCU ESP8266 with cloud storage and mobile app.	Enabled real-time tracking of location, speed, and passenger count.	Limited to institutional use; lacks anomaly detection and commercial scalability.
GPS-Based Vehicle Tracking and Theft Detection Systems using Google Cloud IoT Core & Firebase [Crisgar et al., 2021]	Ineffective vehicle recovery due to delayed or limited GPS/GSM tracking.	Implemented GPS-based tracking using Google Cloud IoT Core, MQTT, Firebase, and PWA interface.	Enabled real-time vehicle tracking, theft detection, and remote monitoring.	Designed for passenger cars; may require hardware adaptation for trucks/fleets.
Fuel Quantity and Quality Detection in Automobiles Using IoT [RS et al., 2024]	Fuel fraud through inaccurate dispensing and poor fuel quality in motorcycles.	IoT-enabled fuel management using strain gauges and quality sensors with real-time display/logging.	Achieved accurate real-time quantity and quality measurement.	Designed for motorcycles; not validated for larger fleets.
Fuel Consumption Classification for Heavy-Duty Vehicles [Mumcuoglu et al., 2023]	Identifying abnormal fuel consumption due to driver behavior or system faults.	Machine learning models using driving data from 57 trucks, considering weight and road slope.	92.2% accuracy; F1 score 0.78 for outlier detection.	Dataset limited to 606 records; lacks real-time integration.
Automatic Fuel Tank Monitoring, Tracking & Theft Detection System [Komal Shoukat Ali et al., 2018]	Inefficiency of manual fuel monitoring.	Fuel sensors, Arduino, GSM, GPS, LabView for data logging.	Real-time monitoring, alerts for theft/leakage.	Tested on basic hardware; lacks advanced analytics and large-scale validation.
Development of an Intelligent Fuel Monitoring and Theft Detection System [Mathi et al., 2024]	Fuel theft and battery anomalies in hybrid/-electric vehicles.	Resistance-based fuel sensor, voltage sensor, GPS, GSM, buzzer, LCD.	Reliable monitoring and anomaly detection.	Lacks advanced sensors for enhanced leak detection.
Unsupervised Machine Learning to Detect Impending Anomalies in Testing of Fuel Economy and Emissions [Fortela et al., 2023]	Impending anomalies in fuel economy and emissions testing of light-duty vehicles.	Unsupervised ML methods (PCA, K-Means, SOM) on 34,602 test samples.	Trend detection in large datasets.	Limited to historical datasets; not validated on real-time operational data.
Smart Fuel Monitoring System for Diesel Generator [Tatababu et al., 2024]	Fuel quantity indication and theft in generators.	Ultrasonic/capacitive sensors, flow sensors, GSM, GPS, intelligent theft detection.	Accurate real-time monitoring, theft prevention.	Findings specific to tested generators; limited generalizability.

Literature Title (APA Citation)	Research Problem	Method	Results	Limitations
Design and Implementation for Trucks Tracking System Using GPS Based on Semantic Web [Nada, 2017]	Inefficient GPS-based truck tracking for fleet management.	GPS tracking system integrated with semantic web platform.	Accurate location tracking and reporting via web interface.	Dependent on GSM/GPS reliability; lacks IoT integration and anomaly detection.
Automatic Fuel Monitoring System [aut, 2019]	Accurate fuel monitoring and theft prevention.	Fuel sensors and GSM module with SMS alerts.	Monitors fuel, triggers alarms, and sends GPS alerts.	Basic sensors; limited scalability and analytics in complex environments.
Interpretable Machine Learning Models for Vehicle Fuel Consumption Anomalies [Barbado and Corcho, 2022]	Detecting/explaining fuel anomalies in fleets to optimize usage.	Unsupervised anomaly detection combined with interpretable ML.	Up to 80% anomalous instances explained; potential fuel reduction up to 88%.	Only analyzed individual feature relevance; not fully integrated in real-time systems.

1.2.1 Research Gaps Identified

A review of existing literature in fuel monitoring and anomaly detection for logistics and fleet management reveals persistent gaps that point to a need for more robust, integrated solutions. Although many systems incorporate IoT elements—such as sensors, GSM, and GPS—these are often limited in scope, typically designed for specific applications like diesel generators or motorcycles, rather than being adaptable for broader fleet use.

For example, systems like the Smart Fuel Monitoring System for Diesel Generators [Tatababu et al., 2024] and the Automatic Fuel Monitoring System [aut, 2019] demonstrate basic real-time monitoring and theft detection capabilities. However, these systems heavily rely on elementary sensor technology and offer minimal integration with advanced analytics or anomaly detection methods. Similarly, the machine learning-based approach in [Mumcuoglu et al., 2023] was limited in its use of data and did not incorporate real-time or unsupervised learning techniques.

While [Barbado and Corcho, 2022] explored the potential of interpretable machine learning and unsupervised anomaly detection, their models were evaluated on constrained datasets and not integrated into real-time operational systems. [Fortela et al., 2023] also demonstrated effective anomaly detection using unsupervised learning in emissions and fuel economy testing, but again, their data was static and gathered in controlled environments. These findings were not validated using on-vehicle, operational data, which limits their applicability in dynamic, real-world conditions.

Additionally, studies such as those by [Joshi et al., 2024] and [Nada, 2017] underscore the need for real-time GPS tracking in fleet operations, but they stop short of presenting a cohesive system that incorporates GPS, fuel monitoring, and intelligent anomaly detection into one unified platform.

The core research gap, therefore, lies in the absence of a fully integrated, real-time, IoT-based system that combines fuel monitoring, GPS tracking, and anomaly detection powered by unsupervised machine learning. Addressing this gap is vital to improving operational efficiency, fuel security, and predictive maintenance in modern fleet management.

1.2.2 Implemented Response to the Identified Gaps

This study developed and evaluated an integrated IoT-based fleet monitoring system that unifies fuel level monitoring, GPS-based vehicle tracking, and unsupervised machine learning for anomaly detection. Unlike previous solutions that focus on isolated functions or static datasets, the developed system operated in real time using vehicle telemetry acquired through the OBD-II interface and GPS module. The prototype was evaluated using controlled-injection trip data and field-test telemetry data to assess fuel monitoring performance, anomaly detection capability, and communication latency.

The system architecture incorporates scalable and cost-efficient components such as

OBD-II acquisition, GPS modules, and cloud-based transmission via mobile networks, ensuring compatibility with partner-fleet operating constraints.

For anomaly detection, the system employs unsupervised, non-parametric machine learning algorithms such as Matrix Profile and Isolation Forest. These models identify outliers within time-series data without the need for labeled training datasets, making them suitable for fleet scenarios where data behavior varies with routes, driver habits, or environmental conditions. Unlike traditional parametric approaches, non-parametric models are more flexible and adaptive because they do not assume any fixed data distribution.

In addition, synchronized HMI, mobile, and dashboard interfaces provide meaningful insights by correlating fuel usage patterns with routes and travel times. These interfaces support real-time notifications such as driver rest alerts, maintenance prompts, and efficiency summaries, enabling data-informed decision-making and operational improvements.

By addressing the limitations of prior research—especially the lack of integration, real-time analysis, and adaptive anomaly detection—this study offers a practical and scalable solution for enhancing fuel management and fleet operations in dynamic environments.

1.3 Problem Statement

Efficient fuel monitoring and route tracking are essential components of sustainable and cost-effective truck fleet operations. However, existing systems face significant challenges, including inaccurate fuel measurement, disconnected data sources, delayed anomaly detection, and limited support for safety-based decision-making, as outlined below.

1. PS1: The Ideal Scenario

- Trucking companies should have real-time access to accurate fuel usage and GPS tracking data to ensure efficient operations, reduce fuel theft, and improve route planning.
- Drivers should receive timely rest alerts and vehicle maintenance reminders based on actual driving and fuel metrics to promote safety, avoid overfatigue, and extend vehicle life.
- Fleet managers should be able to detect unusual fuel consumption patterns automatically, enabling early intervention and optimized scheduling.
- All of these should operate through automated, cloud-based systems that integrate IoT sensors, machine learning, and remote dashboard monitoring.

2. PS2: The Reality of the Situation

- Most trucking operations lack real-time visibility into fuel usage, relying instead on manual readings or delayed reports that are prone to error and manipulation.
- GPS tracking is often siloed from other performance indicators like fuel consumption and driving behavior, making behavior-based analysis or route optimization difficult.
- Anomalous fuel consumption events (e.g., theft, leakage, or inefficiency) are rarely detected until long after they've occurred, resulting in financial loss and operational delays.
- Rest and maintenance schedules are typically based on fixed intervals rather than real-time usage data, increasing the risk of driver fatigue, breakdowns, and missed service opportunities.

- Existing technologies are either too costly, fragmented, or lack contextual intelligence to automatically correlate GPS, fuel, and operational metrics.

3. PS3: The Consequences for the Audience

   - Truck fleet operators face increasing operational costs due to undetected fuel inefficiencies and theft, impacting business sustainability.
   - Driver safety is compromised when rest is not scheduled based on actual driving patterns or system recommendations.
   - Vehicle lifespan shortens and maintenance costs rise when alerts are not tied to real-time usage, resulting in unexpected breakdowns and downtime.
   - Decision-makers are unable to make informed, data-driven decisions due to fragmented systems and the absence of an integrated monitoring and alert platform.
   - These issues collectively hinder operational efficiency, safety, and environmental responsibility, particularly for logistics providers in developing regions lacking advanced fleet management tools.

1.4 Objectives and Deliverables

1.4.1 General Objective (GO)

GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.;

1.4.2 Specific Objectives (SOs)

- SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within $\pm 5\%$.
- SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.
- SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.
- SO4: To design and evaluate a context-aware fuel usage deviation model using non-parametric unsupervised machine learning models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.
- SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.

1.4.3 Final Deliverables

1.5 Implemented Deliverables per Objective

TABLE 1.2 IMPLEMENTED DELIVERABLES PER OBJECTIVE

Objectives	Implemented Deliverables
GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.	1. **Trained Unsupervised Machine Learning Model** • A validated ML model (Matrix Profile or Isolation Forest) trained on SGSA-reflective datasets for detecting anomalous fuel consumption. 2. **Web-Based Monitoring Interface** • Real-time dashboard showing GPS location, fuel level, distance, and operating hours for each fleet vehicle. 3. **Embedded In-Vehicle HMI** • Driver-friendly interface displaying driver credentials (name, PIN, and assigned fleet vehicle), current trip activity, and alert notifications. 4. **Cloud Backend Infrastructure** • Cloud-based data aggregation, processing, and automated alert generation accessible from SGSA's base. 5. **Performance Documentation** • Complete validation results under laboratory and SGSA field conditions.
SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/-diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within ±5%.	1. Calibrated fuel-level acquisition module interfaced through the fleet vehicle ECU/OBD-II connector. 2. Accuracy validation report comparing filtered OBD-II fuel data with dashboard-gauge levels. 3. Embedded firmware for fuel data acquisition, filtering, and preprocessing. 4. Drift, noise, and variance analysis across different tank levels and conditions.

Objectives	Implemented Deliverables
SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.	1. Integrated IoT communication module with LoRa, LTE, GSM 2G, WiFi, and SPIFFS offline replay. 2. Cloud backend with secure API endpoints for receiving fuel and GPS data. 3. Latency benchmarking report under variable network environments. 4. Optimized HTTP transmission logic with buffering and fail-safe retries; MQTT is retained as a future-compatible communication option. 5. Security documentation covering packet integrity and encryption features.
SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	1. Integrated GPS module capturing coordinates and timestamps at 5-second intervals. 2. Data fusion algorithm linking GPS location, fuel levels, and time progression. 3. Route visualization through a map interface on the dashboard. 4. Recommended route engine with truck-ban zone awareness. 5. Database structure storing synchronized fuel–location–time datasets. 6. Behavioral analytics showing fuel usage per route segment.
SO4: To design and evaluate a context-aware fuel usage deviation model using non-parametric unsupervised machine learning models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.	1. Labeled normal and anomalous fuel consumption datasets. 2. Implementation of Isolation Forest and/or Matrix Profile in Python/-MATLAB. 3. Evaluation metrics report including precision, recall, and FPR $\leq$10%. 4. Exportable model (PMML/ONNX) for real-time pipeline integration. 5. Context-aware analytics combining slope, route, load, and time-of-day factors.
SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.	1. Cloud dashboard showing fuel usage, distance, operating hours, and maintenance status. 2. Rule-based alert engine for configurable driver safety and vehicle maintenance notifications, with example thresholds such as 300 km or 6-hour rest reminders and 5,000 km oil maintenance alerts. 3. Physical embedded module: Start/Stop button, buzzer/light indicators for rest and maintenance alerts. 4. Backend logic computing distance-based and time-based safety thresholds. 5. Dashboard filters for alert logs and historical events. 6. Extended safety monitoring: speed tracking and overspeed alerts.

1.6 Significance of the Study

1.6.1 Technical Benefit

This research developed a context-aware, cloud-connected fleet monitoring system that integrates diagnostic-connector fuel acquisition, GPS route tracking, and unsupervised machine learning to enhance real-time monitoring of fuel usage and detection of anomalous consumption events. Fuel levels are measured using digital fuel-level data exposed through the truck's ECU/diagnostic connector, with the system targeting an accuracy of $\pm 5\%$ against known-volume pump-gas events and manufacturer dashboard readings. Real-time fuel and GPS data are transmitted through IoT-based communication to a cloud server and made accessible through the centralized dashboard.

GPS modules embedded in the vehicles log location data every 5 seconds, enabling the system to correlate fuel consumption with time, distance, and route conditions for behavioral tracking and operational analysis. The study implements non-parametric unsupervised learning methods, specifically Matrix Profile and Isolation Forest, to identify deviations in fuel usage without the need for labeled training datasets, allowing scalable and adaptive anomaly detection. In addition, the cloud backend integrates hardware-driven rest alerts and maintenance reminders, leveraging cumulative fuel and distance data to generate intelligent break schedules and service interval notifications. Through these combined capabilities, the system aims to improve technical performance by enhancing measurement accuracy, communication responsiveness, anomaly detection reliability, and overall automation in fleet fuel management and driver safety support.

1.6.2 Social Impact

Through an implementation of automatic rest alerts and preemptive maintenance cues, the study encourages safety and health of the drivers to prevent accidents commonly caused by fatigue as well as breakdowns. The system informs drivers when they are near critical or recommended limits, such as 300 km of continuous driving or extended operating duration, encouraging healthy driving practices and compliance with regulations, especially for long-distance hauls. These practices empower fleet managers to systematically enforce safety policies and promote a culture of responsibility in the industry.

In addition, GPS-based route monitoring and fuel usage logging reinforce accountability and transparency, two essentials in preventing fuel misuse. This helps build trust between management and drivers and supports data-led decision-making in transport logistics.

1.6.3 Environmental Welfare

The system improves operational efficiency by minimizing fuel waste, optimizing route performance, and decreasing unplanned maintenance downtime. Through precise anomaly detection and maintenance forecasting, the platform enables cost-effective fleet operation and longer vehicle life. It also supports environmental sustainability by reducing unnecessary fuel consumption and emissions from inefficient routes, fuel leaks, or late maintenance. By promoting timely service interventions, such as oil changes at every 5,000 kilometers, the study supports cleaner engine performance and reduced environmental impact over time.

1.7 Assumptions, Scope, and Delimitations

1.7.1 Assumptions

1. Technical Infrastructure

    - Consistent internet connection is required to maintain real time data exchange between the IoT sensors, GPS module, and cloud sensor.
    - The system integration between the hardware and cloud based infrastructure operates smoothly without significant technical issues.
    - The system will have access to a stable power supply for an uninterrupted operation of the on board IoT system.
    - Mobile network coverage is able to handle a low latency data communication.

2. Data Accuracy

    - The OBD-II/ECU-based fuel acquisition channel is assumed to provide operationally useful fuel-level readings after filtering and validation against known-volume pump-gas events and manufacturer dashboard readings. Direct dipstick validation was outside the permitted field procedure.
    - The GPS module captures and logs precise coordinates at 5-second intervals, enabling accurate route reconstruction and correlation with fuel consumption and time data.

3. Machine Learning Model Performance

- The unsupervised learning models (e.g., Matrix Profile, Isolation Forest) are capable of identifying anomalous fuel consumption with 90% precision and 10% false positives within the controlled environments evaluated in this study.
- The training and evaluation data of anomaly detection will be representative of normal fuel consumption and truck behavior under different operating conditions.

4. User Interaction

- It is assumed that users (e.g., truck drivers, fleet operators, and system administrators) will interact with the system according to their roles and available tools.
- Truck drivers are not required to use personal mobile devices; instead, they will rely on a simple onboard interface consisting of a small display and buzzer for receiving rest alerts, maintenance reminders, and status notifications. However, a mobile web application was also developed as a supplementary HMI component to support navigation and provide additional convenience for drivers with personal mobile devices.
- Fleet operators and system administrators will interact with the cloud-based dashboard using standard office computers to monitor fuel data, GPS routes, alerts, and maintenance logs.
- Fleet personnel are expected to follow system recommendations (e.g., rest intervals, maintenance reminders) based on real-time fuel, distance, and performance metrics.

5. Operating Environment
    - Environmental conditions, together with power and network breakdowns, will not significantly affect system performance.
    - System maintenance is performed routinely and at normal intervals to ensure operational continuity.

1.7.2 Scope

The scope of the study encompasses the development and implementation of a smart fleet monitoring system that utilizes IoT and machine learning technologies for enhanced fuel usage , route monitoring and vehicle maintenance reminder.

1. **Real-Time Fuel Monitoring**
    a) Deployment of a filtered fuel-level acquisition module interfaced through the truck's ECU/OBD-II diagnostic connector, providing continuous fuel-level monitoring validated against known-volume pump-gas events and manufacturer dashboard readings.
2. **Cloud Based Data Transmission**
    a) The system implements prioritized telemetry transmission through LoRa, LTE, GSM 2G, WiFi, and SPIFFS offline replay, with latency and packet-delivery performance evaluated under field and controlled conditions.
3. **GPS Enabled Route Tracking**

   a) Incorporation of GPS modules for logging vehicle location at a sampling rate of 5 seconds, allowing correlation of fuel consumption with route patterns and driving behavior.

4. **Anomaly Detection via Machine Learning**
   a) Context-aware fuel deviation detection model development by means of unsupervised learning methods (e.g., Isolation Forest, Matrix Profile).
   b) Analysis of precision and false positive rate of the model under the project-native evaluation protocol.

5. **Driver Rest Alerts and Maintenance Reminders**
   a) Integration of an in-vehicle HMI, mobile driver web application, and administrative dashboard for viewing trip state, fuel status, alerts, and cumulative fuel and distance measures.
   b) Production of automated rest alerts (e.g. every 300 km or after 6 hours) and maintenance reminders (e.g. after every 5,000 km for checking oil).

1.7.3 Delimitations

1. **Focus on Fuel Monitoring and Anomaly Detection**
   - The system focuses on fuel level monitoring, detecting consumption anomalies, and providing maintenance guidance based on fuel usage and distance traveled. The prototype incorporates selected OBD-II context variables such as speed, RPM, engine load, throttle, MAF, coolant temperature, runtime, and DTC

status, but does not implement full manufacturer-grade diagnostics or advanced vehicle-health fault isolation.

2. **Hardware and Communication Constraints**

    - The system utilizes the truck's ECU/OBD-II diagnostic connector, an embedded GPS module, and a microcontroller-based HMI node; higher-end industrial sensing platforms are not employed.
    - Real-time data transmission primarily uses IoT communication protocols, enabling low-latency delivery of fuel and GPS data to the cloud backend.
    - A cellular module (LTE & GSM) functions as a fallback communication pathway, activated only during LoRa connectivity loss, and is limited to transmitting essential alerts and critical status messages due to bandwidth and cost constraints.
    - Other wireless communication technologies beyond the selected IoT modules are not implemented within the system.

3. **Limited Fleet Size and Geography**

    - The system testing will be limited to the number of vehicles provided for use by the trucking company within a specific geographic area. Large scale testing and deployment across a wider range of fleet types and terrains are not covered.

4. **User Features**

    - The system includes three interfaces: an embedded in-vehicle HMI, a mobile driver web application, and an administrative dashboard. Driver actions, mobile trip state, and dashboard monitoring are synchronized through the backend.

- The administrative dashboard is limited to essential features such as real-time fuel level monitoring, GPS tracking, driver rest alerts, maintenance reminders, trip history, telemetry logs, alert management, analytics summaries, threshold configuration, user management, and fleet management.

1.8 Description and Methodology

This study employs a quantitative research paradigm combined with a data-centric, system development approach to implement an AI-driven fleet fuel monitoring and anomaly detection system. The methodology integrates theoretical foundations from Chapter 3 on cyber-physical systems and IoT architectures, system design principles from Chapter 4, and practical implementation strategies detailed in Chapter 5. The system leverages real-time vehicle data acquired through the OBD-II/CAN interface, GPS modules for location tracking, and embedded alert subsystems for driver safety. These components feed into a microcontroller that preprocesses signals, synchronizes time-series data, and transmits information to a cloud platform through prioritized LoRa, LTE, GSM 2G, WiFi, and offline-replay channels.

Offline, historical and hybrid datasets, combining fleet telemetry and publicly available vehicle logs, are used to train non-parametric unsupervised machine learning models such as Matrix Profile and Isolation Forest. These models identify anomalous fuel consumption patterns and provide context-aware alerts. In the cloud, a Python-based engine continuously evaluates incoming data, logs anomaly events, and stores synchronized measurements in a centralized database. Users access the information through an embedded HMI, a mobile driver web application, and an administrative dashboard that display real-time truck status, historical fuel usage, route playback, trip state, and operational alerts.

The methodology includes signal preprocessing (smoothing via moving averages), time synchronization using linear interpolation, hybrid dataset construction, and rigorous evaluation metrics (precision, false positive rate, accuracy, detection latency) to validate system objectives, as elaborated in Chapter 5. The approach follows iterative, agile development cycles, allowing continuous refinement of sensor integration, communication protocols, and anomaly detection algorithms, thereby ensuring reliability, scalability, and actionable fleet monitoring insights.

1.9 Research Dissemination Plan

The findings of this study will be disseminated to relevant academic, industrial, and technical stakeholders to promote the adoption and further development of intelligent fleet monitoring technologies. Upon approval of the study, the final manuscript will be submitted to the institutional repository of De La Salle University, including the Animo Repository, in accordance with university research archiving policies. Primary target audiences include the research partner Hino Paranaque Subic GS Auto Inc. (SGSA), fleet managers, logistics operators, academic researchers, engineering students, and faculty members within the field of IoT, transportation systems, and machine learning.

The dissemination process will occur in multiple stages. During the implementation and testing phase, progress updates and preliminary findings will be presented to SGSA representatives and thesis advisers for technical evaluation and operational feedback. Upon completion of the study, the final results, including system performance metrics, anomaly detection accuracy, latency measurements, and operational observations, will be formally presented through the thesis defense at De La Salle University. A digital and printed copy of the approved manuscript will also be submitted to the institutional repository of De La Salle University, Animo Repository, in accordance with institutional requirements. In addition, the proponents may present selected findings in future academic conferences, research exhibits, or technology demonstrations related to IoT systems, intelligent transportation, or artificial intelligence applications.

The dissemination timeline will follow the project completion schedule presented in Section 1.9, including the individual Gantt charts of the proponents. Preliminary technical findings and prototype demonstrations will be shared during the system testing and validation phase. Final dissemination activities, including the thesis defense, manuscript submission, and partner presentation, will occur after the completion of data analysis, evaluation, and manuscript revisions. Any future publication or presentation derived from the study will only use anonymized and non-sensitive operational data to preserve confidentiality and comply with ethical and data privacy standards.

The effectiveness of the dissemination activities will be evaluated through several criteria, including feedback from thesis panel members, adviser evaluations, partner company assessments, and audience engagement during presentations and demonstrations. Additional indicators of dissemination effectiveness include successful acceptance of the manuscript, utilization of recommendations by the partner company, and the potential application or continuation of the proposed system in future research or operational deployments.

De La Salle University

1.10 Conflict of Interest

None.

1.11 Estimated Work Schedule and Budget

1.11.1 Gantt Chart

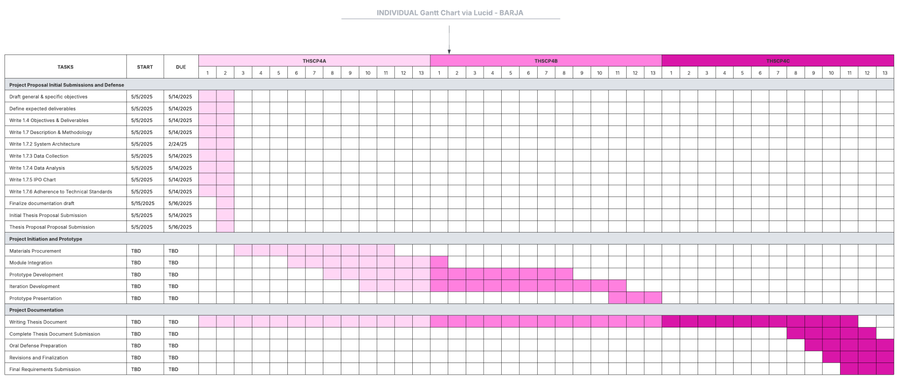

Fig. 1.1 Individual Gantt Chart – Barja

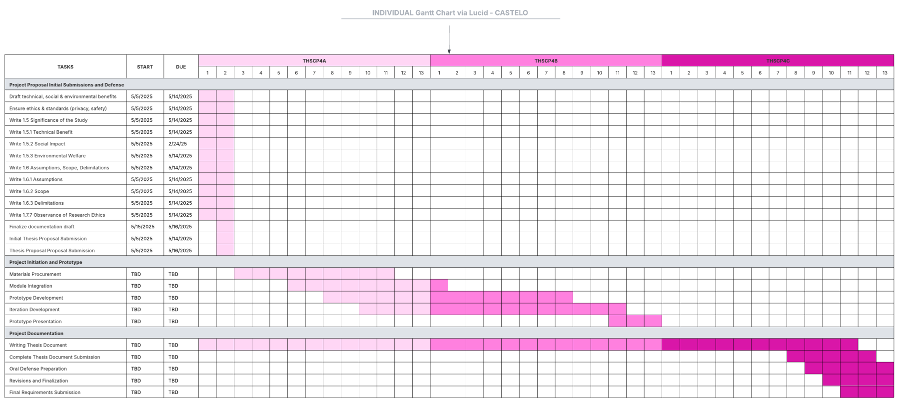

Fig. 1.2 Individual Gantt Chart – Castelo

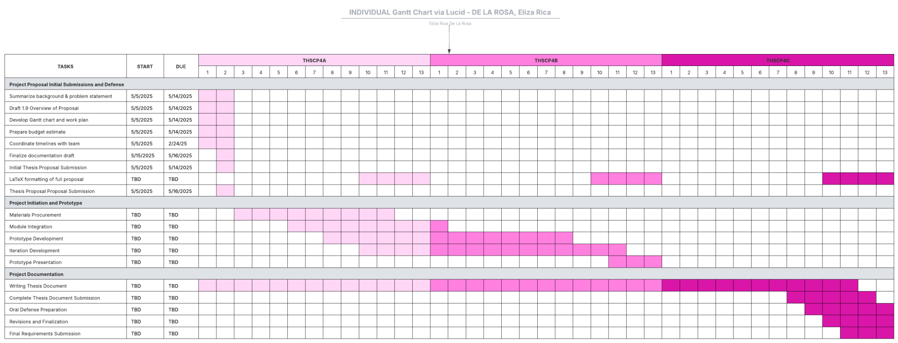

Fig. 1.3 Individual Gantt Chart – De La Rosa

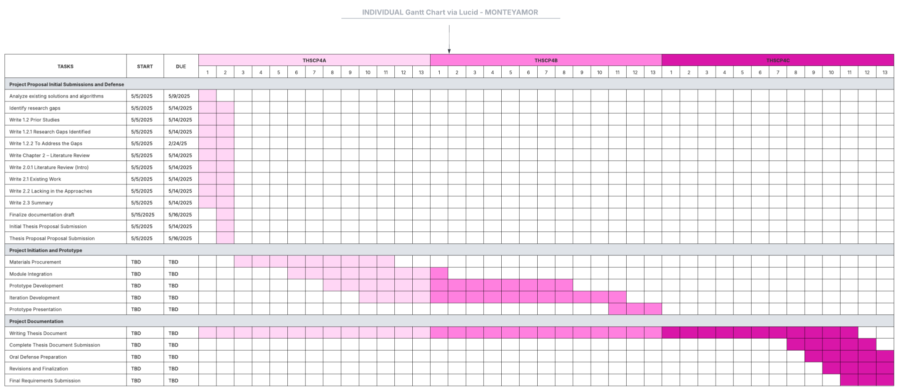

Fig. 1.4 Individual Gantt Chart – Montemayor

1.11.2 Estimated Budget

TABLE 1.3 DETAILED BUDGET BREAKDOWN FOR THE PROJECT

Item / Activity	Unit	Qty	Unit Cost (PHP)	Total Cost (PHP)	Remarks
A. Covered by Hino Motors					
Use of truck for testing	Truck	2	Sponsored	-	Provided by Hino
B. Covered by Research Team					
Hardware Components					
CAN Bus / OBD-II Interface Module		1	1,200	1,200	Reads fuel levels and vehicle parameters from ECU
GPS Module	-	1	650	650	Route tracking and speed monitoring
ESP32 DevKit 30 pins	-	2	350	700	Main microcontroller boards
A7670E	-	1	1,300	1,300	Cellular communication module
MCP2515 CAN Bus Module	-	1	91	91	CAN bus interface module
Buzzer Module	-	1	45	45	Audio alert component
Buttons	-	2	10	20	Driver input controls
Arduino TFT Screen 3.5 in	-	1	648	648	In-vehicle display interface
LoRa Module Ra-02 433 MHz	-	2	218	436	Long-range communication modules
OBD2 16-pin connector	-	1	154	154	Vehicle diagnostic connector
GPS Module NEO-M8N	-	1	600	600	GPS tracking module
Resistors (1x220 ohms, 1x5 kohms)	-	2	1	2	Circuit protection components
SIM card (TNT)	-	1	100	100	Cellular network access
Prototype Case					
3D Model and Printing	-	1	4,500	4,500	Prototype enclosure fabrication
Total Estimated Budget				**10,446**	

1.12 Overview of the Thesis Paper

This chapter provides an overview of the study, which addresses inefficiencies in logistics and trucking operations while tackling safety concerns related to fuel monitoring, vehicle tracking, and anomaly detection. Modern trucking operations involve complex coordination, and current approaches—manual fuel measurements, fragmented GPS tracking, and delayed anomaly detection—are insufficient for ensuring operational efficiency and driver safety. This work developed and evaluated an integrated, IoT-based intelligent monitoring platform that acquires digital fuel-level data through the truck's ECU/diagnostic connector, with a four-stage filter cascade (median-of-31, EMA, dead-band, direction-aware step rule) implemented on an ESP32-based in-truck HMI, GPS, and cloud-deployed machine learning models to provide real-time insights and actionable alerts.

The problem statement highlights a fragmented monitoring environment with limited automation, reactive maintenance scheduling, and the inability to detect unusual fuel consumption proactively. The objectives of the research focus on developing a real-time system that reads fuel levels from the truck's ECU via CAN bus, tracks location and speed using GPS, and identifies anomalies in consumption patterns using unsupervised machine learning models (Matrix Profile and Isolation Forest). The system is supported by a cloud-based dashboard, which provides fleet managers with fuel usage insights, rest alerts, route tracking, and preventive maintenance notifications. This approach promotes operational efficiency, enhances driver safety, and reduces fuel wastage, making it a technically and environmentally relevant solution.

Building on this foundation, Chapter 2: Literature Review examines previous research on IoT-based fleet monitoring systems, GPS-enabled tracking technologies, and machine learning methods for anomaly detection. The review compares existing methods, identifies gaps in real-time deployment, system integration, and context-aware alerting, and evaluates technology innovations and algorithmic limitations. This analysis establishes the academic basis for the study and demonstrates how the developed system advances the state of the art in smart fleet management by providing a fully integrated, real-time, and context-aware monitoring solution.

Chapter 2

LITERATURE REVIEW

2.1 Systematic Literature Review

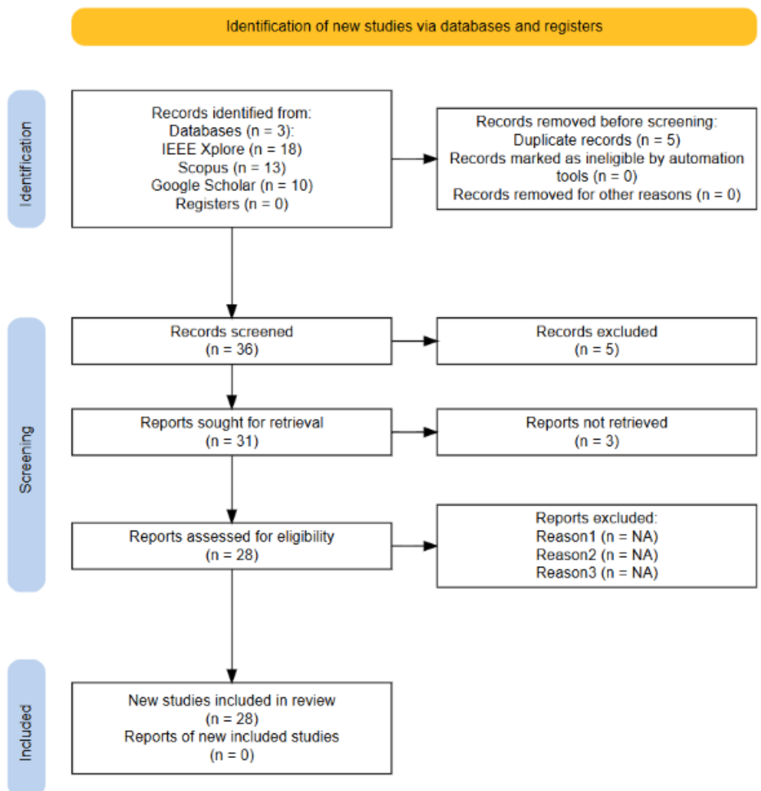

Fig. 2.1 PRISMA Diagram

The PRISMA diagram illustrates the systematic process used for selecting studies for review through screening and identification. The search identified a total of 41 records from three databases: Scopus, IEEE Xplore, and Google Scholar. Five (5) duplicate records were excluded from the 41 records, resulting in 36 records screened. The evaluation of 36 records resulted in five exclusions due to records being outdated/irrelevant, which reduced the number of reports needed for retrieval to 31. Three (3) reports were not retrieved upon retrieval, resulting in 28 reports assessed for eligibility. The final review included 28 studies only, while no new studies were reported.

2.2 Existing Work

2.2.1 Vehicle Tracking and Fuel Sensing Technologies

Fuel Monitoring Systems

Various fuel monitoring systems have been proposed, discussed, and developed in existing studies, emphasizing real-time fuel level, theft indication, and enhanced operational efficiency. In the work of Joshi et al. [Joshi et al., 2024], a system combines ESP32 and Arduino UNO microcontrollers with the ThingSpeak cloud-based platform to measure fuel levels, control transactions, and enable predictive maintenance at fuel stations. Similarly, Ahmed et al. [Ahmed et al., 2017] developed a fuel management system using Raspberry Pi and both ultrasonic and chemical Etape sensors to acquire real-time fuel level data, displayed through a web application.

In the automotive domain, several studies aim to enhance the accuracy of fuel management through sensor integration. The digital system by Sakthimohan et al. [M, 2021] employs ultrasonic and flow sensors with an Arduino UNO to display real-time measurements on an LCD, providing a replacement for imprecise analog gauges. Meanwhile, Farzana et al. [Farzana, 2020] proposed a motion-based fuel monitoring technique that integrates vehicle motion parameters with fuel level sensor readings, achieving 80% success in accurately determining fuel levels and 100% detection success in identifying fuel theft and refills via a smartphone application.

Studies focused on fuel theft detection, such as those by Tatababu et al. [Tatababu et al., 2024] and Mathi et al. [Mathi et al., 2024], utilize capacitive or ultrasonic sensors, GPS tracking, and GSM-based alerts to detect tampering or fuel loss and notify operators immediately. In the Intelligent Fuel Monitoring and Theft Detection System using GPS & GSM, Mathi et al. [Mathi et al., 2024] also monitored battery charge (SoC), with applicability for both hybrid and conventional vehicles, making it suitable for logistics and fleet management.

Ali et al. [Komal Shoukat Ali et al., 2018] presented an automatic fuel tank monitoring, tracking, and theft detection system integrating Arduino-based fuel circuits with GSM/GPS modules and LabVIEW for database management. This system replaces manual fuel tracking with automated alerts and volume monitoring, offering a cost-effective solution for transportation industries. Overall, these systems indicate a trend toward inexpensive, sensor-based, remotely accessible fuel monitoring technologies with high priority on accuracy, security, and real-time data analysis.

GPS and Vehicle Telematics

Global Positioning System (GPS) and vehicle telematics technologies have become essential components in modern vehicle monitoring and fleet management systems. These technologies allow real-time tracking, remote diagnostics, and performance analysis by collecting and reporting data such as location, speed, fuel consumption, and engine status. A common trend in these studies is the combination of GPS and GSM modules to facilitate real-time vehicle tracking, usually complemented by cloud-based or mobile-supported platforms. Joshi et al. [Joshi et al., 2024] proposed a system enabling real-time monitoring for logistic management using sensors, GPS tracking, and cloud-based data analytics. Similarly, Alquhali et al. [Alquhali et al., 2019] and Nada [Nada, 2017] employed GPS and GSM modules to send live location data, show vehicle tracks on electronic maps, and record location history for future reference and analysis. Vigneshwaran et al. [Vigneshwaran et al., 2020] presented a GPS-GSM-based bus tracking system supporting student notifications for bus arrival, emergency notifications, and route alerts, enhancing safety and saving waiting time. The system further offered a framework for real-time fuel monitoring with implications for broader fleet management applications.

2.2.2 LoRa Technology

LoRa (Long Range) is a wireless communication technology engineered for Internet of Things (IoT) and machine-to-machine (M2M) communication. It supports low-power, battery-operated embedded systems that transmit small amounts of data over long distances [Orange Connected Objects & Partnerships, 2016]. Traditional cellular networks such as GSM, 2G, 3G, and 4G prioritize high data throughput, which can result in inefficient power consumption for IoT applications that require infrequent, small data transmissions [Majeed, 2015, Lee et al., 2007]. Unlike cellular networks or short-range protocols such as Wi-Fi and Bluetooth, LoRa is optimized for applications where energy efficiency and communication range are more important than high throughput [Devalal and Karthikeyan, 2018].

The LoRa communication architecture follows a structured flow of information so that small data packets can reach cloud-based systems efficiently:

- **LoRa Node or End Device:** This component includes the sensors or applications where sensing and control take place. It collects telemetry data and transmits them as small, optimized data packets.
- **LoRa Gateway:** The gateway receives packets from end nodes and acts as a bridge between LoRa nodes and the internet.

- **Cloud Backend:** Once the gateway forwards data through a backhaul connection, the information reaches a network or application server where it can be processed, visualized on dashboards, or stored for further analysis.

2.2.3 LoRa Gateway

LoRa gateways are multi-channel, multi-modem transceivers capable of simultaneous reception and decoding of LoRa packets from numerous end nodes. Each gateway acts as a bridge that connects LoRa nodes to the internet by capturing transmitted node data and forwarding them through standard backhaul connections such as Wi-Fi, Ethernet, or cellular data. Data from different gateways are processed through a network server, which functions as the central intelligence of the LoRa architecture. This server performs security authentication, filters messages, manages adaptive data rates, and relays appropriate acknowledgments back to the gateways [Orange Connected Objects & Partnerships, 2016].

Gateway deployment varies according to coverage requirements. Micro gateways are typically deployed in public networks to provide broad city-wide or nationwide connectivity, while pico gateways are used in dense or challenging environments to improve network capacity and service quality. To support these deployment scenarios, LoRa base stations may use omni-directional or multi-sector antenna configurations [Devalal and Karthikeyan, 2018].

2.2.4 Technical Performance Metrics and Environmental Factors

The operational effectiveness of LoRa networks depends on the interaction between transceiver hardware configuration and environmental conditions. According to Chengdu Ebyte Electronic Technology, the communication distance of a LoRa module is influenced by transmit power, receiver sensitivity, and antenna gain [Chengdu Ebyte Electronic Technology Co., Ltd., 2024]. In open rural areas, a standard LoRa layout can achieve communication links of approximately 15 to 20 kilometers. In dense urban environments, however, coverage may decrease to approximately 1 to 5 kilometers because of building penetration losses and multipath propagation effects [Chengdu Ebyte Electronic Technology Co., Ltd., 2024].

Manalu et al. [Manalu et al., 2023] evaluated the performance of a LoRa-based vehicle tracking system using a HopeRF-RFM95W transceiver module operating at 915 MHz on a moving vehicle. The study analyzed link-quality parameters in a Non-Line-of-Sight (N-LoS) suburban environment at speeds of 10 km/h, 20 km/h, and 30 km/h. The evaluation focused on how buildings, vegetation, and other wireless signals affect key performance metrics. Received Signal Strength Indicator (RSSI) measures absolute signal power and degrades behind thick barriers, vegetation, or dense buildings. Signal-to-Noise Ratio (SNR) measures signal clarity against electromagnetic interference and may drop significantly because of disruptions from pedestrians, structures, vehicles, and Wi-Fi towers. Packet loss may also increase sharply near major structural obstacles, showing that LoRa performance in mobile fleet applications must account for physical barriers and N-LoS conditions [Manalu et al., 2023].

2.2.5 Real-World Implementation: LoRa-Based Telemetry and Fleet Management Systems

The long-range and low-power characteristics of LoRa make it suitable for geographic areas with limited commercial telecommunication infrastructure. Common applications include smart agriculture, pasture health monitoring, and isolated industrial operations. However, reliance on a single low-power wide-area network (LPWAN) protocol may introduce communication risks when asset-tracking nodes move between remote rural spaces and dense urban areas.

To reduce blind spots, industrial systems may use hybrid communication architectures. The Meitrack LoRa Fleet Tracking Solution illustrates this approach in large-scale agricultural operations, such as orchards and palm plantations, where GSM coverage may be unreliable or unavailable [Meitrack,]. In such environments, multi-modem tracking hardware can use a dual LoRa and GPRS switching configuration based on network availability. While operating in cellular dead zones, localized sensor payloads, including ultrasonic fuel-level data, driver authentication logs, and safety alerts, can be routed through LoRa to a local base station. This architecture helps preserve asset visibility when standard cellular infrastructure is unavailable [Meitrack,].

2.2.6 GSM and LTE Technologies

The Global System for Mobile Communication (GSM) is a widely recognized standard for digital cellular technology. It originated from a group formed in 1982 to develop a unified European mobile telephone standard for cellular radio systems operating at 900 MHz [The International Engineering Consortium,]. GSM is known for broad geographic coverage and supports voice calls and basic data services through second-generation cellular communication. Long-Term Evolution (LTE), by contrast, is a wireless broadband communication standard that succeeded 3G and became a cornerstone of 4G mobile networks. LTE development began as a Third Generation Partnership Project (3GPP) effort in 2004 to provide a new radio access technology focused on packet-switched data [Miyim and Wakili, 2019].

LTE was developed to address increasing demand for mobile broadband services by improving network capacity, data rates, quality of service, and latency. These improvements support mobile wireless connectivity that aligns with performance benchmarks established by the International Telecommunication Union (ITU). After LTE was established in 2009, further performance requirements led to LTE-Advanced (LTE-A), which satisfied International Mobile Telecommunications (IMT) requirements for fourth-generation networks and achieved peak data rates of up to 1 Gbps [Miyim and Wakili, 2019, Kanchi et al., 2013].

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Fig. 2.2 Globe GSM and LTE Coverage Map

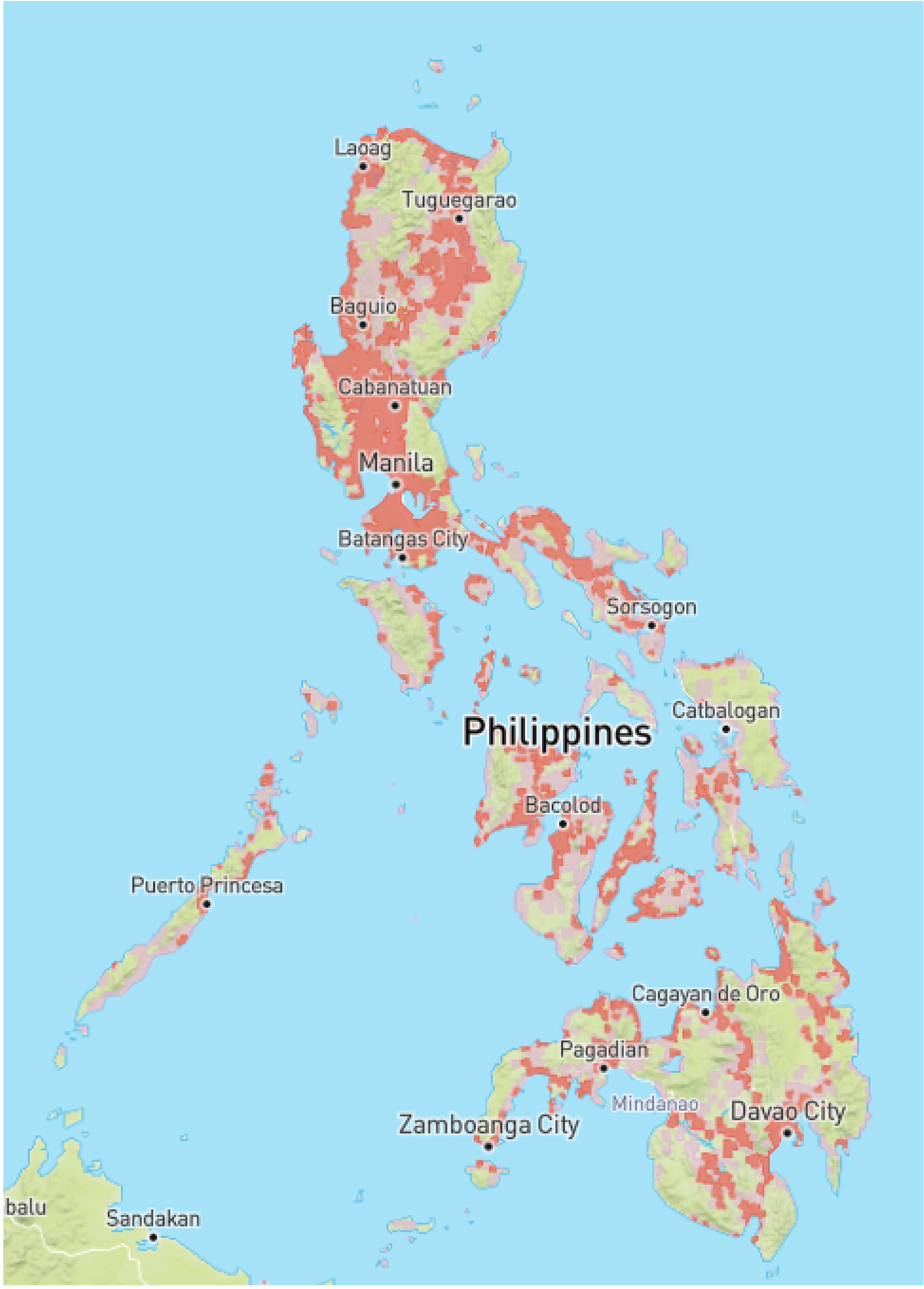

Fig. 2.3 Smart GSM and LTE Coverage Map

TABLE 2.1 COMPARISON BETWEEN GSM AND LTE

Feature	**GSM (2G)**	**LTE (4G)**
Network Architecture	Circuit-switched	Packet-switched (All-IP)
Primary Focus	Voice and text services	High-speed mobile broadband
Coverage	Extensive broad geographical reach, including rural and remote areas	High-capacity and density-dependent coverage that requires closer proximity to modern cell towers
Frequency Band	900 MHz and 1800 MHz	Bands 1 to 25 for Frequency Division Duplex and bands 33 to 41 for Time Division Duplex
Signal Propagation	Excellent because of lower frequencies	Variable because higher frequencies generally have shorter range
Infrastructure Needs	Lower base station density	Higher base station density
Data Capability	Low, supporting basic data and messaging	Very high, supporting multimedia and internet access

The coverage maps in Figures 2.2 and 2.3 illustrate GSM and LTE coverage patterns in the Philippines based on GSMA Intelligence coverage data [GSMA, 2026]. Comparing the two networks, Globe shows a noticeably denser and more contiguous spread of both GSM and LTE coverage, particularly through the Visayas and Mindanao, where its colored patches connect more continuously along coastal and lowland routes. Smart's coverage, while still following the same population centers, appears comparatively more fragmented, with more isolated patches and gaps between cities, especially in the southern islands.

In both cases, LTE coverage appears as a smaller subset within the broader GSM footprint. This indicates that 4G service is generally concentrated in and around cities and main highway corridors, while 2G/GSM extends farther into surrounding rural areas as a fallback connectivity layer. This difference is important for fleet monitoring systems because mobile assets may move between high-capacity LTE areas and wider but lower-bandwidth GSM fallback zones [Opensignal, , GSMA, 2026].

The Philippine cellular landscape is transitioning away from legacy networks. In August 2025, the NTC issued Memorandum Circular No. 003-09-2025, mandating phased decommissioning of 2G and 3G services, with nationwide 3G switch-off by 31 December 2026 and 2G on a separate, slower schedule because rural areas still depend on it. The circular also bars new type approval for devices whose primary capability is 2G or 3G. Two design implications follow: LTE must be the primary wide-area path, since it is the network being expanded rather than retired; and any 2G fallback is transitional, which motivates retaining LoRa and on-device buffering where neither LTE nor 2G is dependable [China Academy of Information and Communications Technology, 2025].

2.2.7 Anomaly Detection

Several studies on anomaly detection in vehicular systems aim to improve operational safety, combat theft and fraud, and optimize fuel efficiency. Various technological means, such as GPS tracking, machine learning, and time series analysis, have been used to identify abnormal vehicle behaviors. Crisgar et al. [Crisgar et al., 2021] developed a vehicle tracking system using Google Cloud IoT Core and Firebase to monitor vehicle movements and engine activity for theft detection. Unauthorized ignition or unexpected movement beyond a 50-meter radius is flagged as potential theft, with real-time notifications pushed to a Progressive Web App, enabling users to respond quickly.

Studies focusing on fuel efficiency in heavy-duty vehicles (HDVs) through anomaly detection models include Turan et al. [Turan et al., 2024], who proposed an ensemble of unsupervised machine-learning models (Isolation Forest, Autoencoder, k-NN Regression) combined via weighted majority voting to detect abnormal fuel consumption patterns based on road slope and speed variations. Mumcuoglu et al. [Mumcuoglu et al., 2023] used decision tree ensembles to classify fuel usage as normal or anomalous, achieving 92.2% accuracy and an F1 score of 0.78. Both studies indicate that repeated detection of outliers can highlight anomalies such as mechanical issues or inefficient driving.

Aquize et al. [Aquize et al., 2017] applied Self-Organizing Maps (SOM) to detect abnormal fuel consumption in RFID-registered vehicles in Bolivia. By learning conventional consumption patterns, the SOM model achieved an 80% confidence rate in flagging suspicious activity indicative of fraud. Semenov et al. [Semenov et al., 2024] employed DBSCAN clustering and Local Outlier Factor (LOF) in a three-level telematics platform to forecast and detect truck breakdowns in real-time. This system tracks multiple vehicle parameters, labeling them as normal or outliers based on deviation from learned patterns, minimizing surprise breakdowns and providing early driver alerts. Aquize et al. [Aquize et al., 2017] proposed a fuel anomaly detection framework based on Self-Organizing Maps (SOM) using RFID-registered vehicle data collected from fleet operations in Bolivia. The SOM algorithm learned normal fuel consumption patterns and identified suspicious deviations associated with potential fuel fraud. The study achieved approximately 80% confidence in detecting anomalous fuel usage and demonstrated the effectiveness of unsupervised learning approaches for identifying irregular fuel consumption behavior without requiring labeled datasets.

Similarly, Kaleem Habib and Arif Iqbal Umar [Habib and Umar, 2015] utilized data mining techniques to calculate and detect anomalies in fuel expenses by analyzing historical fuel transaction records and consumption patterns. Their approach identified abnormal expenditure behaviors that could indicate fuel theft, reporting errors, or operational inefficiencies. The findings demonstrate the value of data-driven anomaly detection methods in supporting cost control and fraud prevention within fleet management environments.

Garcia Vega [Garcia Vega, 2022] emphasized time series anomaly detection for vehicle monitoring, developing an iterative approach to detect unusual driving patterns through preprocessing, subsequence segmentation, scoring, clustering, and expert validation. Applied in an industrial context, it identified over 120 validated anomalies, helping diagnose mechanical failures or unsafe driving behavior. The study highlights the importance of visual interpretability tools and recommends future improvements via real-time data processing.

Because published fleet fuel-anomaly work (Barbado and Corcho, 2022) [Barbado and Corcho, 2022] uses boxplot/IQR-based unsupervised detection as an interpretable baseline before applying explainable models, the present study compares Isolation Forest and Matrix Profile against a rule-based / IQR baseline under leave-one-trip-out evaluation.

This baseline serves both as a fairness check (to show that the non-parametric models genuinely add value over a thresholding heuristic) and as an ablation reference in Chapter 5.

2.2.8 Machine Learning Applications

Barbado and Corcho (2022) [Barbado and Corcho, 2022] developed an explainable machine-learning framework for vehicle fuel-consumption anomalies using real-world telematics data from diesel and petrol fleet vehicles. Their study emphasised that anomaly detection should not only identify abnormal fuel consumption but also explain the operational causes behind the anomaly, using XAI techniques on top of an unsupervised detection layer. They first identified candidate anomalies via a univariate boxplot/IQR outlier method and then trained a contextual model that incorporates domain knowledge such as vehicle subgroup, route type, and driver behaviour. This is directly relevant to the present thesis because fuel anomalies in Philippine logistics must be interpreted with operational context - the same context (driver, route, vehicle, fuel-consumption baseline) is encoded in the Isolation Forest Model C feature set used in Chapter 5.

Recent advances in machine learning have been integrated into vehicle systems to improve fuel efficiency, detect anomalies, and enhance operational transparency. In fuel monitoring and anomaly detection, machine learning methodologies extract actionable insights from complex, high-dimensional vehicle data. Fortela et al. [Fortela et al., 2023] applied unsupervised machine learning algorithms (Principal Components Analysis, K-Means Clustering, Self-Organizing Maps) to detect anomalies in fuel economy and emission data from light-duty vehicles (2015–2023). These methods revealed clusters where CO emissions were positively linked to fuel economy, contrary to expected patterns, highlighting irregularities caused by specific fuel test procedures. Barbado and Corcho [Barbado and Corcho, 2022] combined unsupervised anomaly detection with domain knowledge and interpretable machine learning (XAI) to explain potential causes of fuel consumption anomalies. Their models demonstrated up to 88% potential fuel savings on larger datasets and across various user roles.

Barbado and Corcho [Barbado and Corcho, 2022] further demonstrated that interpretable machine learning models such as Explainable Boosting Machines (EBM) can be used not only to detect anomalous fuel consumption but also to explain the factors contributing to those anomalies. Their methodology combines unsupervised anomaly detection, domain knowledge, and explainable artificial intelligence (XAI) metrics to provide actionable recommendations for fleet managers and operators. The study highlights the importance of model transparency in fleet monitoring systems, allowing stakeholders to understand the causes of abnormal fuel usage rather than merely identifying anomalous events.

More recent work sharpens the methodological basis for unsupervised, on-vehicle detection. Cherdo et al. [Cherdo et al., 2023] applied small recurrent and convolutional neural networks to automotive CAN sensor time series, demonstrating that bus data of the kind exposed through an OBD-II interface is amenable to unsupervised anomaly detection, although their models still require gradient-based training. Surveying the domain, Hirth et al. [Hirth et al., 2025] formalize anomaly detection in trucks as identifying deviations across multivariate signals at each time step of a stream, and observe that deep architectures impose retraining and computational burdens that complicate continuous deployment. These observations motivate the two non-parametric methods adopted in this study: Isolation Forest [Liu et al., 2008], which isolates anomalous individual observations without modeling a distribution of normal behavior, and the Matrix Profile [Yeh et al., 2016], which detects anomalous subsequences and has been extended to live data streams [Lu et al., 2022] and to multiple sensor channels [Yeh et al., 2024]. The mathematical formulation of both methods is presented in Chapter 3.

TABLE 2.2 SUMMARY OF EXISTING WORKS

Study	**Technology Used**	**Key Features**	**Strengths**	**Limitations**
Tatababu et al. (2024) [Tatababu et al., 2024]	Ultrasonic/Capacitive sensors, GSM, GPS, Smartphone/Web interface	Theft detection, fuel level alert, real-time remote access	Accurate for remote generator monitoring, GPS-based tracking	Findings specific to the tested generator systems and environments
Joshi et al. (2024) [Joshi et al., 2024]	IoT, GPS, RFID, Cloud Computing	Real-time vehicle/asset tracking, data analytics, cloud-based monitoring	High tracking accuracy, improved logistics efficiency, responsive alert system	Limited by hardware and logistics test case set
Vigneshwaran et al. (2020) [Vigneshwaran et al., 2020]	GPS, GSM	Real-time bus tracking, emergency alert system, communication to students, route updates	Improves safety, scheduling, and passenger information	Not highly scalable; lacks detailed integration with modern telematics platforms
Patil & Patil (2024) [Patil and Patil, 2024]	IoT, GPS, GSM, Accelerometer, IR Sensors	Tracking system with SMS-based communication; remote ignition cut-off mechanism and vehicle password protection	Comprehensive vehicle security, real-time alerts, detection of tampering/impacts	Low adaptability to harsh conditions like GPS/GSM signal loss in remote areas
Fortela et al. (2023) [Fortela et al., 2023]	Unsupervised Machine Learning: PCA, K-Means, SOM	Detects hidden trends and impending anomalies; analyzes correlation between CO emissions and fuel economy	Effectively identifies non-obvious patterns within clusters; highlights flaws in standardized testing procedures	Lacks integration with real-time or operational vehicle data; conclusions depend on historical datasets

Study	Technology Used	Key Features	Strengths	Limitations
Barbado & Corcho (2022) [Barbado and Corcho, 2022]	Unsupervised Anomaly Detection; Explainable Boosting Machine (EBM)	Provides fuel anomaly detection and cause explanations, tailored recommendations for fleet operators	High interpretability via XAI metrics; role-based recommendations; potential fuel savings up to 88%	Performance depends heavily on quality of telematics data or feature relevance
Aquize et al. (2017) [Aquize et al., 2017]	Self-Organizing Maps (SOM)	Detects fraudulent fuel usage based on RFID data patterns	80% accuracy applied to real socio-economic problem	Used only simple SOM model; accuracy can improve with more context variables
Liu et al. (2008) [Liu et al., 2008]	Isolation Forest	Ensemble of isolation trees that identify anomalies through random partitioning	Non-parametric anomaly detection with linear-time complexity and low memory requirements	No labeled data required; superior AUC compared with ORCA and LOF; computationally efficient
Yeh et al. (2016) [Yeh et al., 2016]	Matrix Profile (STAMP)	Z-normalized nearest-neighbor similarity profile for motif and discord discovery	Exact subsequence anomaly detection with minimal parameter tuning	Deterministic, interpretable, and near parameter-free
Lu et al. (2022) [Lu et al., 2022]	DAMP Streaming Matrix Profile	Real-time Matrix Profile computation for high-volume streaming data	Detects discords in continuously arriving data streams at very large scales	Suitable for online and real-time deployment
Yeh et al. (2024) [Yeh et al., 2024]	Multidimensional Matrix Profile	Extension of Matrix Profile for multivariate time-series anomaly detection	Simultaneously analyzes multiple sensor channels	Competitive performance across numerous benchmark datasets and highly interpretable results
Hirth et al. (2025) [Hirth et al., 2025]	Deep Learning-Based Truck Anomaly Detection (Survey)	Comprehensive review and taxonomy of anomaly detection methods for trucks	Defines anomaly detection in multivariate truck telemetry streams	Provides truck-specific insights and identifies current research directions

Study	**Technology Used**	**Key Features**	**Strengths**	**Limitations**
Cherdo et al. (2023) [Cherdo et al., 2023]	Small RNN and CNN Models	Unsupervised anomaly detection using automotive CAN-bus sensor data	Demonstrates feasibility of on-vehicle anomaly detection under limited hardware resources	Effective for OBD-II/CAN sensor environments and embedded deployment

2.3 Comparison of Commercial Fleet Management Systems

Beyond the academic literature, six widely deployed commercial fleet management platforms were examined to establish the detection methodologies available to operators in practice: Geotab, Samsara, Fleetio, Wialon, Fleet Complete, and Navixy. Each platform's own product documentation was used as primary evidence of its capabilities. Despite mature telematics infrastructure, fuel-loss detection across these systems reduces to fixed or manually configured rules. Samsara's documentation describes its fuel theft detection as a configurable alert on any sudden fuel drop of five percent or more [Samsara, 2026]. Navixy requires operators to configure rate-of-change drain thresholds per sensor and can be set to ignore drains while a vehicle is moving [Navixy, 2026]. Geotab and Fleetio approach the problem from the fuel-card transaction side, flagging exceptions such as purchases exceeding tank capacity or fueling locations inconsistent with vehicle position [Geotab, 2024, Fleetio, 2026]. Wialon detects drain events from fuel-level sensor data and commonly relies on operator review for verification [Wialon, 2026], while Fleet Complete's public documentation centers on fuel usage reporting without specifying an adaptive detection method [Fleet Complete, 2026]. Across all six, no platform learns a vehicle's normal consumption behavior, adapts to operating context, or is designed specifically for Philippine fleet conditions.

TABLE 2.3 COMPARISON OF COMMERCIAL FLEET MANAGEMENT SYSTEMS

System	Data Sources	Fuel Monitoring Method	ML for Fuel Anomalies?	Detection Methodology	Limitations
Geotab [Geotab, 2024]	OBD/telematics device, fuel cards, GPS	Fuel-up records and engine data reporting	No (rule checks)	Capacity mismatch, purchase greater than capacity, and location discrepancy	Transaction-rule logic; no learned consumption behavior
Samsara [Samsara, 2026]	Telematics gateway, fuel sensor/OBD, fuel cards, GPS	Real-time fuel level and usage reports	No (threshold)	Configurable sudden-drop alert (≥5%)	Static threshold; blind to gradual loss; one rule for all conditions
Fleetio [Fleetio, 2026]	Fuel card transactions and telematics integrations	Transaction-based fuel management	No	Exception alerts for location mismatch and volume greater than tank capacity	Sees only fill-up events; blind between refuels
Wialon [Wialon, 2026]	Fuel level sensors, GPS trackers, optional cameras	Sensor-based fuel charts and drain/refill detection	No	Sensor-threshold drain events with human/video verification	Threshold-based and manually reviewed; reactive
Fleet Complete [Fleet Complete, 2026]	Telematics device, GPS, vehicle data, fuel reports	Fuel usage and fleet reporting	No specified adaptive model	Reporting-based fuel monitoring and exception review	Public documentation does not specify adaptive anomaly detection
Navixy [Navixy, 2026]	Fuel level sensors, OBD/CAN, GPS	Drain/refill event detection from sensor data	No	Per-sensor rate-of-change drain thresholds; option to ignore in-motion drains	Manual tuning; in-motion exclusion can hide moving theft or leaks

2.4 Lacking in the Approaches

While existing technologies in vehicle tracking, fuel sensing, and anomaly detection form the foundation upon which developments can be built, they suffer from critical limitations that prevent their use in real-time, fleet-wide operations. Research works such as Krishnasamy [Krishnasamy, 2019] have employed basic sensor technologies without any validation of fuel measurements in real-time driving environments or estimation of accuracy parameters relative to manual readings. Besides, current systems are limited to specific uses and are not tested for scalability or integration with fleet vehicles like trucks such as institutional/organizational level of motorcycles in the study of RS et al. [RS et al., 2024]. Some works do consider machine learning techniques for anomaly detection, but such techniques are mostly based on static and post-processed datasets, with no real-time on-vehicle analysis. Very few implementations include unsupervised machine learning and advanced context-aware modeling, and precision benchmarks are usually left unreported, if at all addressed.

Across the surveyed commercial systems, including Geotab, Samsara, Fleetio, Wialon, Fleet Complete, and Navixy, several operational capabilities remain insufficiently addressed. First, closed-loop in-cab driver feedback is largely missing because most platforms push alerts to a manager dashboard, whereas this thesis pushes alerts to a driver-facing HMI inside the truck cab with rest/snooze acknowledgments captured in the `alerts` table. Second, offline-tolerant telemetry with explicit replay accounting is rarely documented in academic prototypes or commercial fuel-monitoring systems; this thesis tags every telemetry row with `channel_used`, including `offline_replay`, which lets the backend distinguish live data from replayed data. Third, multi-interface state reconciliation is not typically addressed, while this thesis uses partial unique indexes to enforce one-active-trip-per-truck/driver across three concurrent client surfaces: HMI, mobile, and dashboard.

Ultimately, there is no prior system that provides a full solution that brings together a cloud-connected solution for real-time fuel monitoring with integration of GPS tracking and anomaly detection. Further, it completely lacks features such as automatic rest alerts and predictive maintenance reminders, which are necessary for safe, efficient, and compliant operations. Such shortcomings create an opportunity for a more integrated, intelligent, and real-time solution, and this is what the thesis proposes to address.

TABLE 2.4 IDENTIFIED GAPS, LIMITATIONS, AND PROPOSED SOLUTIONS

Study / Gap	**What they did**	**Limitation**	**How this thesis responds**
Area of precision	Existing models lack context-awareness and show lower precision. Models presented in the study of Barbado & Corcho [Barbado and Corcho, 2022] performed up to 80% of anomalous instances explained.	Prior work does not establish the same precision and false-positive benchmark required for this deployment.	(SO4) To design and evaluate a context-aware fuel usage deviation model using non-parametric unsupervised machine learning models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.

Study / Gap	**What they did**	**Limitation**	**How this thesis responds**
Barbado & Corcho (2022) [Barbado and Corcho, 2022]	Used real-world telematics and interpretable ML to detect and explain fleet fuel-consumption anomalies; combined box-plot/IQR pre-filtering with contextual XAI models.	Focused on fleet-level analytics and explanation; did not produce an embedded HMI + OBD-II + multi-channel-communication + live-alert prototype that closes the loop in the vehicle cab.	This thesis integrates OBD-II fuel acquisition (Toyota Mode 22 PID 21/29), embedded HMI, GPS tracking, four-channel communication fallback, backend dashboard, mobile driver web app, and unsupervised anomaly detection (IF Model C + Matrix Profile) in one working prototype evaluated under LOTO.
Absence of a comprehensive system integrating fuel anomaly detection with GPS-based tracking, and predictive maintenance and alerts	Previous works, including those by Joshi et al. [Joshi et al., 2024] and Nada [Nada, 2017], focus on real-time tracking using GPS/GSM.	These works lack integration with anomaly detection methods for analyzing fuel consumption patterns or predicting vehicle maintenance issues.	(SO3, SO4, SO5) To capture and log GPS location data via an embedded module, correlating route information with fuel and time data for behavior tracking; design and evaluate a context-aware fuel usage deviation model using non-parametric unsupervised machine learning models; develop and integrate rest alerts and maintenance reminders into a cloud-based dashboard.

Study / Gap	What they did	Limitation	How this thesis responds
No Philippine transportation context	Reviewed studies and platforms do not model local conditions such as Metro Manila congestion, Port of Manila utilization, the 2014 Manila truck ban, and fuel fraud costs [Japan International Cooperation Agency, 2025, Patalinghug et al., 2015a, Llanto, 2017, Asian Development Bank, 2016].	Models may misclassify congestion-shaped idling and consumption as anomalies when Philippine operating conditions are absent.	(GO, SO4) Collect and train on operational data from the partner fleet, Hino Paranaque Subic GS Auto Inc., so congestion-shaped idling and consumption are learned as baseline behavior.
One detection rule for all trucks	Commercial systems use single fleet-wide settings [Samsara, 2026] or manual per-sensor tuning [Navixy, 2026]; prior studies were validated on one vehicle class, generators [Tatababu et al., 2024], or motorcycles [RS et al., 2024].	Shared thresholding cannot represent each truck's normal consumption profile.	(SO4) Learn per-vehicle baselines that update from each truck's own operating history, removing dependence on a shared manual threshold.
Detection of sudden drops only	Threshold and transaction logic are commonly used to detect sudden losses [Samsara, 2026, Fleetio, 2026]; Navixy can ignore in-motion drains [Navixy, 2026].	These approaches may miss slow siphoning and gradual leaks.	(SO4) Apply the Matrix Profile for anomalous-subsequence or discord detection of gradual loss, paired with Isolation Forest for sudden point anomalies.
No real-time analytics or live alerts	Machine learning studies were validated offline [Fortela et al., 2023, Barbado and Corcho, 2022]; deep models burden streams [Hirth et al., 2025, Cherdo et al., 2023]; and Wialon relies on human review [Wialon, 2026].	Prior approaches do not provide live anomaly scoring and alert delivery across operational interfaces.	(SO2, SO5) Deploy a cloud pipeline scoring live OBD-II/GPS streams with Isolation Forest and a streaming Matrix Profile [Lu et al., 2022], issuing live alerts to the dashboard, HMI, and mobile application.

2.5 Summary

With this chapter, it was discovered that there have been various studies on vehicle tracking, fuel sensing, and anomaly detection. The PRISMA diagram shows the systematic approach that was used in the identification of the studies, out of which the 41 initial records were narrowed down to 28 after excluding the duplicates, outdated/irrelevant, and not retrieved records. Moreover, the existing studies presented that though a lot has been done to develop individual systems for fuel level monitoring, GPS tracking, and anomaly detection, most of these works are either narrowly focused or operate in isolation. Many rely on basic sensor integrations without advanced analytics, or implement machine learning models without connecting them to real-time, in-vehicle data streams.

Numerous studies have exhibited the application of different sensor-based systems for fuel monitoring in real-time and detecting theft, in many cases involving microcontrollers, GSM, and GPS modules. Likewise, research on vehicle telematics illustrates the application of GPS in route tracking and fleet management. In the field of anomaly detection, some methods based on unsupervised machine learning and clustering techniques have been utilized to detect abnormal fuel consumption patterns and flag fraudulent or inefficient activity. Yet, most of these researches are based on special scenarios, are not integrated with real-time operational data, or depend on historical data.

While advances have been made in each of these domains, a significant void exists in creating a holistic, real-time solution that integrates fuel level measurement, GPS tracking, and intelligent anomaly detection through machine learning. Current systems exist in silos, are vehicle type- or use case-specific, and do not commonly incorporate predictive maintenance or driver feedback mechanisms.

This research addresses these gaps through an integrated IoT-based system tailored for trucks. The system combines real-time fuel monitoring, GPS tracking, and machine-learning-based anomaly detection to improve operational efficiency, driver safety, and maintenance planning.

Chapter 3

THEORETICAL CONSIDERATIONS

This chapter establishes the theoretical foundations underpinning the proposed IoT-based fleet fuel monitoring and anomaly detection system. It presents the essential concepts, mathematical models, and scientific principles that guide the development of the system. These theories act as the main foundation of the study, supporting the methodological decisions described in Chapter 5. By understanding the underlying theories—ranging from cyber-physical systems to signal processing and machine learning—the relevance and rigor of the chosen approach become clear.

3.1 Overview

The methodologies used in the system are grounded in established theories in embedded systems, time-series processing, IoT architectures, unsupervised machine learning, and vehicular telematics. These theoretical concepts justify the design choices made in the development of real-time fuel monitoring components, GPS-based tracking, and anomaly detection models. This chapter explains the foundations behind the signal models, communication processes, and machine learning algorithms that make accurate and robust system performance possible.

3.2 Theoretical Foundations

3.2.1 Cyber-Physical Systems (CPS)

The proposed solution functions as a Cyber-Physical System, integrating embedded sensors, microcontroller-based processing, cloud computation, and real-time communication.

CPS theory emphasizes:

- the tight coupling between physical signals and digital processing,
- deterministic timing and synchronized data flow,
- continuous monitoring and computational feedback loops.

These principles justify the system's architecture, where fuel-level signals, GPS data, and operational states are processed and transmitted in real time.

3.2.2 Sensor Theory and Signal Measurement

Fuel-level readings extracted through the CAN bus/OBD-II interface follow digital measurement theory. Sensor signals can exhibit noise, quantization error, and environmental fluctuation. Fuel-level readings may be stabilized using smoothing methods such as moving-average, median, and exponential moving-average filters. A Simple Moving Average (SMA) is a standard reference smoothing method defined as:

$$x_t^{\text{smooth}} = \frac{1}{N}\sum_{i=0}^{N-1} x_{t-i} \tag{3.1}$$

This equation is included as the theoretical baseline for smoothing; however, the final implementation used a median + EMA + dead-band + direction-aware filter cascade because fuel-tank slosh and sender chatter include both impulse noise and slow drift. The implemented cascade is described mathematically in the following signal-filtering section and operationally in Chapter 5.

3.2.3 OBD-II and CAN-based Vehicle Data Acquisition

Vehicle data can be acquired through a standardised diagnostic interface rather than an added in-tank sensor. The underlying transport is the Controller Area Network (CAN), a message-based broadcast bus standardised in ISO 11898 in which contention is resolved by non-destructive bitwise arbitration: priority is encoded in the message identifier, and the highest-priority frame proceeds without loss of data or time. Standard CAN uses 11-bit identifiers and extended CAN uses 29-bit identifiers.

On-Board Diagnostics II (OBD-II) operates over CAN as a request–response protocol, with its application layer defined by SAE J1979 / ISO 15031-5 and multi-frame transport by ISO 15765-4. It distinguishes standardised current-data PIDs (Mode 01), which are mandated and uniform across manufacturers, from enhanced PIDs (Mode 22), which are manufacturer-specific and use proprietary scaling. Because Mode 22 scaling is not publicly standardised, any value read through it must be validated empirically, introducing measurement uncertainty.

Heavy-duty commercial vehicles conventionally use SAE J1939, a higher-layer protocol over 29-bit CAN identifiers in which Parameter Group Numbers (PGNs) group signals and Suspect Parameter Numbers (SPNs) define each signal's bit position, scaling, and units. Under J1939, fuel level is a standardised SPN broadcast periodically without an explicit request. The theoretical contrast is therefore between a polled, proprietary request–response model (OBD-II Mode 22) and a push-based, standardised broadcast model (J1939): the former trades standardisation and immediacy for broad device compatibility. Analog fuel senders accessed through either path are subject to slosh-induced fluctuation and quantisation error, which motivates the filtering theory developed next [International Organization for Standardization, 2015b, SAE International, 2023, International Organization for Standardization, 2021, SAE International, 2016, SAE International, 2017, International Organization for Standardization, 2024a, Robert Bosch GmbH, 1991].

3.2.4 Signal Filtering for Fuel-Level Stabilisation

Raw signals from physical sensors carry impulse noise, low-frequency disturbance, and quantisation error; no single filter addresses all three, so a multi-stage cascade is used, each stage grounded in distinct theory.

Median filter. A running median is a non-linear order-statistic filter that replaces each sample with the median of a sliding window of length W:

$$x_t^{\text{med}} = \text{median}(x_{t-W+1}, \ldots, x_t). \tag{3.2}$$

Its breakdown point is 50%; up to half the window may be corrupted by impulse noise before the output is affected, whereas the arithmetic mean has a 0% breakdown point.

Exponential moving average. An EMA is a first-order recursive IIR low-pass filter:

$$y_t = \alpha x_t^{\text{med}} + (1-\alpha) y_{t-1}, \qquad 0 < \alpha \leq 1, \tag{3.3}$$

in which past samples decay geometrically and the smoothing factor α sets a time constant $\tau = -1/\ln(1-\alpha)$; a smaller α gives heavier smoothing at the cost of responsiveness. The simple moving average, by contrast, is a finite-impulse-response filter optimal for white noise but without impulse robustness and requiring the full window in memory.

Dead-band (hysteresis). A dead-band publisher emits a new value only when the smoothed signal departs from the last published value by more than a threshold:

$$\text{publish}(y_t) \Leftrightarrow |y_t - y_{\text{pub}}| \geq \delta. \tag{3.4}$$

Direction-aware gating. An asymmetric step rule treats increases and decreases differently according to the physical behaviour of the measured quantity. In the implemented fuel cascade, a downward update is accepted at a smaller threshold than an upward update, while upward movement during motion is suppressed unless a stationary refuel signature is present:

$$y_{\mathrm{pub},t} = \begin{cases} y_t, & y_t \leq y_{\mathrm{pub},t-1} - \delta_{\downarrow}, \\ y_t, & y_t \geq y_{\mathrm{pub},t-1} + \delta_{\uparrow} \ \wedge \ v_t = 0 \ \wedge \ T_{\mathrm{stop}} \geq 60\ \mathrm{s}, \\ y_{\mathrm{pub},t-1}, & \text{otherwise.} \end{cases} \tag{3.5}$$

Here, $\delta_{\downarrow}$ represents the downward fuel-change threshold and $\delta_{\uparrow}$ represents the larger upward refuel threshold. A warm-up guard suppresses output until the median window is populated, since earlier samples are unrepresentative.

3.2.5 Classification and Evaluation Metric Theory

Detectors are evaluated as binary classifiers. From the confusion matrix (TP, FP, TN, FN):

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3.6}$$

$$\text{FPR} = \frac{FP}{FP + TN} \tag{3.7}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.8}$$

$$L_d = T_{\text{detected}} - T_{\text{actual}}. \tag{3.9}$$

Accuracy is uninformative under class imbalance, where it is dominated by the majority class [Powers, 2020]. A Receiver Operating Characteristic (ROC) curve plots true-positive rate against false-positive rate, and its area equals the probability that a random positive outranks a random negative [Fawcett, 2006]. Under heavy imbalance, however, ROC-AUC is optimistic; the Precision–Recall curve is more informative, its area collapsing toward the positive prevalence so that a modest value reflects the base rate rather than poor detection [Davis and Goadrich, 2006, Saito and Rehmsmeier, 2015]. Detection latency is $L_{\text{det}} = T_{\text{detected}} - T_{\text{actual}}$, where T_{actual} is when the anomaly occurred and T_{detected} is when it was flagged; the Numenta Anomaly Benchmark scores latency with a time-decaying reward [Lavin and Ahmad, 2015]. For binomial proportions estimated from few positives, the Wilson score interval is preferred over the normal-approximation (Wald) interval [Wilson, 1927]:

$$\frac{\hat{p} + z^2/(2n) \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + z^2/n}. \tag{3.10}$$

Leave-one-group-out cross-validation, here leave-one-trip-out, holds out each group entirely to prevent leakage from within-group correlation [Arlot and Celisse, 2010]. Finally, evaluation against injected synthetic anomalies provides exact ground truth but bounds performance on the injected classes only, since these may not represent the full distribution of real events [Chandola et al., 2009].

3.2.6 Time-Series Synchronization Theory

Sensor streams often arrive at irregular timestamps. Synchronizing them requires interpolation theory. For two samples at timestamps T_i and T_{i+1}, the interpolated point at t_j is computed as:

$$x_j = x_i + \left(\frac{x_{i+1} - x_i}{T_{i+1} - T_i}\right)(t_j - T_i) \tag{3.11}$$

This ensures aligned and consistent time-series data across fuel level, GPS speed, and distance.

3.2.7 IoT Communication Theory

The communication subsystem relies on Internet of Things (IoT) principles and long-range wireless communication theory. LoRa operates using Chirp Spread Spectrum (CSS) modulation, characterized by:

- long-range transmission,
- low power consumption,
- robustness to interference.

Theoretical latency between sender and cloud receiver is expressed as:

$$L = T_{\text{received}} - T_{\text{sent}} \tag{3.12}$$

Understanding latency is essential for analyzing responsiveness and real-time performance.

3.2.8 GPS Temporal and Spatial Theory

GPS logging follows Global Navigation Satellite System (GNSS) principles. To evaluate consistency of GPS sampling, the inter-arrival interval is computed as:

$$t_i = T_i - T_{i-1} \tag{3.13}$$

Maintaining a stable interval ensures accurate reconstruction of routes and distance traveled.

3.2.9 Unsupervised Anomaly Detection Theory

The anomaly detection component relies on non-parametric, unsupervised machine learning, which identifies unusual behavior in multivariate time-series data without labeled training samples. Following the taxonomy adopted for vehicle anomaly detection [Hirth et al., 2025], three anomaly types are distinguished: point anomalies, or individual observations that are anomalous with respect to the data, such as a sudden fuel drop while stationary; contextual anomalies, which are anomalous only in context, such as high consumption at zero speed; and collective or subsequence anomalies, where a sequence of individually unremarkable observations forms an anomalous pattern, such as gradual siphoning. The system assigns point and contextual anomalies to Isolation Forest, operating on engineered contextual features, and collective anomalies to the Matrix Profile.

3.2.9.1 Matrix Profile Theory

Matrix Profile computes similarity between subsequences within a time series. For a time series T of length n and window size m, the profile value is:

$$P[i] = \min_{j \neq i} \| T_{i:i+m-1} - T_{j:j+m-1} \| \tag{3.14}$$

The subsequences with the highest $P[i]$ values, known as discords, are likely anomalies. This method is ideal for detecting sudden fuel drops or siphoning patterns.

3.2.9.2 Isolation Forest Theory

Isolation Forest isolates anomalies through random partitioning. An anomaly score is expressed as:

$$s(x) = 2^{-\frac{E(h(x))}{c(n)}} \tag{3.15}$$

Where:

- $E(h(x))$ is the expected path length for observation x,
- $c(n)$ normalizes the tree depth by dataset size.

Higher scores indicate higher likelihood of anomalous fuel behavior.

3.3 Summary

This chapter presented the theoretical foundation for the system's methodology. It discussed the key concepts supporting real-time sensing, signal preprocessing, GPS logging, IoT communication, and unsupervised anomaly detection. By establishing these theoretical pillars, this chapter clarifies the scientific basis that makes the system's design viable, accurate, and aligned with established models in engineering and data science.

Chapter 4

DESIGN CONSIDERATIONS

This chapter presents the design considerations that guided the development of the implemented IoT- and AI-driven fuel monitoring and anomaly detection system. Whereas Chapter 3 established the theoretical foundations of IoT architectures, cyber-physical systems, signal processing, and unsupervised anomaly detection, the present chapter translates those theories into practical engineering decisions that were directly implemented in Chapter 5.

These design considerations serve as the structural "columns" of the system: grounded in accepted standards, ethical guidelines, and engineering best practices, but tailored to support the procedural steps carried out in Chapter 5. The discussion begins with a comprehensive account of the technical standards that ensure reliability, safety, interoperability, and data security, followed by software and hardware design decisions. Ethical considerations—including privacy, responsible AI, and driver safety—are also presented.

4.1 Standards

4.1.1 Adherence to Technical Standards

To ensure reliability, accuracy, and safety, the prototype used commercially available modules and followed relevant design guidance for automotive electrical transients, embedded control, secure transmission, software architecture, and data protection. Full product-level certification was outside the scope of the undergraduate prototype. These standards and guidance documents directly influence the system architecture, data handling, and model implementation workflows outlined in Chapter 5.

1. **Standards for Sensors and Hardware**

The system integrates sensors and modules that employ standard communication protocols—**CAN bus/OBD-II, UART, I2C, and GPIO**—consistent with IEEE 1451 smart transducer guidelines [IEEE Standards Association, 2024a]. The CAN bus/OBD-II fuel data acquisition aligns with industry telematics conventions [International Organization for Standardization, 2020], while the GPS module adheres to global navigation and timing standards [NASA EarthData, 2025].

IEC 61131-2 and ISO 7637-2 were used as references for embedded-control and automotive transient considerations [International Electrotechnical Commission, 2017, International Organization for Standardization, 2011b]. Commercial off-the-shelf modules with available CE/FCC-marked variants were selected where possible; however, full product-level certification was outside the scope of this undergraduate prototype [European Commission, 2022, Federal Communications Commission, 2025].

2. **Data Acquisition and Signal Processing**

The fuel-level signal is stabilised by a multi-stage filter, a median stage for impulse rejection followed by an exponential moving average for smoothing, rather than a single moving average.

Time alignment of fuel, GPS, speed, and engine readings follows ISO/IEC 19510 principles for structured time-series data [International Organization for Standardization, 2013]. These standard-aligned preprocessing steps feed into the anomaly detection models described in Chapter 5.

3. **Communication and Data Transmission**

Communication follows TCP/IP standards under IETF and IEEE 802.11 [Postel, 1981, IEEE Standards Association, 2025]. Telemetry transmission is implemented through HTTPS REST endpoints, while MQTT remains a compatible future alternative for broker-based deployment. End-to-end encryption via TLS/SSL aligns with ISO/IEC 27001 cybersecurity requirements [International Organization for Standardization, 2022a].

These communication strategies form the basis for the system's message-handling and cloud-upload mechanisms in Chapter 5.

4. **Machine Learning and Model Validation**

The Matrix Profile and Isolation Forest models are implemented using open-source libraries, and their evaluation follows ISO/IEC 29119 software quality standards [International Organization for Standardization, 2022b]. Validation metrics—confusion matrices, ROC curves, precision–recall graphs—provide statistical grounding for model evaluation, as presented in Chapter 5.

5. **Software Architecture and Data Management**

The modular software design complies with ISO/IEC/IEEE 42010 architectural principles [International Organization for Standardization, 2022c]. ACID-compliant storage and GDPR-/ISO 27018-aligned data protection protocols [International Organization for Standardization, 2025] ensure data integrity and privacy. All processed datasets follow FAIR principles.

4.1.2 Observance of Research Ethics

Ethical considerations guide the responsible deployment of the system, especially given the safety-critical and privacy-sensitive context of transportation fleets. Three major ethical domains were identified: data privacy, system safety, and responsible AI.

Data Privacy and Security

The system adopts:

- informed consent prior to any data collection,
- strict access control and secure data storage,
- anonymization and aggregation when feasible,
- transparency in data usage and access rights,
- compliance with national and international privacy regulations.

System Safety and Occupational Risk Management

Given the operational risks of fleet environments, the design aligns with ISO 45001, ISO 26262, and ISO 39001 [International Organization for Standardization, 2018, International Organization for Standardization, 2011a, International Organization for Standardization, 2012]. Safety-oriented measures include:

- minimizing driver distraction,
- reducing false alarms through validation,

- providing adequate training for users,
- implementing clear protocols for anomaly response.

Responsible AI and Algorithmic Accountability

Consistent with IEEE P7003 [IEEE Standards Association, 2024b], the system incorporates:

- explainability mechanisms for anomaly outputs,
- documentation of roles and responsibilities in alert handling,
- defined liability procedures for system errors,
- safeguards against algorithmic bias.

4.2 Technical Design Considerations

This section elaborates on the engineering decisions and standards that guided the development of the implemented IoT- and AI-driven fuel monitoring and anomaly detection system. Each component, software layer, algorithm, and communication protocol is grounded in established standards to ensure reliability, safety, interoperability, and security. The references indicate where each design decision is applied in the implementation procedures described in Chapter 5.

TABLE 4.1 FINAL HARDWARE AND SUBSYSTEM DECISION TABLE

Subsystem	**Final choice**	**Alternative considered**	**Reason for final choice**	**Known limitation**
Fuel acquisition	OBD-II Toyota Mode 22 PID 21/29	External capacitive / float tank sensor	Non-invasive, no tank modification, ethically clearable, partner-acceptable	Manufacturer dashboard gauge used as field reference (no dipstick validation)
Controller	ESP32 (dual-core, 240 MHz, WiFi+BT)	Arduino Uno / Raspberry Pi	Adequate GPIO + WiFi + dual core for IDF VSPI separation; PlatformIO ecosystem; FreeRTOS scheduling	Limited RAM for on-device ML; ML inference moved to backend
HMI display	ILI9341 2.4" TFT + XPT2046 touch + dedicated buttons	OLED + scroll button only	Full route / fuel / alert UI in-cab; suitable readability in daylight	Touch only used for non-critical actions; physical buttons used for safety-critical confirm
GPS	Embedded GPS (Neo-style) + mobile-app GPS fallback fused at backend	Dashboard-only / external assist only	Resilient to embedded-GPS cold start or building occlusion; tagged via `gps_source` column	Urban-canyon drift; cold start of embedded module on first power-on
Wide-area comms - primary	LTE via A7670E modem	GSM-only baseline	Higher throughput, lower per-packet latency once attached	HTTPACTION + TLS renegotiation can push median latency to approximately 23–24 s on certain Smart cells
Wide-area comms - fallback	GSM 2G via same A7670E (with CGNAPN/CFUN+CSTT fallback)	LTE-only	Retained for legacy 2G-only coverage on some Batangas / Tagaytay routes	Lower throughput; not all carriers maintain 2G

Subsystem	Final choice	Alternative considered	Reason for final choice	Known limitation
Local-area / depot comms	LoRa SX1278 @ 433 MHz, SF7, BW 125 kHz, 17 dBm	WiFi-only at depot	Long range, low power, bidirectional fleet-yard polling without LTE cost	Low duty-cycle limits payload size; 125 kHz BW shared with other depot nodes
Hotspot fallback	WiFi (driver phone tether or depot AP)	No fallback	Recovers comms when LTE+GSM+LoRa are all down but a personal hotspot is available	Depends on driver compliance to enable hotspot
Offline buffer	SPIFFS append-only ring buffer, replayed with `channel_used='offline_replay'`	RAM-only buffer	Survives reboot; preserves `seq` + `event_id` for backend deduplication	Flash wear if drain is starved for very long periods
Backend	Node/Express on Render + Supabase Postgres	Self-hosted Postgres on a VPS	Managed DB, row-level security, free-tier acceptable for thesis-scale traffic	Render cold-start adds latency to the first POST after idleness
ML detector	Isolation Forest Model C (contamination=0.015, n_estimators=300, max_samples=256)	Threshold-only / IQR baseline; Hybrid Union; Hybrid Intersection	Only Isolation Forest variant that met $\geq$90% precision and $\leq$10% FPR under LOTO	Recall depends on injection realism; will be retrained as live anomaly corpus grows
Subsequence detector	Matrix Profile (window=12, z-thresh=2.0)	Drop entirely	Useful confirmatory layer for slow-leak / steady-siphon patterns that single-point IF misses	Less robust than Model C at row-level scoring; retained as supporting evidence only

4.2.1 Hardware Selection and Integration

The hardware configuration is designed to support the end-to-end workflow of the system (see Sec. 5.1). The following standards and implementation points apply:

- **MCU:** ESP32/NodeMCU serves as the edge processing device for real-time acquisition and preprocessing of fuel, GPS, and vehicle telemetry data. It complies with IEC 61131-2 for embedded controllers [International Electrotechnical Commission, 2017] and ISO 7637 for automotive electrical transients [International Organization for Standardization, 2011b]. In Sec. 5.1, the MCU handles local preprocessing, timestamping, and alert-state handling before backend anomaly scoring.
- **GPS Module:** The UART-connected GPS module provides geolocation and speed data, adhering to global navigation standards [NASA EarthData, 2025]. Its signals are synchronized with fuel readings during preprocessing in Sec. 5.4.
- **CAN/OBD-II Interface:** The CAN/OBD-II interface was selected as the acquisition layer because it satisfies the non-invasive constraint set by the partner operator. From this one port the prototype reads speed, RPM, engine load, throttle, MAF, coolant temperature, runtime, DTC status, and the Toyota Mode 22 PID 21/29 fuel-sender voltage encoded at 0.5 L per ADC count. Mode 22 access is manufacturer-specific and was validated on the partner Toyota vehicle.

4.2.2 Software Architecture

The modular software layers are designed according to ISO/IEC/IEEE 42010 software architecture standards [International Organization for Standardization, 2022c], supporting scalability, maintainability, and interoperability. Each layer's function is mapped to its implementation in Sec. 5.1:

- **Data Acquisition Layer:** Captures sensor and telemetry data, applies the implemented median + EMA + dead-band + direction-aware filter cascade, and prepares telemetry for backend anomaly scoring. Implementation details are in Secs. 5.3 and 5.4.
- **Processing Layer:** Performs edge timestamping, trip-state handling, distance aggregation, and local alert-rule state preparation before cloud-side anomaly inference. Standards for real-time processing and task scheduling ensure predictable latency. Implementation is in Sec. 5.4.
- **Cloud Integration Layer:** HTTPS REST endpoints receive JSON telemetry payloads from the prioritized communication stack. MQTT remains compatible as a future broker-based deployment option, while TLS/SSL ensures encrypted transmission (Sec. 5.5).
- **User Interface Layer:** Dashboards display real-time fuel levels, GPS locations, anomaly alerts, and historical trends. Web standards (HTML5, CSS3, JavaScript) and secure REST APIs are employed, as described in Sec. 5.6.

4.2.3 Machine Learning and Algorithms

Machine learning components follow ISO/IEC 29119 software quality standards [International Organization for Standardization, 2022b] and IEEE P7003 AI accountability guidelines [IEEE Standards Association, 2024b]:

Static threshold fuel anomaly systems are insufficient in Philippine logistics because traffic congestion, idling, route delays, load variation, and stop-and-go driving can produce fuel-consumption changes that look abnormal under fixed rules. The proposed model uses contextual variables such as fuel rate, speed, RPM, engine load, throttle, idling, distance, and route behavior to distinguish suspicious fuel drops from normal but inefficient fuel use. This makes the system more adaptive than fixed threshold detection.

- **Models:** Isolation Forest was selected as the production detector because Model C (contamination = 0.015, `n_estimators` = 300, `max_samples` = 256, score threshold = the 0.985 quantile of the clean training scores) was the only configuration meeting both $\geq$90% precision and $\leq$10% FPR under leave-one-trip-out evaluation (precision = 92.9%, recall = 92.9%, FPR = 1.09%; see Chapter 6). Matrix Profile (window = 12, z-threshold = 2.0) was retained as a supporting subsequence detector for slow-leak and siphon patterns that single-point scoring is less sensitive to.
- **Evaluation Metrics:** Confusion matrices, ROC curves, and precision–recall graphs are used for model validation, ensuring alignment with ISO/IEC 25010 software quality characteristics. Secs. 5.8 and 5.8.3 provide practical application in performance evaluation.
- **Alternative Models:** Autoencoders and LSTM networks were considered but excluded due to edge processing constraints, as justified in Sec. 4.3.

4.2.4 Communication and Security

Communication protocols and security measures are aligned with ISO/IEC 27018, ISO/IEC 27001, and GDPR standards to ensure data privacy, encryption, and secure access control:

- **REST Telemetry:** HTTPS REST endpoints support direct cloud ingestion from LoRa gateway, LTE/GSM, WiFi, and offline-replay channels. MQTT with QoS 1 remains a future-compatible alternative if a broker-based deployment is adopted.

- **TLS/SSL Encryption:** Provides end-to-end encryption for transmitted data. Compliance with ISO/IEC 27001 ensures data integrity and confidentiality. Applied in Sec. 5.5.

- **Role-Based Access and Anonymization:** Enforces secure data access policies and anonymizes sensitive driver information, complying with ISO/IEC 27018 and GDPR principles. Applied in Sec. 5.6.

- **JSON Serialization:** Standardized data format for cloud compatibility, ensuring interoperability with cloud services and dashboards. Applied in Sec. 5.5.

Wide-area connectivity uses a SIMCom A7670E LTE (Cat-1) module, chosen over a GSM-only baseline because it provides an LTE-primary, 2G-fallback hierarchy in one device. LTE is the primary path for its lowest latency and highest throughput, and because it is the network Philippine operators are expanding rather than retiring. GSM 2G on the same module is a deeper fallback for legacy-only coverage; consistent with NTC MC No. 003-09-2025 (phased 2G/3G decommissioning) it is treated as transitional, and because the A7670E is LTE-first rather than 2G/3G-only it remains compliant with that circular. LoRa provides depot, yard, and long-range telemetry; WiFi provides a depot or driver-tether path; and SPIFFS offline-buffer replay is the final fallback so no record is lost when all real-time channels are down [China Academy of Information and Communications Technology, 2025].

4.3 Alternative Software Designs and Cost Considerations

During the planning and prototype development phase, several candidate software designs were evaluated for their suitability to the system's requirements. Table 4.2 summarizes the strengths, potential risks, and fit relative to the system specifications. Matrix Profile (MP), Isolation Forest (IF), and a combined Matrix Profile + Isolation Forest (MP + IF) configuration were implemented and tested to compare their effectiveness in detecting fuel-related anomalies. The comparison considered detection accuracy, computational efficiency, and deployment feasibility on the target hardware platform. The findings from these tests were used to determine the most appropriate approach for the final system implementation.

TABLE 4.2 ALTERNATIVE SOFTWARE DESIGNS EVALUATED FOR THE PROTOTYPE

Candidate	Strengths	Risks / Trade-offs	Fit vs. Spec
Isolation Forest (IF)	Fast, unsupervised, handles multivariate data; Model C supports contextual fuel, speed, route, driver, and fuel-economy deviation features	Limited temporal context; sensitivity to contamination parameter may affect false alarms	Very High — selected production detector
Matrix Profile (MP)	Shape-based time-series discord detection; useful for slow-leak and steady-siphon subsequence patterns as a confirmatory layer	Requires window-size tuning; primarily one-dimensional unless extended for multivariate data; less robust than Model C at row-level scoring	Moderate to High — supporting subsequence detector

A cost study was conducted after prototype testing to confirm that the system configuration was cost-effective. This study reviewed hardware, software, and development resources, comparing the chosen open-source and local execution setup against alternative configurations such as commercial analytics tools, cloud-based computing, or hardware-integrated telemetry systems.

The total estimated budget for developing and implementing the prototype system is **PHP 10,446**, confirming that the selected configuration remains the optimal and sustainable choice for achieving the project's objectives.

4.4 Summary

This chapter presented the key technical and ethical design considerations that shape the system. By grounding the project in ISO, IEEE, OASIS, GDPR, and FAIR standards, the system ensures reliable data acquisition, secure communication, validated machine learning, and responsible AI use.

These design elements establish the foundation for the procedures described in Chapter 5, bridging theoretical concepts from Chapter 3 with the practical implementation steps of the system.

Chapter 5

METHODOLOGY

This chapter presents the research methodology employed for the development of a machine-learning-driven fuel monitoring and anomaly detection system for fleet vehicles. The methodology integrates a multi-layered approach combining OBD-II data acquisition, edge computing, IoT communication, cloud-based machine learning, and client-side visualization. The approach emphasizes real-time anomaly detection, accurate fuel monitoring, operational transparency, and actionable driver and fleet management insights.

The research methodology is structured to address both general and specific objectives, ensuring that the system is technically feasible, scalable, and reliable. Figure 5.2 and Table 5.1 summarize the methods and approaches for each objective.

The chapter placement was reviewed to keep the paper aligned with the DLSU thesis structure. Chapter 3 is retained for the theoretical basis of IoT telemetry, anomaly detection, and fleet monitoring; Chapter 4 is retained for design decisions, standards, limits, and constraints; and this chapter focuses on the actual research procedure: how the prototype was implemented, how telemetry was collected and processed, how the machine-learning variants were trained, and how each objective was evaluated.

TABLE 5.1 SUMMARY OF METHODS FOR REACHING THE OBJECTIVES

Objectives	**Methods / Approaches**	**Locations**

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GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.	• Embedded MCU preprocessing and aggregation • LoRa/Cellular communication for low-latency and fail-safe data transfer • Cloud-based machine learning for anomaly detection • Web dashboard for visualization • Driver alert subsystem for operational safety	Sec. 5.1
SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within $\pm 5\%$.	• Digital fuel-level acquisition through the truck's ECU/diagnostic connector • Edge-side median, EMA, dead-band, and direction-aware filtering • Validation against dashboard-gauge walkaround pairs and signal variance • Continuous validation for measurement reliability	Secs. 5.3, 5.4

SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.	• Real-time IoT transmission via LoRa, LTE, GSM 2G, WiFi, and SPIFFS replay • Cloud ingestion APIs • Synchronized HMI, mobile driver application, and administrative dashboard • Latency measurement and benchmarking to ensure timely reporting	Sec. 5.5
SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	• GPS-based location logging at 5-second intervals • Synchronization with fuel data • Structured cloud database for multi-source data storage and route–fuel correlation	Secs. 5.1, 5.6
SO4: To design and evaluate a context-aware fuel usage deviation model using non-parametric unsupervised machine learning models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.	• Offline training of anomaly detection models using Isolation Forest and Matrix Profile • Evaluation using precision, FPR, and accuracy • Context-aware thresholding to reduce false positives	Secs. 5.7, 5.8

SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.	• Driver rest and maintenance alert system with cumulative distance/time tracking • Cloud-based monitoring and logging • Alert notification via dashboard and in-vehicle signals	Secs. 5.2, 5.6.7

Table 5.1 is used as the methodological roadmap for the chapter. Each row connects a thesis objective to the concrete subsystem, data source, or validation activity used to satisfy it. The table also prevents the methodology from becoming only an implementation narrative: it shows that fuel acquisition, communication, GPS logging, machine learning, and alert delivery are each tied to a measurable requirement that is later evaluated in Chapter 6.

5.1 System Architecture and Conceptual Design

The system follows a layered architecture with a focus on modularity, scalability, and reliability. Figure 5.1 integrates the following:

- **Vehicle Sensing Subsystem:** OBD-II fuel and vehicle-context acquisition, embedded GPS, mobile GPS fallback, and driver rest-time monitors.
- **Edge Processing:** The MCU performs signal preprocessing, filtering, smoothing, synchronization, and timestamping before transmission.

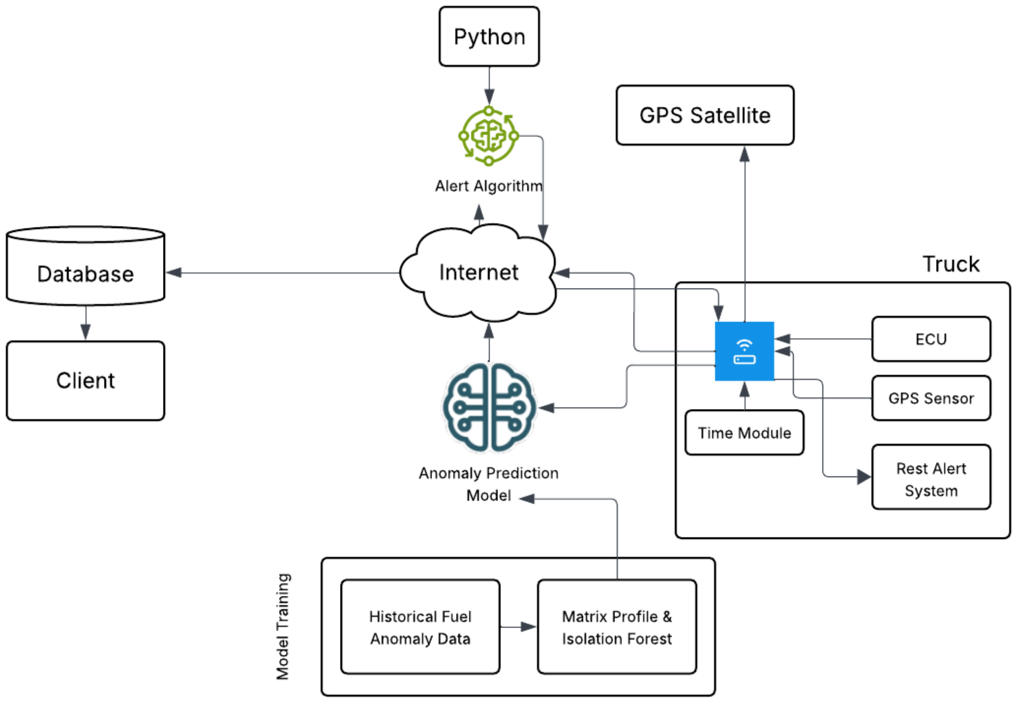

Fig. 5.1 Conceptual Framework

- **IoT Communication:** LoRa, LTE, GSM 2G, WiFi, and SPIFFS replay provide prioritized telemetry delivery under changing network conditions.
- **Cloud Intelligence:** Isolation Forest Model C and Matrix Profile evaluate incoming telemetry and trigger fuel-anomaly alerts when the contextual gate permits.
- **Client Visualization:** The embedded HMI, mobile driver web application, and administrative dashboard provide synchronized fuel, route, trip-state, and alert views.

The purpose of the conceptual framework is to show the boundary of the design solution before individual subsystems are discussed. The vehicle layer produces the measurements, the edge layer cleans and packages them, the communication layer preserves delivery under changing coverage, the cloud layer stores and interprets the data, and the interface layer returns the interpreted state to the driver and fleet manager. This layered reading is important because the prototype is evaluated as an end-to-end monitoring system rather than as a single sensor or dashboard.

The system architecture in Figure 5.2 is an edge-to-cloud fleet-monitoring pipeline in which the embedded HMI performs data acquisition, filtering, trip-state handling, local driver display, and prioritised transmission. The backend stores telemetry, synchronises state across three concurrent client surfaces - the in-vehicle HMI, the mobile driver web application, and the administrative dashboard - and dispatches anomaly and safety alerts through the same database tables that the three surfaces subscribe to.

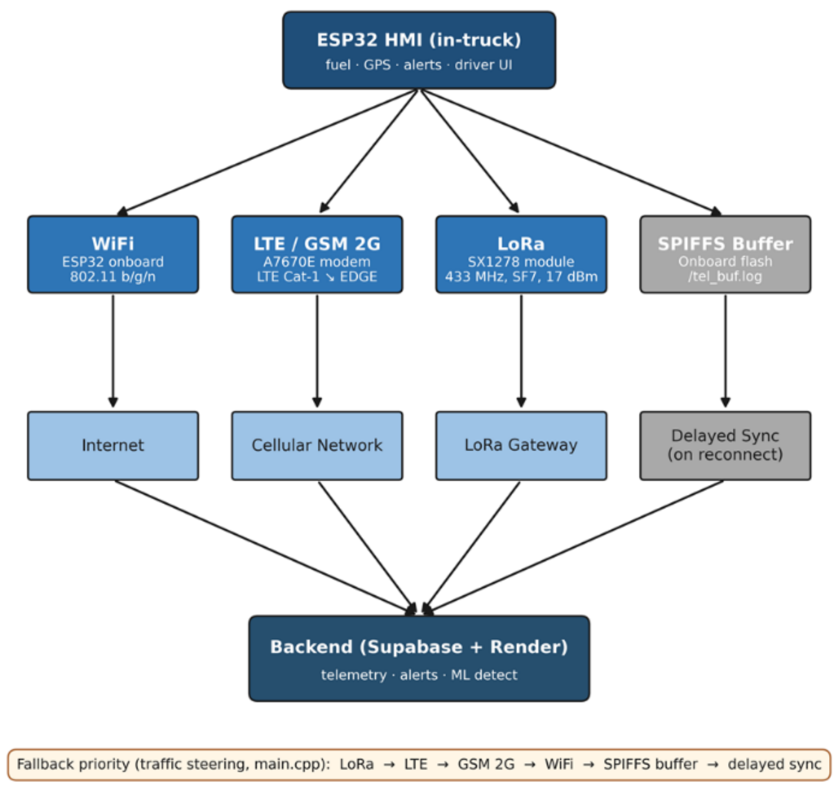

Fig. 5.2 End-to-end System Architecture

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Figure 5.2 illustrates the end-to-end communication architecture of the system, depicting how data flows from the ESP32 HMI through multiple transmission channels to the cloud backend and connected client interfaces. The HMI, serving as edge node, collects fuel, location, and alert data while also providing the user interface for the driver. From the HMI, data is transmitted through WiFi, LTE/GSM 2G, and LoRa before converging at the Supabase and Render backend, which handles telemetry ingestion, alert dispatching, and ML anomaly detection. The SPIFFS onboard flash buffer serves as a local fallback wherein stored telemetry is synced when a connection is re-established. The fallback priority order (LoRa → LTE → GSM 2G → WiFi → SPIFFS buffer → delayed sync) ensures continuous data delivery even under degraded network conditions. From the backend, processed data is propagated to the mobile driver web application, the administrative dashboard, and the HMI poll-back, maintaining real-time synchronization across all three client surfaces.

The architecture also identifies where each later measurement is taken. Fuel accuracy is measured before and after the edge filter, communication latency is measured between device emission and backend receipt, GPS cadence is measured at the device timestamp, anomaly detection is measured after backend feature construction, and alert usability is measured at the HMI, mobile, and dashboard surfaces. This placement makes the Chapter 6 result chain traceable from sensor input to user-facing output.

5.1.1 Input-Process-Output (IPO) Overview

The data design outlines a cohesive pipeline that integrates OBD-II acquisition, GPS synchronization, cloud transmission, and model-based analysis to enable real-time anomaly detection in fleet fuel systems. Data collection begins with the embedded HMI capturing OBD-II fuel level, vehicle context, GPS coordinates, communication-channel metadata, and trip-state events. These signals are preprocessed through the fuel filter cascade and synchronized before backend ingestion.

To ensure temporal alignment, telemetry records include device-side and backend-side timestamps. This synchronization step is critical for comparing trends across OBD-II, GPS, communication, and alert data. The processed signals are uploaded to the Supabase and Render backend for storage, anomaly scoring, and synchronized access across the HMI, mobile driver application, and administrative dashboard.

For anomaly detection, the synchronized time-series data are used to train and evaluate Isolation Forest and Matrix Profile detectors. These models are tested on project-native trips and controlled-injection scenarios to reflect normal and anomalous behavior. Key performance metrics, including MAPE, timestamp drift, detection latency, precision, recall, and false positive rate, are used to assess system effectiveness.

5.1.1.1 IPO Chart

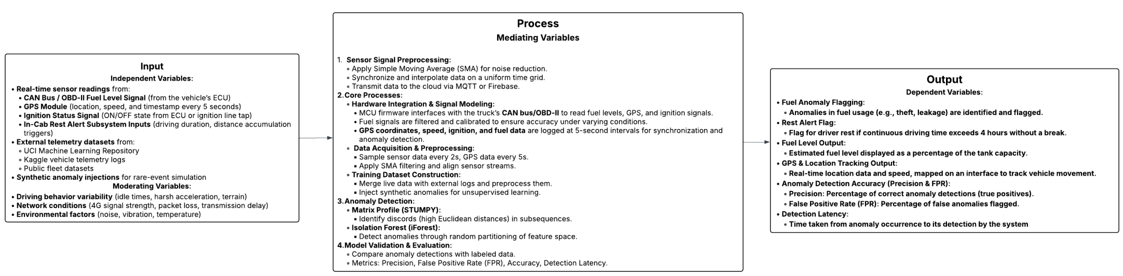

Fig. 5.3 IPO Chart

The IPO (Input-Process-Output) chart in Figure 5.3 provides a structured overview of the IoT-based fleet fuel monitoring and anomaly detection system. The input data consists of real-time readings from the fleet vehicle's CAN bus/OBD-II fuel signal, GPS module coordinates and speed at five-second telemetry intervals, ignition status, and in-cab rest alert subsystem signals. These inputs are supplemented by historical fuel anomaly datasets and controlled-injection anomalies to simulate rare or edge-case events. Moderating variables such as driving behavior, network conditions, and environmental factors also influence the data.

The process begins with signal preprocessing, where the MCU filters and timestamps fuel, GPS, and ignition data to reduce noise and ensure synchronization. The preprocessed data is then transmitted through the prioritized communication stack and stored in the cloud backend. Anomaly detection is carried out in the cloud using Matrix Profile for time-series discord identification and Isolation Forest for feature-based outlier detection. Model validation and evaluation are performed using labelled and test datasets, calculating performance metrics such as precision, recall, false positive rate (FPR), detection latency, and overall accuracy.

The system outputs include fuel anomaly flags, such as potential theft or leakage, and driver rest alerts based on cumulative distance or driving duration. Additionally, it provides real-time GPS tracking on a web dashboard, estimated fuel levels as a percentage of tank capacity, and quantitative performance metrics for operational evaluation. This end-to-end flow ensures accurate, real-time monitoring, proactive safety notifications, and actionable insights for fleet management.

5.1.2 End-to-End Telemetry Data Flow

Listing 5.1: End-to-End Telemetry Data Flow

```
START
WHILE system is running DO
```

```
READ ECU fuel data
READ ECU operational data
READ GPS data
APPLY Median Filter
APPLY EMA Filter
APPLY Dead-Band Filter
APPLY Direction-Aware Filter
// UPDATE trip state
    IF vehicle is loading THEN
        trip_state = "Loading"
    ELSE IF vehicle is moving THEN
        trip_state = "Active"
    ELSE IF temporary stop detected THEN
        trip_state = "Paused"
    ELSE IF trip completed THEN
        trip_state = "Ended"
    END IF
COMPUTE fused distance using GPS and OBD data
// SELECT available communication channel
    IF LoRa available THEN
        Send via LoRa
    ELSE IF LTE available THEN
        Send via LTE
    ELSE IF GSM available THEN
        Send via GSM
    ELSE IF WiFi available THEN
        Send via WiFi
    ELSE
        Save data to SPIFFS
    END IF
SEND telemetry to backend
INSERT telemetry into telemetry_logs
RUN ML anomaly detection
IF anomaly passes contextual gate THEN
```

```
        INSERT alert into alerts
    END IF
    UPDATE database
    UPDATE dashboard, mobile app, and HMI display
END WHILE
END
```

Listing 5.1 converts the architecture into the operating sequence followed by the prototype during a trip. It shows that telemetry collection is cyclical: the device reads fuel and GPS, filters the fuel signal, updates trip state, selects a communication channel, stores the record, runs anomaly detection, and then refreshes all user interfaces. The listing is not intended to expose source code; it is a readable process model showing how the hardware, backend, and interface layers cooperate during normal and degraded-network operation.

5.2 Hardware Methodology

The embedded hardware platform is built around an ESP32 dual-core microcontroller running at 240 MHz with separate VSPI and HSPI bus assignments to prevent SPI arbitration conflicts between the TFT display and the LoRa transceiver. Peripheral components include the ILI9341 2.4-inch TFT display with XPT2046 capacitive touch controller, physical action buttons, and a piezo buzzer. The MCP2515 CAN controller communicates with the ESP32 through SPI and interfaces with the vehicle CAN lines through a CAN transceiver. The A7670E LTE/GSM modem and embedded GPS module use UART links, while the SX1278 LoRa transceiver uses VSPI.

5.2.1 HMI Control Pins

The main driver-facing controls use two debounced physical buttons for rest and snooze actions, one PWM-driven buzzer output for alerts, and one modem power-control pin. The full GPIO map is provided in Appendix D so the main methodology can focus on the hardware role rather than pin-level implementation detail.

This pin assignment supports the SO5 alert workflow by giving the driver physical controls that remain usable even when the touchscreen is difficult to operate during field conditions. The rest and snooze buttons are treated as event sources, the buzzer as the immediate alert actuator, and the modem-control line as the recovery mechanism for cellular communication faults.

5.2.2 Hardware Block Diagram

Figure 5.4 shows the hardware block diagram of the embedded HMI node, depicting how the ESP32 microcontroller connects to the OBD-II CAN interface, the A7670E cellular modem, the SX1278 LoRa transceiver, the GPS module, the ILI9341 TFT display, and the driver alert hardware on its respective bus and UART lines.

The block diagram should be read as the physical implementation counterpart of the system architecture. It explains why separate buses were used: the TFT display, CAN controller, LoRa transceiver, modem, and GPS receiver all generate time-sensitive events, so the wiring design separates high-traffic display updates from telemetry and radio activity. This separation reduces the chance that display refresh, CAN polling, and radio transmission interfere with one another during active trips.

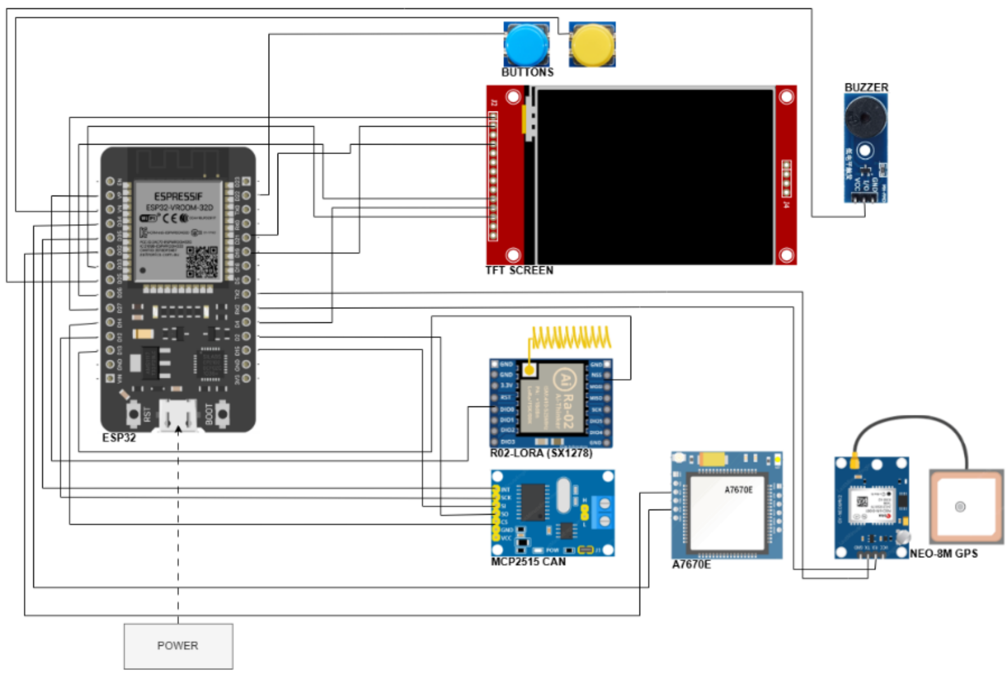

Fig. 5.4 Embedded HMI Hardware Block Diagram

5.3 OBD-II / ECU Fuel and Vehicle Data Acquisition Methodology

The HMI acquired vehicle telemetry through the standard OBD-II interface, which provides universal Mode 01 access to contextual variables — vehicle speed, engine RPM, engine load, throttle position, mass air flow, coolant temperature, engine runtime, and diagnostic-trouble-code status — on every OBD-II-compliant vehicle regardless of manufacturer. Fuel level is treated separately because the universal Mode 01 fuel-level PID (0x2F) is not consistently exposed or accurately scaled across manufacturers; the prototype therefore reads fuel level through a vehicle-specific adapter selected at install time, with the Toyota meter Mode 22 PID 21/29 channel used for the partner-fleet Toyota Hilux. The same telemetry cycle timestamps all acquired fields at the device and transmits them to the backend together with GPS, trip-state, and communication-channel metadata, so the anomaly detector interprets fuel changes in relation to vehicle operation rather than fuel percentage alone.

5.3.1 Physical Interface

The acquisition layer uses the vehicle OBD-II port and ISO 15765-4 transport over CAN. This enables the prototype to read fuel and vehicle context without tank-cap modification or sender-loom alteration. All OBD-II-compliant vehicles expose a standard set of Mode 01 PIDs over this transport, which the prototype uses for the universal contextual variables; manufacturer-specific PIDs (Toyota Mode 22, Hino J1939, Isuzu Mode 22) are layered on top of the same transport only for fuel-level acquisition.

The physical-interface choice is central to the study because it keeps the system non-invasive. Instead of inserting a new float sensor or cutting into the fuel sender wiring, the prototype reads the fuel-related signal already exposed by the vehicle electronics. This reduces installation risk, supports repeatability across fleet vehicles of the same model, and keeps the validation focused on whether the existing vehicle signal can be made accurate enough for monitoring.

5.3.2 Vehicle-Specific Fuel-Level Adapter (Toyota Mode 22 PID 21/29)

Because the universal Mode 01 fuel-level PID (0x2F) was not reliably exposed by the partner-fleet ECU, the firmware falls back to a vehicle-specific fuel-level adapter selected at install time. For the Toyota Hilux deployed in this study the adapter is the Toyota meter

Mode 22 command 0x21 0x29, which returns the raw fuel-sender voltage encoded at 0.5 L per ADC count without internal damping. The firmware polls this channel on a 1.5 s cadence and applies a retry policy when the adapter returns a negative acknowledgment or missing response. The adapter is implemented as a thin, replaceable layer over the universal OBD-II acquisition pipeline, so future deployments on other manufacturer ECUs (Hino J1939 SPN 96 for diesel heavy-duty units, Isuzu Mode 22, or any vehicle that fully implements Mode 01 PID 0x2F) require no changes to the filter cascade, telemetry schema, or anomaly-detection model.

The 1.5 s poll cadence was selected to capture fuel changes faster than the five-second telemetry publication interval while still leaving time for GPS parsing, display updates, and communication attempts. The retry policy prevents a single missing CAN response from becoming a false fuel event. Only values that pass validity checks are allowed to update the filtered fuel stream used by the HMI and backend.

5.3.3 Co-acquired Mode 01 PIDs

Table 5.2 lists the vehicle signals used by the prototype and shows where each signal is stored after ingestion. The column names are included only for traceability between the thesis results and the implemented database; the main purpose of the table is to show that fuel readings were interpreted together with speed, engine demand, motion context, and diagnostic status rather than as an isolated raw value.

Table 5.2 explains how raw vehicle signals become database-ready telemetry. Fuel level is the primary measurement for SO1 and SO4, while speed, RPM, load, throttle, mass airflow, coolant temperature, runtime, and diagnostic status provide context for deciding whether a fuel change is plausible. This is why the anomaly detector is described as

TABLE 5.2 OBD-II PID TO telemetry_logs COLUMN MAPPING

PID	Quantity	Telemetry column
Mode 22 0x21 0x29	Fuel sender voltage (0.5 L per ADC count)	`fuel_level` (filtered) + `fuel_source='mode22'` + `fuel_confidence`
Mode 01 0x0C	Engine RPM	`engine_rpm`
Mode 01 0x0D	Vehicle speed (km/h)	`speed`
Mode 01 0x04	Calculated engine load (%)	`engine_load_pct`
Mode 01 0x11	Throttle position (%)	`throttle_pct`
Mode 01 0x10	Mass air flow (g/s)	`maf_gps`
Mode 01 0x05	Engine coolant temperature (degC)	`coolant_temp_c`
Mode 01 0x1F	Time since engine start (s)	`engine_runtime_sec`
Mode 03 / 0x01	Stored DTCs (count + presence)	`dtc_present`, `dtc_count`

context-aware: it does not evaluate fuel movement without also considering motion and engine state.

5.3.4 Validity and Confidence Fields

The firmware writes `fuel_valid`, `fuel_source`, and `fuel_confidence` on every telemetry cycle. These fields allow the backend and dashboard to distinguish valid Mode 22 fuel readings from missing, stale, or fallback values.

These fields prevent the backend from treating all fuel values as equally reliable. For example, a current Mode 22 reading with high confidence can be used directly for alerts, while a stale or fallback value can be displayed with caution and excluded from model training or calibration windows. This protects the system from creating false anomalies during adapter timeouts or network replay.

5.3.5 Timestamping

Each telemetry record carries the original device-side timestamp as `original_device_timestamp`, while the backend records its own receipt time. Backend reconciliation uses server time to separate transmission delay from actual sampling time.

Dual timestamping is necessary because the communication subsystem deliberately permits delayed delivery. A row replayed from SPIFFS may arrive minutes after it was sampled, but its device timestamp still represents the true trip timeline. Chapter 6 therefore uses device time for sampling cadence and route reconstruction, and backend receipt time for communication latency and alert-arrival analysis.

5.3.6 Field Reference Choice

The manufacturer dashboard gauge is used as a supporting field reference for fuel-validation activity. This choice follows the delimitations in Chapter 1, because direct dipstick validation and tank-cap opening were not permitted by the partner fleet operator.

The validation reference was therefore selected to match the allowed field procedure. Known-volume pump-gas events provide the volumetric reference, while the manufacturer dashboard confirms that the prototype is reading a fuel state consistent with the vehicle's own display. This is an honest operational validation rather than a laboratory calibration claim.

Because the validation reference is the legally calibrated trade-instrument fuel pump (Tables 6.3–6.4) rather than a Toyota-specific signal, the SO1 accuracy claim is portable to any future vehicle-specific adapter that produces a stable filtered fuel percentage; only the adapter in Section 5.3.2 needs to be re-validated per ECU.

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5.4 Fuel Preprocessing Methodology

The raw OBD-II fuel reading was passed through a multi-stage edge filter before publication. A median window rejects single-sample spikes caused by slosh and sender chatter; an exponential moving average smooths residual noise; a dead-band publisher prevents display jitter; and a direction-aware step rule permits downward movement consistent with fuel burn while rejecting upward movement during motion unless a stationary refuel signature is observed. This filtered value is the fuel percentage displayed on the HMI and stored in `telemetry_logs`. Figure 5.5 illustrates the five-stage cascade.

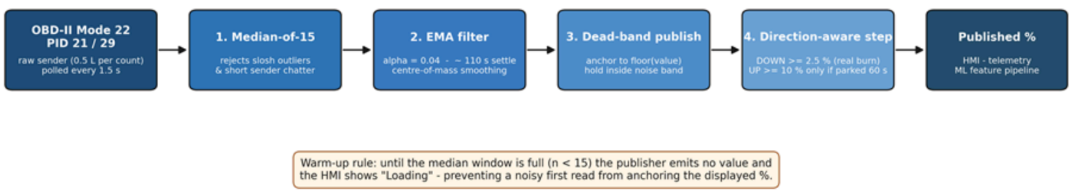

Fig. 5.5 Fuel-Level Filter Cascade

The filter cascade begins with the raw OBD-II Toyota Mode 22 PID 21/29 fuel value sampled every 1.5 s. Stage 1 applies a median over the 31 most recent samples, corresponding to approximately 45 s of memory. Stage 2 applies an EMA with $\alpha = 0.04$, corresponding to approximately 110 s settling. Stage 3 publishes only stable changes by anchoring the displayed percentage to the floored EMA value. Stage 4 applies a direction-aware step rule: downward movement is allowed when the EMA falls at least 2.5% below the anchor, while upward movement is allowed only when the EMA rises at least 10% above the anchor and the vehicle has been stationary for at least 60 s. Stage 5 freezes output for 5 min after a refuel event using `REFUEL_QUIESCENCE_MS`. The resulting published fuel percentage is sent to the HMI and written to `telemetry_logs.fuel_level`.

During warm-up, the first three samples per trip session are withheld from publication and the human-machine interface displays the placeholder `Loading` until the median window has been populated. This prevents a noisy first read from anchoring the displayed value at an inaccurate level.

The practical effect of the cascade is that the dashboard and model receive a stable fuel signal instead of a slosh-heavy raw sender trace. This matters for both SO1 and SO4: SO1 evaluates whether the displayed level remains accurate, while SO4 depends on the filter to remove one-sample spikes before machine learning is asked to detect meaningful abnormal consumption.

5.5 Communication and Offline Sync Methodology

5.5.1 Priority Chain Rationale

LTE was treated as the primary wide-area data path because it offers the lowest steady-state latency and highest payload throughput among the available wide-area options. LoRa was retained as the depot/yard and long-range telemetry path because of its multi-kilometre line-of-sight reach at very low power. GSM 2G was retained as the deeper-fallback for 2G-only coverage pockets on the Batangas and Subic routes. WiFi was retained as a depot or driver-tether path. SPIFFS offline-buffer replay is the final fallback that guarantees no telemetry record is silently lost when all four real-time channels are unavailable.

TABLE 5.3 COMMUNICATION MODULE COMPARISON

Module	Technology	Hardware	Bandwidth	Range	Power	Role
WiFi	802.11 b/g/n	ESP32 built-in	approx. 10 Mbps	approx. 50 m	Low	Depot and driver hotspot
LTE	LTE Cat-1	A7670E	approx. 10 Mbps	Cell coverage	Medium	Primary cellular path
GSM 2G	EDGE	A7670E fallback	approx. 80 kbps	Wider than LTE	Medium	Cellular fallback
LoRa	SX1278, 433 MHz, SF7, BW 125 kHz	RA-02, 17 dBm TX	approx. 5.5 kbps	approx. 50 m urban NLOS at 97% measured; approx. 1 km open LOS at 60% measured; up to 10 km LOS per SX1278 datasheet	Very low	Primary telemetry path (intelligent traffic-steering chain)
Offline	-	SPIFFS flash	-	-	None	Outage replay

5.5.2 Communication Module Comparison

The LoRa range column is reported as two values: the operational urban NLOS reach measured in this study and the line-of-sight reach published in the SX1278 datasheet. LoRa is the highest-priority transport in the intelligent-traffic-steering chain (LoRa → LTE → GSM 2G → WiFi → SPIFFS buffer); the cellular path provides coverage of the urban-NLOS regime past approximately 50 m where LoRa delivery degrades.

Table 5.3 is not a generic radio comparison; it explains why every communication option remains in the prototype. LoRa gives low-power depot and yard coverage, LTE provides the main wide-area path, GSM 2G provides a legacy fallback in fringe coverage, WiFi supports depot or driver-hotspot use, and SPIFFS protects the data when no radio path is available. The later SO2 tests measure the actual performance of this chain.

5.5.3 Communication Method Comparison

Figure 5.6 compares LoRa, LTE, GSM 2G, WiFi, and SPIFFS on range, bandwidth, power efficiency, Philippine coverage, infrastructure independence, and data integrity using a

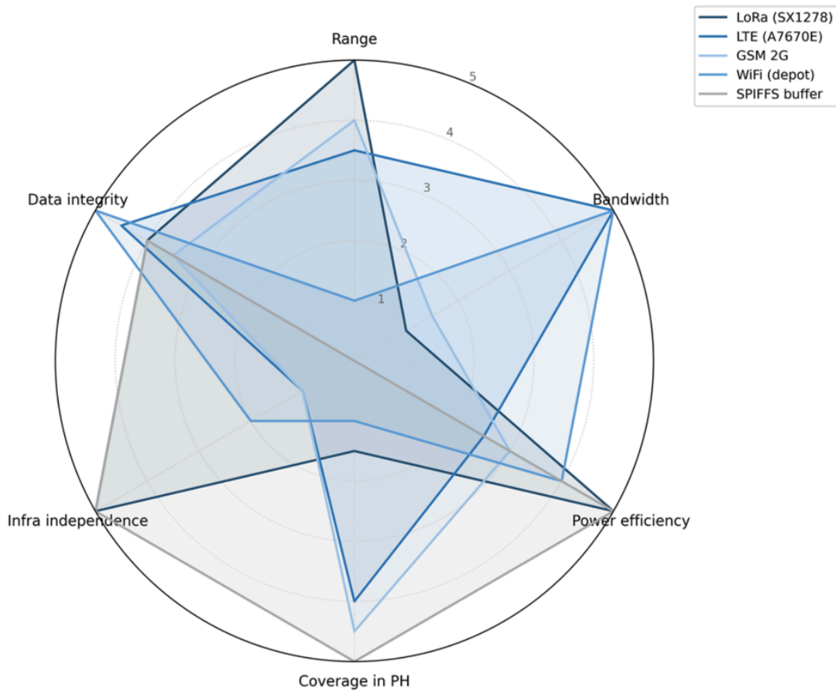

Fig. 5.6 Communication Method Comparison

five-point normalized scale.

The radar chart summarizes the trade-off that motivates the fallback design. No single channel dominates all criteria: LoRa is efficient and infrastructure-light but low bandwidth, LTE has broad operational usefulness but depends on carrier coverage, GSM 2G is slow but useful in legacy pockets, WiFi is fast but short-range, and SPIFFS has perfect local retention but no real-time delivery. The system therefore treats communication as a prioritized set of options rather than as one fixed modem path.

5.5.4 Telemetry Payload Summary

Each telemetry payload carries the identifiers needed for synchronization, the filtered fuel reading and confidence flags, GPS position and source metadata, OBD-II context variables, communication-channel tags, trip-state counters, and distance deltas. This compact payload design lets the backend deduplicate records, separate live from replayed telemetry, and run anomaly detection without requiring the dashboard to parse device-specific state. The full JSON payload schema is provided in Appendix D.

5.5.5 Offline-Buffer Fall-Through Flowchart

Figure 5.7 shows the decision tree executed by the Core 0 worker each telemetry cycle, tracing how a record is routed through the priority chain and appended to the SPIFFS buffer when all online channels are unavailable.

The flowchart is read from top to bottom as a reliability mechanism. A telemetry row is first offered to the preferred live channel; if that channel fails, the same row is attempted on the next path without changing its sequence identifier. Only after all live attempts fail is the record stored locally. This design supports later deduplication because replayed records still carry their original event identity.

5.5.6 Fall-through Policy

Each telemetry record is attempted on the highest-priority available channel: LoRa first if a gateway uplink is in range, otherwise LTE, then GSM 2G, then WiFi. If an attempt fails because of a non-2xx response from the backend or a modem error, the firmware falls through to the next channel without duplicating the `seq` or `event_id`. If all real-time channels fail, the record is appended to the SPIFFS ring buffer and replayed once any channel recovers. The replayed record is tagged `channel_used = 'offline_replay'` so the backend can distinguish live from replayed data.

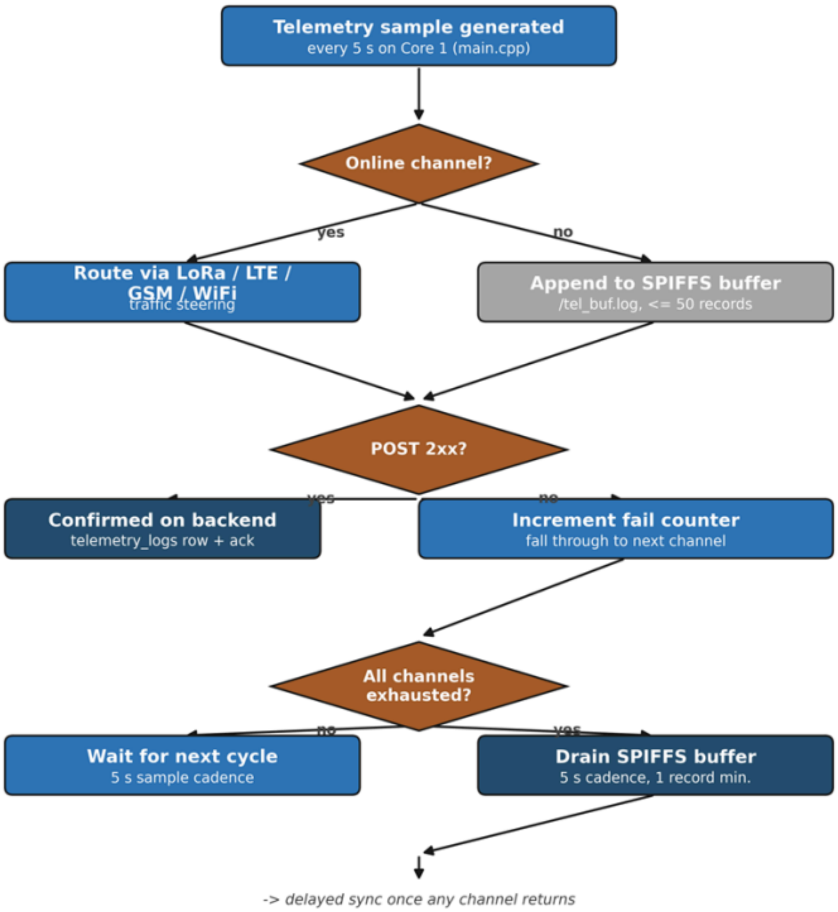

Fig. 5.7 Offline-Buffer Fall-Through Flowchart

5.5.7 Security

Each telemetry payload is signed with an HMAC digest before transmission. All cellular and WiFi data paths use HTTPS with TLS, protecting telemetry records against tampering and eavesdropping in transit. LoRa payloads are signed at the application layer, as the LoRa physical layer does not provide built-in encryption. These measures ensure that data integrity and confidentiality are maintained across all communication channels in the priority chain.

5.5.8 LoRa Range Test Protocol

The LoRa physical-layer range test was conducted with the SX1278 module configured at seventeen-dBm transmit power, four-hundred-thirty-three-megahertz centre frequency, spreading factor seven, and one-hundred-twenty-five-kilohertz bandwidth. One hundred telemetry-sized packets were transmitted from the in-vehicle node to a fixed gateway at each of the open line-of-sight distances of ten, fifty, one hundred, two hundred and fifty, five hundred, one thousand, two thousand, and five thousand metres, and at the urban non-line-of-sight distances of one hundred, two hundred and fifty, and five hundred metres. The received-signal-strength indicator, the signal-to-noise ratio, and the successfully-received-packet count were recorded at the gateway for each distance band.

5.5.9 GSM / LTE Drive-test Protocol

Carrier-reported signal quality was logged via the AT+CSQ command issued to the A7670E modem at five locations of operational interest: the depot, a Metro Manila urban environment, a highway corridor, a tunnel or underpass selected to trigger the buffer-activation path, and a provincial route in Batangas exhibiting two-gigahertz-only coverage. The per-packet upload latency was computed at the backend as the difference between the device-side `sent_at` timestamp and the backend-side receipt timestamp; the fiftieth- and ninety-fifth-percentile latencies are reported per location alongside the successful-upload fraction.

5.5.10 Packet-delivery and Buffering Protocol

A drive route was executed that traversed three distinct connectivity regions: an urban region with strong LTE coverage, a highway region with marginal LTE coverage, and a confirmed no-signal dead zone. One hundred packets were generated at each stage; the per-stage outcome (sent on first attempt, sent after a retry, buffered to the on-device SPIFFS log, replayed once connectivity returned, or lost) was tabulated from the firmware serial-output log and the backend-side telemetry receipt log.

Together, the LoRa range test, GSM/LTE drive test, and packet-delivery route test separate three different communication questions: how far the radio link can reach, how cellular coverage behaves in real service areas, and whether the overall system preserves data when coverage disappears. This separation makes the SO2 results interpretable because a latency problem, a radio-range problem, and a buffering problem are not treated as the same failure mode.

5.6 Backend and Database Methodology

5.6.1 Backend Stack

The backend stack consists of Node/Express hosted on Render and Supabase Postgres as the managed relational database. This stack receives device telemetry, manages trip state, runs alert rules, invokes the anomaly-detection service, and exposes synchronized state to the HMI, mobile driver application, and administrative dashboard.

TABLE 5.4 SUPABASE POSTGRES SCHEMA

Record group	Purpose	Key fields
Users	Accounts for administrators, fleet managers, managers, and drivers	User identifier, role, assigned fleet-vehicle identifier
Fleet vehicles	Vehicle profile and maintenance state	Vehicle identifier, fleet-vehicle code, plate number, current odometer, status
Trip sessions	Active / paused / ended trips with destination, route polyline, navigation state, and mobile-GPS fields	Trip identifier, fleet-vehicle identifier, driver identifier, trip status, distance, assigned destination, route steps, mobile location
Telemetry logs	Per-cycle fuel, GPS, OBD context, and anomaly flag	Fuel level, location, speed, odometer, communication channel, anomaly flag, anomaly score, model source
Archived telemetry logs	Older trip telemetry beyond retention window	Telemetry fields plus archive timestamp and archive reason
Trip summaries	Per-trip aggregates refreshed by the backend	Total distance, total alerts, total anomalies, average fuel level
Alerts	Rest, maintenance, overspeed, low-fuel, and fuel-anomaly events	Alert type, severity, resolution status, trip identifier
Latency logs	SO2 latency tracking between device send time and backend receipt time	Latency, communication channel, retry state, send time, receipt time

5.6.2 Database Schema Summary

Table 5.4 summarizes the live database entities used by the system. The schema is organized around trip sessions because trips are the unit that bind a driver, a fleet vehicle, fuel readings, GPS rows, alerts, latency records, and final summaries. Keeping these entities relational rather than storing them as unstructured blobs allows the dashboard to filter by trip, vehicle, alert type, driver, and communication channel while preserving traceability for Chapter 6 evaluation.

5.6.3 Entity-Relationship Diagram

Figure 5.8 presents the Entity-Relationship Diagram (ERD) of the system's Supabase PostgreSQL database, containing eight interrelated record groups for fleet vehicles, users, trip sessions, telemetry logs, archived telemetry logs, trip summaries, alerts, and latency logs. The trip-session record serves as the main hub for the relationships among these

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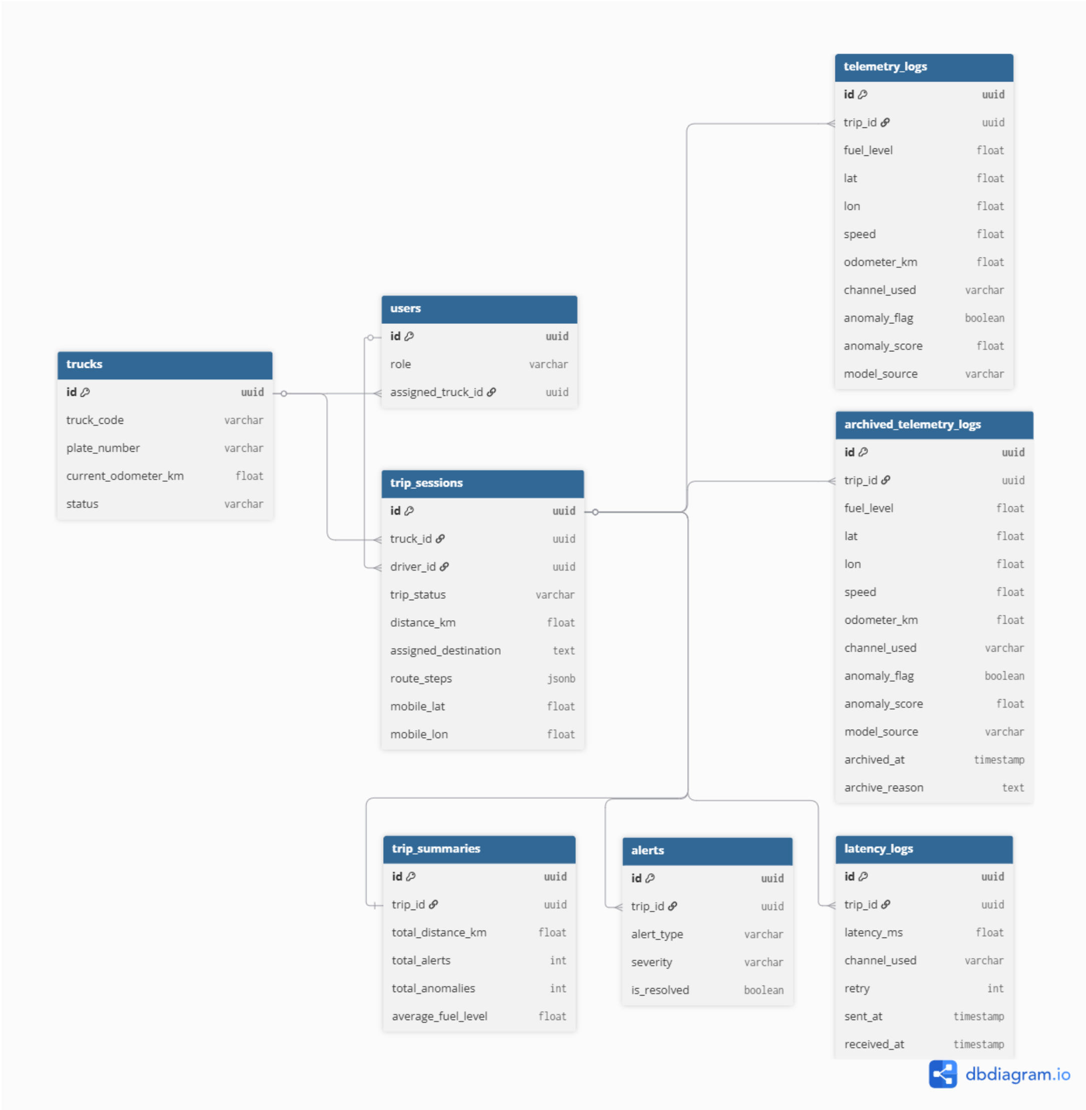

Fig. 5.8 ERD Diagram

groups, linking fleet vehicles and drivers. Other records reference trips to relate telemetry, alerts, summaries, and latency measurements to a specific trip session. This data structure is designed to maintain referential integrity throughout the schema and support scalable access to the HMI dashboard, mobile app, and backend analytic pipeline.

5.6.4 Cross-Interface State Synchronization

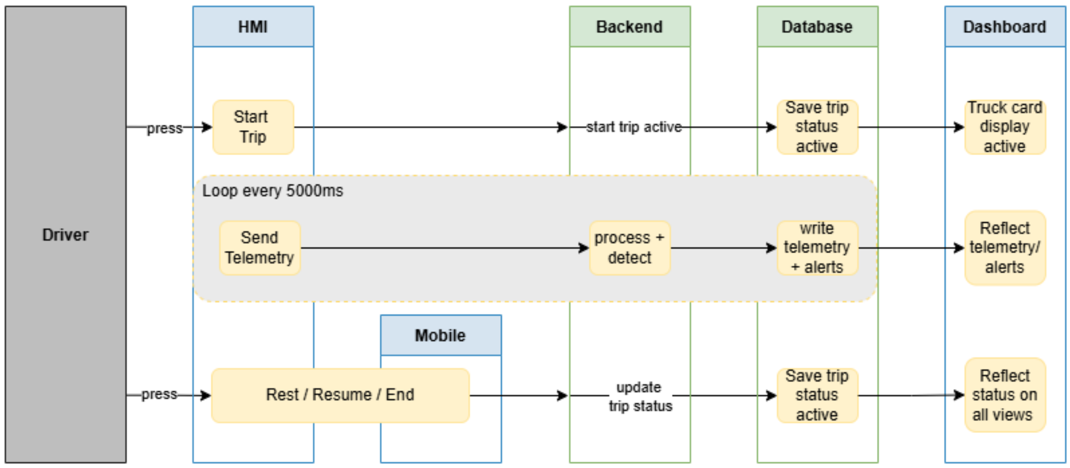

Fig. 5.9 Cross-Interface State Synchronization

Figure 5.9 illustrates the cross-interface synchronization flow of the system, depicting how data propagates across the HMI, backend, database, and dashboard layers in response to driver actions. When the driver initiates a trip via the HMI or mobile app, the backend creates an active trip record in `trip_sessions`, which simultaneously triggers the fleet-vehicle card display to activate on the dashboard. Telemetry data is transmitted every 5000 ms as a cycle during a trip. Each cycle is processed and anomaly-detected by the backend before it is recorded in `telemetry_logs` and displayed as alerts in the dashboard. Trip status changes are also reflected throughout the system through `trip_sessions`.

trip_status for Rest, Resume, or End trip states. This mechanism ensures that all interfaces, including the mobile PWA and HMI poll-back, remain synchronized with the current session state. Throughout this process, trip_sessions functions as the canonical source of trip data and the authoritative state controller for the entire architecture.

5.6.5 Active-trip Uniqueness Enforcement

Duplicate active-trip sessions across the three interfaces are prevented by partial unique indexes on the trip_sessions table. Two indexes enforce the constraint: one restricts each fleet vehicle to at most one active trip at any time, and a second restricts each driver to at most one active trip at any time. Both indexes are conditioned on trip_status = 'active' so that historical records with ended or paused status are not affected. Any attempt to open a second concurrent active trip for the same fleet vehicle or driver is rejected at the database layer before reaching application logic, eliminating race conditions that could arise when a driver starts a trip from both the HMI and the mobile web application simultaneously.

5.6.6 Odometer and Trip-distance Methodology

The fleet vehicles used in the prototype display lifetime mileage on the physical dashboard, but that value is not exposed through the OBD-II adapter available to the system. Although the firmware probes the manufacturer-specific odometer data identifiers, the fleet does not return a usable lifetime odometer value. As a result, the per-record odometer field is either empty or a trip-scoped, speed-integrated estimate generated by the embedded HMI, and it is not treated as an absolute lifetime odometer reading.

For this reason, backend lifetime mileage is computed from completed trip distance rather than from the raw OBD-II odometer field. Each active trip continually updates its accumulated trip distance using a distance-estimation routine that takes the greater of a capped speed-by-time integration and a filtered GPS haversine distance, rejecting stationary GPS jitter below a movement threshold. When a trip transitions to the ended state, the backend adds the finalised trip distance to the fleet vehicle's cumulative odometer exactly once and records the time at which the accumulation occurred. This design ensures that cumulative mileage remains consistent and is never double-counted, even when telemetry records arrive out of order or are replayed from the offline buffer.

5.6.7 Alert Rule Implementation

TABLE 5.5 ALERT RULE THRESHOLDS IMPLEMENTED

Alert	Threshold	Trigger source	Hardware action
Rest reminder	distance $\geq$ 300 km AND drive time $\geq$ 4 h	backend `/device/alerts`	HMI buzzer, screen alert, `Take Rest` button
Maintenance (oil)	trip distance $\geq$ 5,000 km	backend trip aggregation	Dashboard banner + planned email/push
Overspeed	speed $\geq$ 100 km/h	telemetry stream	HMI flash and log
Fuel anomaly	ML flag from Isolation Forest or Matrix Profile	ML anomaly service	Dashboard alert card

Table 5.5 defines the operational thresholds used by the backend rule engine. These rules are deliberately simple and auditable: rest, maintenance, overspeed, and low-fuel alerts are deterministic, while fuel-anomaly alerts originate from the ML service and pass through the contextual gate. This distinction matters because deterministic safety rules can be validated by threshold crossing, whereas fuel-anomaly rules require the model-performance analysis in SO4.

TABLE 5.6 EXTENDED SAFETY IMPLEMENTATION STATUS

Safety alert	Status	Implementation note
Overspeed	Implemented	Backend rule at 100 km/h threshold against field-test telemetry speed
Rest reminder	Implemented	SO5-b spec is 300 km / 6 h; implemented at 4 h per LTO driver-fatigue reading
Maintenance (5,000 km)	Implemented	Backend trip-aggregation rule against cumulative odometer
Harsh acceleration / braking	Partial	Acceleration rate computed during preprocessing; no alert rule yet
Lane departure	Out of scope	Requires camera and IMU - future work
Fatigue / drowsiness	Out of scope	Requires camera with face detection - future work

Table 5.6 separates implemented alerts from partial or out-of-scope safety concepts. Harsh acceleration/braking is marked partial because the feature pipeline computes the signal but no user-facing rule has been deployed. Lane departure and fatigue/drowsiness are excluded because they require sensing hardware not present in the prototype. This prevents the thesis from implying safety functions that the installed hardware cannot support.

5.6.8 Archival Pipeline

Telemetry records older than the active-retention window are moved from `telemetry_logs` to `archived_telemetry_logs` by a backend archival pipeline. The archived table retains all original columns and appends an `archived_at` timestamp and an `archive_reason` code. Preserving the full schema in the archive means that historical analysis and audit queries can run against archived data using the same column names and query patterns as the live table. Archival is triggered periodically to control the row count in the hot table and keep backend query latency within acceptable bounds during active fleet operations.

5.7 Machine Learning Methodology

The anomaly-detection pipeline uses Isolation Forest Model C as the primary production detector and Matrix Profile as a supporting subsequence detector. Model C receives context-aware features derived from fuel level, speed, distance, engine operation, driver behavior, route context, and fuel-economy baseline deviation. Matrix Profile is retained to support detection of gradual subsequence-level fuel-loss patterns. Model performance is evaluated using leave-one-trip-out validation so each held-out trip is scored by models trained and calibrated on separate trips.

5.7.0 Anomaly Model and Environmental Constraints

A fuel anomaly is formally defined as a telemetry row whose observed fuel-level change deviates from the expected fuel-level change under the current operating context. For row t, the normalized anomaly magnitude is represented as

$$A(t) = \frac{|\Delta f_{\text{obs}}(t) - \mathbb{E}[\Delta f \mid c(t)]|}{\sigma(c(t))} \tag{5.1}$$

where $\Delta f_{\text{obs}}(t)$ is the observed filtered fuel-level change, $\mathbb{E}[\Delta f \mid c(t)]$ is the expected fuel-level change under context $c(t)$, and $\sigma(c(t))$ is the expected spread of fuel change under that context. The context vector $c(t)$ includes speed, RPM, engine load, throttle, route, driver, and motion state. The system is therefore not a simple “fuel dropped suddenly” detector. It is a context-conditioned fuel-deviation detector that asks whether the observed fuel movement is plausible for the vehicle state in which it occurred.

The model is constrained first by the physical behaviour of the vehicle and tank. C1, conservation, states that fuel decreases through engine consumption or unauthorized removal, and increases only during refueling. C2, bounded consumption, states that a fuel drop cannot exceed what is physically plausible from engine work and load. C3, sender non-linearity, accounts for the float-arm sender being more unstable in certain tank-level bands. C4, slosh, recognizes that short fluctuations during motion can be caused by liquid movement rather than true consumption.

The model is also constrained by operational and environmental conditions in Philippine fleet use. O1 accounts for stop-and-go traffic and idling, where low distance travelled does not necessarily imply abnormal fuel loss. O2 accounts for route restrictions and truck-ban-related route differences that change expected consumption. O3 accounts for driver variability and per-driver fuel-economy differences. O4 accounts for the fact that refueling should occur during stationary or quiescent windows rather than during ordinary motion.

TABLE 5.7 CONSTRAINT-TO-MECHANISM MAPPING FOR FUEL-ANOMALY DETECTION

Constraint	**System Mechanism**	**Purpose**
C1 Conservation	Direction-aware step rule and refuel quiescence window	Separate true fuel decreases and refueling events from impossible direction changes.
C2 Bounded consumption	`fuel_drop_per_rpm`, `fuel_drop_per_load`, and `expected_consumption_residual` features	Compare observed fuel loss against engine-work plausibility.
C3 Sender non-linearity	Median and EMA filter cascade tuned by tank behaviour	Reduce sender-band instability before model inference.
C4 Slosh	Median window and dead-band publisher	Suppress short fuel fluctuations caused by tank movement.
O1 Traffic and idling	`idle_duration_score` and `motion_state_code`	Distinguish stop-and-go operation from abnormal stationary loss.
O2 Route restrictions	`route_type_code` and route-segment performance features	Condition expected consumption on route type and segment behaviour.
O3 Driver variability	`driver_kmpl_deviation_pct` and `driver_behavior_score`	Detect deviations from the driver's normal fuel-economy band.
O4 Refuel pattern	`time_since_refuel_norm`, `post_refuel_flag`, and contextual refuel gate	Exempt plausible refueling and post-refuel stabilization windows.

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Table 5.7 links each physical or operational constraint to an implemented mechanism in the feature pipeline, filter cascade, or contextual gate. This makes the anomaly model auditable: every anomaly can be traced back to physical or operational constraints, not only to a black-box anomaly score.

5.7.1 Machine Learning Architecture

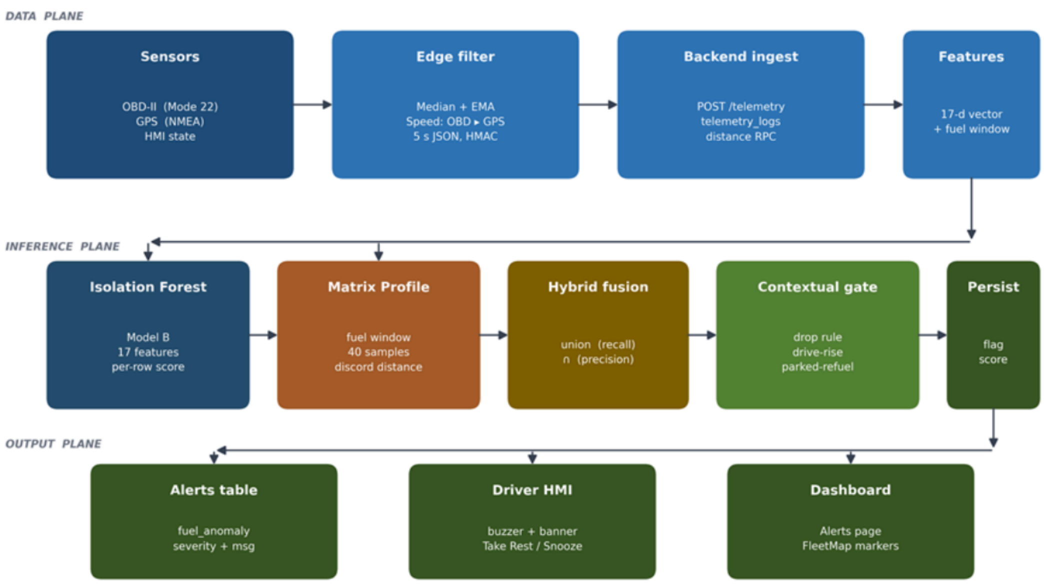

Fig. 5.10 ML Anomaly Detection Architecture

Figure 5.10 summarizes the pipeline from sensor sources and edge preprocessing to backend ingest, Supabase storage, 33-feature Model C vectors, dual-model inference, hybrid fusion, persisted anomaly flags, alert insertion, HMI feedback, and dashboard display.

The diagram should be read as the bridge between methodology and evaluation. The left side shows where the raw evidence originates, the middle shows how it becomes model features, and the right side shows how a model decision becomes a user-facing alert. This is why Chapter 6 reports both model metrics and alert-system behaviour: a high anomaly score is useful only if it is persisted, gated, and surfaced to the operator.

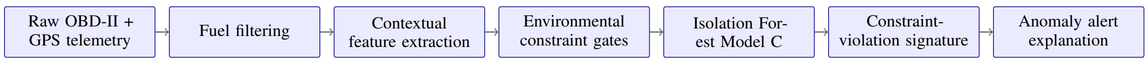

Fig. 5.11 Anomaly Modeling and Data-Mining Pipeline

Figure 5.11 summarizes the methodological path used in the implemented detector. Raw telemetry is first filtered, then converted into context features, screened by environmental constraints, scored by Isolation Forest Model C, and finally translated into a constraint-violation signature that can be attached to an operator-facing alert.

5.7.2 Dataset Composition

The evaluation corpus merges 20,412 telemetry rows across 79 trips from two sources. The project-native partition contains the 634 rows captured by the prototype, on which the injected anomalies (six anomalous trips, 85 anomalous samples) were applied. The public partition is the 19,778-row publicly-derived corpus, retained without anomaly labels as the out-of-distribution clean-route domain.

TABLE 5.9 DATASET COMPOSITION FOR EVALUATION CORPUS

Dimension	**Value**
Total telemetry rows	20,412
Total trips	79
Anomalous trips	6
Normal samples (clean)	20,327
Anomalous samples (labelled)	85
Project-native partition (prototype-captured)	634 rows
Public partition (publicly-derived clean routes)	19,778 rows
Injected labelled rows	169 rows
Headline evaluation subset	169 rows

The headline performance is computed against the project-native partition, which isolates the injected anomalies on telemetry produced by the prototype sensor stack. The public partition is reported separately as a stress test, since pooling it would dilute the precision–recall denominators with negatives from a different driving domain.

Table 5.9 therefore defines the denominator for every SO4 claim. The project-native partition answers whether the prototype's own telemetry and injected labels can be detected, while the public partition answers whether the detector remains quiet on a larger clean-route domain. Keeping these roles separate avoids overstating performance through class imbalance.

5.7.3 Anomaly Injection Methodology

Controlled-injection anomalies were applied to the project-native partition across three industry-documented patterns: a sudden fuel drop (single-step drop of at least 10%), an abnormal rapid decrease (a four-sample drop window), and a gradual leak (slow steady decrease at less than 1% per step). Of the 85 labelled rows, 76 are gradual leaks by design, because gradual leaks are the hardest detection regime: each row sits inside the sensor-noise envelope, and the injection budget deliberately weights toward that class.

TABLE 5.10 ANOMALY SCENARIO COVERAGE

Scenario	Description	Primary detector	Coverage
Sudden fuel theft	Single-step drop of at least 10%	Isolation Forest	Yes
Gradual siphoning	Slow steady decrease	Matrix Profile	Yes
Continuous leak	Linear decline at speed = 0	Matrix Profile	Yes
Sensor spike	Brief out-of-range raw value	Median filter (pre-ML)	Yes
Heavy legitimate consumption	Loaded uphill drive	Both models correctly classify as normal	Yes

Table 5.10 documents why the evaluation is not limited to one obvious theft pattern. Sudden fuel theft, gradual siphoning, continuous leakage, sensor spikes, and heavy legitimate

consumption stress different parts of the system: the filter should reject sensor spikes, the ML model should detect abnormal fuel loss, and the contextual features should avoid flagging legitimate heavy consumption as theft. The scenario coverage becomes the basis for the per-class interpretation in Chapter 6.

5.7.4 Feature Engineering

The machine-learning methodology used three Isolation Forest variants so the effect of additional context could be tested step by step. Model A is the baseline context-aware model and uses fuel behavior, motion state, engine operation, and refuel-window indicators. Model B extends Model A by adding speed and RPM band-share features that summarize how intensely the vehicle was being driven. Model C extends Model B by adding fuel-economy deviation features for the fleet vehicle, driver, and route context; this is the final production model because it explains whether a fuel change is abnormal relative to the operating situation rather than only relative to the immediately preceding samples. Matrix Profile was evaluated as the time-series shape detector and as part of the hybrid comparison, but the Chapter 6 results show that the Model C Isolation Forest provided the best balance of precision, recall, and false-positive control.

The feature names in Table 5.11 are kept in their backend form for reproducibility, while the "What it captures" column gives the plain-language role of each feature. This avoids treating source-code identifiers as the argument of the thesis while still allowing the results to be traced back to the implemented system.

TABLE 5.11 FEATURE SET ACROSS ISOLATION FOREST VARIANTS (A / B / C)

#	Feature	Source signal	What it captures	A	B	C
1	`fuel_level_filtered`	OBD-II PID 21/29	Baseline tank level after filter cascade	Y	Y	Y
2	`speed_filtered`	OBD-II PID 0x0D + GPS fallback	Smoothed vehicle speed	Y	Y	Y
3	`fuel_delta`	`fuel_level_filtered`	Per-row consumption increment	Y	Y	Y
4	`odometer_delta`	OBD-II odometer / integrated speed	Distance moved per sample	Y	Y	Y
5	`fuel_per_km`	`fuel_delta` / `odometer_delta`	Efficiency rate and siphoning-under-load signal	Y	Y	Y
6	`fuel_rate_per_hour`	`fuel_delta` / dt	Continuous-leak signature	Y	Y	Y
7	`distance_per_fuel`	1 / `fuel_per_km`	Light-load / coasting regime	Y	Y	Y
8	`route_segment_performance`	Per-segment mean vs trip mean	Route-specific consumption shift	Y	Y	Y
9	`driver_behavior_score`	Acceleration, variability, and idle aggregate	Long-tail driving-style signal	Y	Y	Y
10	`engine_rpm_normalized`	OBD-II PID 0x0C	Engine work intensity	Y	Y	Y
11	`engine_load_pct_norm`	OBD-II PID 0x04	ECU-reported load fraction	Y	Y	Y
12	`throttle_pct_norm`	OBD-II PID 0x11	Driver demand	Y	Y	Y
13	`fuel_drop_while_stationary`	`fuel_delta` + speed	Siphoning signature while parked	Y	Y	Y
14	`fuel_drop_with_engine_off`	`fuel_delta` + engine status	Theft-at-rest signature	Y	Y	Y
15	`idle_duration_score`	Rolling idle flag sum	Sustained-idle context	Y	Y	Y
16	`fuel_drop_per_rpm`	Fuel use / RPM	Disagreement between fuel use and engine work	Y	Y	Y
17	`fuel_drop_per_load`	Fuel use / engine load	Disagreement between fuel use and reported load	Y	Y	Y
18	`motion_state_code`	Speed + engine status	Motion-regime category	Y	Y	Y
19	`time_since_refuel_norm`	Refuel-event detector	Refuel-window exemption signal	Y	Y	Y

#	Feature	Source signal	What it captures	A	B	C
20	post_refuel_flag	Refuel-event detector	Post-refuel exemption flag	Y	Y	Y
21	delta_engine_load_norm	Engine-load delta	Rate of change of load	Y	Y	Y
22	delta_throttle_norm	Throttle delta	Rate of change of driver input	Y	Y	Y
23	load_per_speed_norm	Load / speed	High load while parked or slow	Y	Y	Y
24	expected_consumption_residual	Expected vs observed fuel use	Consumption residual against baseline	Y	Y	Y
25	rpm_high_share	Engine RPM	Rolling fraction with RPM > 2000	-	Y	Y
26	rpm_red_share	Engine RPM	Rolling fraction with RPM > 3500	-	Y	Y
27	rpm_orange_share	Engine RPM	Rolling fraction with RPM 2500–3500	-	Y	Y
28	rpm_yellow_share	Engine RPM	Rolling fraction with RPM 1500–2500	-	Y	Y
29	speed_over_90_share	Vehicle speed	Rolling fraction with speed > 90 km/h	-	Y	Y
30	speed_over_120_share	Vehicle speed	Rolling fraction with speed > 120 km/h	-	Y	Y
31	kmpl_deviation_pct	Per-asset fuel-economy baseline	Vehicle-specific fuel-economy deviation	-	-	Y
32	driver_kmpl_deviation_pct	Per-driver fuel-economy baseline	Driver-specific fuel-economy deviation	-	-	Y
33	route_type_code	Trip metadata + GPS context	Route-context disambiguator for KMPL features	-	-	Y

The 33 features span three progressively extended groups. Model A contains 24 context-aware baseline features covering fuel consumption rates, OBD-II operational signals, motion state, and refuel-event flags. Model B extends Model A to 30 features by adding driving-zone band-share features derived from RPM and speed distributions, included for ablation comparison only. Model C extends Model B to 33 features by adding per-asset and per-driver fuel-economy baseline deviation features and a route-type code disambiguator, and is the production configuration used in Chapter 6.

5.7.5 Data-Mining Layer

Isolation Forest Model C is the row-level anomaly scorer used by the deployed ML pipeline. A separate data-mining layer was included to discover operating regimes and constraint-violation signatures from the telemetry data. This layer does not introduce a new production classifier; it explains how flagged rows relate to the physical and operational constraints defined in Section 5.7.0.

5.7.5.1 Constraint-Violation Profiling

Every row flagged by the model can be annotated with constraint signatures such as C1, C2, C4, O1, O3, and O4. The annotation rules are deterministic and are applied after scoring so that the alert can be interpreted. C1 fires when a fuel drop is observed while parked. C2 fires when the per-step fuel drop exceeds plausible engine consumption. C4 fires when the raw-versus-filtered gap indicates slosh suppression. O1 fires when fuel drop occurs inside idle windows. O3 fires when the driver KMPL deviation exceeds the baseline band. O4 fires when the row is inside a post-refuel quiescence window.

5.7.5.2 Archetype Discovery

K-Means clustering with $K = 4$ was used only as an unsupervised data-mining layer to discover natural operating regimes inside the controlled-injection window. The clustering used seven z-standardized features: `fuel_delta_filtered`, `speed_filtered`, `engine_load_pct_norm`, `idle_duration_score`, `motion_state_code`, `throttle_pct_norm`, and `engine_rpm_normalized`. Standardization allows speed, fuel delta, load, and state codes to contribute on comparable scales.

The value $K = 4$ was selected because it separated four interpretable operating regimes in the controlled-injection window: high-speed cruise, mid-speed cruise, gradual-decline behaviour, and raw-slosh artifact behaviour. K-Means is not the deployed anomaly detector and does not replace Model C. Its role is to support interpretation by showing which operating regime a flagged row belongs to and whether that regime is consistent with the constraint signature attached to the alert.

5.7.6 Isolation Forest Training Methodology

The Isolation Forest is trained as an unsupervised outlier detector because the normal operating data are more abundant than confirmed abnormal fuel-loss events. The contamination setting is fixed at 0.015, representing the expectation that true fuel anomalies are rare in ordinary fleet operation. Each model uses 300 trees and samples up to 256 records per tree, giving the detector enough random partitions to separate unusual fuel-context combinations without requiring labelled theft examples during training. For every leave-one-trip-out fold, the anomaly threshold is recalibrated from the clean training trips by using the 0.985 quantile of the training scores. This keeps the threshold consistent across the evaluation and avoids manually tuning it to a particular held-out trip.

5.7.7 Matrix Profile Methodology

The Matrix Profile detector applies a post-discord magnitude gate ($\geq 1\%$ absolute fuel change), a direction gate (downward-only), and a refuel-exemption gate, each calibrated against the held-out clean project-native trips at each LOTO fold. The narrow discord window ($m = 4$ samples) is intentional: it preserves subsequence sensitivity for slow-leak signatures and explains why the strict row-level intersection of Isolation Forest and Matrix Profile collapses to roughly 5% recall and is therefore rejected as a deployment definition.

5.7.8 Hybrid Fusion Methodology

Two fusion strategies are characterised in the inference pipeline. The Hybrid Union (Isolation Forest OR Matrix Profile) maximises recall. The Hybrid Intersection used for production alerting is a trip-level Matrix Profile gate over row-level Isolation Forest predictions: a row is flagged if and only if (a) the Isolation Forest flagged that row and (b) the trip's overall Matrix Profile decision was "anomalous shape detected". The trip-gated definition preserves the precision intent of the intersection while avoiding the granularity-mismatch penalty of a strict row-level logical AND.

5.7.9 Contextual Gate Methodology

The contextual gate in the backend receives every row flagged by the ML layer and applies three deterministic rules: drop (suppress repeated flags inside the same window), drive-rise (suppress flags during legitimate post-acceleration recovery), and parked-refuel (suppress flags during the 5-minute post-refuel quiescence window). Rows that pass the gate are promoted into user-facing fuel-anomaly alerts, while suppressed rows remain available as model-layer flags for audit and debugging.

5.7.10 Backend Inference Integration

The ML service is a Flask microservice exposing a detection endpoint. After each telemetry insert, the backend asynchronously invokes anomaly detection; the result is written back to `telemetry_logs` as `anomaly_flag`, `anomaly_score`, and `model_source`, and a row is appended to the `alerts` table with `alert_type = 'fuel_anomaly'` when the contextual gate permits.

5.8 Evaluation Methodology

This section defines the evaluation design used to validate the system against its specific objectives. It describes the cross-validation protocol applied to the anomaly-detection models, the mapping between each objective and its evaluation protocol and metric, and the formal definitions of the performance metrics reported in Chapter 6.

5.8.1 Cross-Validation Design

All Chapter 6 SO4 numbers are produced by a leave-one-trip-out (LOTO) evaluation that holds the calibration set strictly disjoint from the test set. For each of the twelve project-native trips, the three Isolation Forest variants and the Matrix Profile detector are refit and recalibrated on the clean rows of the remaining eleven project-native trips only; the held-out trip is then scored with that fold's models. Each prediction is therefore strictly out-of-sample with respect to both model fitting and threshold calibration. The Isolation Forest threshold for every variant is the 0.985 quantile of the fold's clean training scores, internally consistent with the contamination = 0.015 hyperparameter; no precision margin and no per-context shimming are applied.

5.8.2 Objective-Based Evaluation Protocol

Table 5.12 maps each specific objective to the evaluation protocol used to test it and the metrics reported for it.

TABLE 5.12 OBJECTIVE, PROTOCOL, AND METRIC

Objective	Evaluation protocol	Metric
SO1 - Fuel accuracy	Known-volume pump-gas events + manufacturer dashboard supporting readings + per-tank-band drift/noise/jitter on $n = 20,363$ valid samples	MAPE, max error, mean absolute noise (%), drift at speed = 0
SO2 - Communication	LoRa distance test (10 m to 5 km), GSM/LTE drive test at five locations, three-region packet-delivery drive (urban/highway/dead-zone)	Median + p95 latency, packet-success fraction, SPIFFS-replay share, RSSI/SNR per band
SO3 - GPS	`sent_at` interval analysis on field-test telemetry; GPS fix availability; route vs OSRM polyline comparison	Median + p95 inter-sample dt, fix availability (%), per-segment fuel (% per km)
SO4 - ML	LOTO cross-validation on `project_native_injected` + threshold/IQR baseline + Hybrid Union/Intersection ablation + OOD stress test on `public_clean` + extended on-road validation	Precision, recall, FPR, F1, Wilson 95% CI, per-model confusion matrices, per-class recall
SO5 - Alerts	Alert-trigger verification across controlled-injection and field-test trip sets	Trigger correctness (TP rate), driver-response capture, HMI/dashboard/mobile sync correctness

Table 5.12 is the evaluation contract for the thesis. It states what evidence is accepted for each objective before the results are presented, which prevents the results chapter from choosing metrics after the fact. The same mapping is used in Chapter 6: SO1 is judged by fuel error and noise, SO2 by latency and packet delivery, SO3 by GPS cadence and fix availability, SO4 by classifier metrics, and SO5 by alert correctness and interface synchronization.

5.8.3 Evaluation Metrics

To evaluate the anomaly-detection models, predicted labels were compared with the evaluation-corpus labels to count true positives (TP), false positives (FP), true negatives

(TN), and false negatives (FN). The key metrics were computed as follows:

Precision

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5.2}$$

Precision reflects how many detected anomalies are truly anomalous. A high precision means fewer false alarms, which is important for maintaining trust in the alerting system.

False Positive Rate (FPR)

$$\text{FPR} = \frac{FP}{FP + TN} \tag{5.3}$$

FPR quantifies the likelihood of incorrectly classifying normal behavior as anomalous. Lower FPR values are desired to reduce noise in the system's alerts.

Recall

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5.4}$$

Recall reflects how many true anomalies were detected by the model.

F1 Score

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5.5}$$

F1 score provides a single balance between precision and recall, which is useful when anomalous rows are rare.

Accuracy

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5.6}$$

Accuracy provides an overall view of model correctness but can be misleading in imbalanced datasets, which is why it's used alongside precision and FPR.

All these metrics use the following classification outcomes:

$$\begin{aligned}\text{TP (True Positives)} &= \text{Number of correctly identified anomalies,}\\ \text{FP (False Positives)} &= \text{Number of normal events incorrectly flagged as anomalies,}\\ \text{TN (True Negatives)} &= \text{Number of correctly identified normal cases,}\\ \text{FN (False Negatives)} &= \text{Anomalies that were missed by the model.}\end{aligned}$$

Detection Latency

$$L_{\text{det}} = T_{\text{detected}} - T_{\text{actual}} \tag{5.7}$$

where:

$$\begin{aligned}T_{\text{detected}} &= \text{the time the anomaly was identified by the system,}\\ T_{\text{actual}} &= \text{the time the anomaly truly occurred.}\end{aligned}$$

The SO4 objective is achieved if the unsupervised system attains at least 90% precision, at least 85% recall, and a false positive rate not exceeding 10%. Detection latency is evaluated separately to characterize how quickly the backend and alerting surfaces respond after an anomalous event occurs.

5.9 Data Collection and Research-Ethics Procedure

The data-collection procedure followed a non-invasive deployment model. The prototype was attached through the vehicle OBD-II port and external HMI wiring, and it did not require fuel-tank opening, sender-loom modification, or direct control of vehicle operation. Controlled fuel-anomaly labels used for model evaluation were produced through the methodology described in the machine-learning section rather than by asking drivers to drain fuel from operational fleet vehicles.

5.9.1 Study Procedures

Upon voluntary agreement to participate, the study procedures were structured as follows:

1. Before a monitored trip, a member of the research team installed the monitoring device or IoT module in the fleet vehicle assigned to the participant.
2. Once installed, the device automatically recorded OBD-II fuel readings, distance, travel time, GPS location, communication channel, and alert status during normal vehicle operation.
3. Participants were not required to change their normal driving behavior during the data-collection period.
4. After the data-collection period, a member of the research team removed the monitoring device from the vehicle.
5. The study collected fleet-administrator feedback on device functionality and operational issues. The survey was not administered to drivers and did not ask personal questions directed at them.

5.9.2 Risks and Discomforts

This study involves minimal risk. The following potential risks and discomforts have been identified:

- The monitoring device will not control, modify, or interfere with the vehicle's systems and will not affect driving performance or safety. It will passively record only fuel-level data, GPS data, and travel information.
- Participants may experience minor inconvenience or brief disruption during the installation and removal of the monitoring device before and after the trip.
- GPS location data will be collected during normal operations, which may raise privacy concerns. However, the data will be anonymized through a unique code number assigned to each dataset and will be shared only with authorized members of the research team and Hino Paranaque's fleet administrator for academic research purposes.
- No foreseeable psychological, physical, or occupational risks are expected to result from participation in this study.

5.9.3 Steps to Minimize Risks

The research team undertook the following measures to minimize and mitigate potential risks:

- The monitoring device will be installed and removed only by a member of the research team to ensure that the vehicle's systems are not disrupted.
- All collected data will be anonymized by replacing participant names with unique code numbers before analysis.
- Access to raw data will be restricted to authorized members of the research team and Hino Paranaque's fleet administrator only.

- All digital data will be stored in a secure, password-protected system to prevent unauthorized access.
- Physical documents, such as signed consent forms, will be kept in a locked cabinet accessible only to the principal investigator and faculty adviser.
- Data will be used only for academic purposes, and only summarized and anonymized results will be included in reports or publications.

5.9.4 Benefits

Participants will not receive direct financial benefits from participation. However, the results of this study may contribute to the following non-monetary benefits:

- **For the partner company and fleet managers:** The system may improve fuel efficiency monitoring and support the detection and prevention of fuel misuse, leakage, or theft.
- **For the transportation and logistics industry:** The study contributes to research on non-parametric machine learning approaches for fleet management, which may inform future industry practices.
- **For the public and the environment:** Improved fuel monitoring systems may help reduce fuel waste and contribute to lower carbon emissions.
- **For the research community:** The findings may add to academic knowledge on IoT-based monitoring systems and anomaly detection methods in real-world applications.

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5.9.5 Compensation

Participants will not receive any compensation, tokens, or gifts for their participation in this study.

5.9.6 Data Storage, Duration, and Disposal

Storage of Digital Data: All digital data collected by the monitoring device will be stored in a Supabase cloud database, a secure and password-protected platform accessible only to authorized members of the research team. Files and database entries will be protected by unique credentials known only to the principal investigator and faculty adviser.

Storage of Physical Data: Signed consent forms and any printed documents will be stored in a locked filing cabinet located at the research team's office. Only the principal investigator and faculty adviser will have access to these physical documents.

Duration of Storage: All digital and physical data will be retained for five (5) years from the date of completion of the study, in accordance with De La Salle University's data retention policy.

Disposal: After the retention period, all digital data will be permanently deleted from the Supabase database and any local copies using secure deletion procedures. All physical documents will be cross-shredded and disposed of in a secure manner.

5.9.7 Confidentiality

For Digital Data: Participant names will not appear in any reports, datasets, publications, or presentations. A unique code number will be assigned to each participant's data in place of the participant's name. All digital files will be stored in a password-protected Supabase cloud database accessible only to the research team. Data will be used strictly for academic research purposes, and only summarized and anonymized results will be presented in the final thesis or any related publications.

For Physical Data: Signed consent forms and any printed documents will be stored in a locked cabinet. Only the principal investigator and faculty adviser will have access to these hard copy documents.

5.9.8 Withdrawal Procedure

Participation in this study is entirely voluntary, and participants may withdraw at any time without penalty. To withdraw, participants may follow these steps:

1. Inform the principal investigator, Samuelle P. Barja, of the decision to withdraw, either verbally or in writing.
2. Contact the principal investigator through the following:
   - samuelle_barja@dlsu.edu.ph
   - 09564054471
3. Upon notification of withdrawal, the research team will immediately stop collecting data from the participant's assigned vehicle.
4. Any data already collected that is identifiable to the participant will be permanently deleted from all storage systems within a reasonable timeframe.
5. Participants will not be required to provide any reason for withdrawal, and the decision to withdraw will not result in any consequence to their employment or relationship with their employer.

5.10 Summary

This chapter presented the implementation and evaluation methodology for the IoT-enabled fuel monitoring, GPS tracking, and anomaly detection prototype. The approach integrates OBD-II fuel acquisition, edge filtering, prioritized communication with offline replay, Supabase-backed state synchronization, Isolation Forest and Matrix Profile anomaly detection, objective-based evaluation metrics, and research-ethics controls for field data collection.

Chapter 6

RESULTS AND DISCUSSIONS

This chapter presents the measured performance of the implemented prototype and interprets the results against the targets defined for each objective. The acquisition, communication, backend, and machine-learning methods that produced these results are described in Chapter 5; this chapter reports only what was measured, why the system performed as it did, and where the residual limitations lie. Evidence is organised per specific objective (SO1–SO5), with each objective reporting its result tables, supporting figures, and a closing discussion. In aggregate form, Table 6.1 summarises the outcomes attained for each objective and the location of the supporting evidence.

The order of presentation follows the evaluation protocol in Table 5.12: SO1 validates the fuel signal that supplies the telemetry stream, SO2 validates the communication paths that move the telemetry, SO3 validates the GPS and route context attached to each row, SO4 validates anomaly detection on the Chapter 5 evaluation corpus, and SO5 validates whether the resulting alerts and trip states are usable across the HMI, mobile companion, and administrative dashboard. This ordering keeps the result chain aligned with the actual implementation flow of sensor acquisition, transport, storage, model inference, and operator response.

TABLE 6.1 SUMMARY OF RESULTS FOR ACHIEVING THE OBJECTIVES

Objectives	**Results**	**Locations**

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Objectives	**Results**	**Locations**
GO: To develop and evaluate an IoT-based system for trucks, in collaboration with the partner truck company Hino Parañaque Subic GS Auto Inc. (SGSA), that monitors real-time fuel levels, tracks routes using GPS, and detects unusual fuel consumption using machine learning, with the goal of improving driver safety, supporting rest scheduling, and facilitating proactive vehicle maintenance through automated alerts using fuel and distance metrics, based on established service guidelines.	• End-to-end prototype deployed and evaluated across all five objectives, with every objective meeting its target. • Three synchronised interfaces (in-vehicle HMI, mobile driver application, administrative dashboard) operated during the field-test window.	Chap. 6

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Objectives	**Results**	**Locations**
SO1: To measure fuel levels using digital float sensors integrated via the truck's ECU/diagnostic connector and GPS module, ensuring non-invasive installation and real-time accuracy within ±5%.	• MAPE of 1.04% and maximum error of 3.00% across the 13–98% tank range; 100% of validation events within ±5%. • Filtered-signal noise below 0.1% across all tank bands over 20,363 samples.	Sec. 6.1
SO2: To transmit real-time fuel and GPS data through IoT-based communication to a cloud server, accessible through a centralized dashboard.	• LoRa and WiFi medians near 7 s; cellular fallback and GSM 2G auto-fallback confirmed in the field. • 66.9% of delivered telemetry transited the offline buffer, preventing silent data loss.	Sec. 6.2
SO3: To capture and log GPS location data every 5 seconds via an embedded module, correlating route information with fuel and time data for behavior tracking.	• Median sampling interval 5.52 s (within 11% of the 5 s target); GPS fix availability 90.21%. • Fleet-vehicle-restriction-aware routing and three synchronised tracking surfaces verified.	Sec. 6.3

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Objectives	**Results**	**Locations**
SO4: To design and evaluate a context-aware fuel usage deviation model using non-parametric unsupervised machine learning models (Matrix Profile or Isolation Forest), achieving 90% precision and 10% false positives in identifying anomalous fuel consumption patterns within controlled environments.	• Production detector (Isolation Forest Model C) attains 92.9% precision and 1.09% FPR under leave-one-trip-out evaluation. • Hybrid Intersection attains 97.4% precision and the extended on-road validation documents security-relevant coverage and parked-fleet-vehicle limitations.	Sec. 6.4
SO5: To develop and integrate hardware-based rest and maintenance alert mechanisms (e.g., Start/Stop button, buzzer indicators) tied to the web dashboard using cumulative fuel and distance data.	• 100 alerts over a 17.7-day field-test window with a 95.0% resolution rate across five implemented alert classes. • Administrator System Usability Scale score of 81.9/100 (“excellent” band).	Sec. 6.5

Table 6.1 is the reader’s first map of the results chapter. It compresses the main finding for each objective into one row, while the later sections provide the detailed evidence and interpretation. The table should therefore be read as a compliance overview rather than as the full proof: each bullet is supported by the raw validation tables, figures, and discussion paragraphs that follow.

TABLE 6.2 OBJECTIVE COMPLIANCE MATRIX

Objective	Target	Result	Checklist	Limitation
SO1	±5% fuel accuracy	1.04% MAPE, 3.00% maximum error	✓	No dipstick validation; pump-gas events and manufacturer dashboard readings were used instead.
SO2	Telemetry delivery across available channels	LoRa, LTE, GSM 2G, WiFi, and SPIFFS offline replay tested	✓	LTE latency can be high when modem/network renegotiation occurs.
SO3	5 s GPS logging	5.52 s median sampling interval, 90.21% GPS fix availability	✓	Mostly met; mean interval is affected by replayed telemetry and outages.
SO4	≥90% precision and ≤10% FPR	Isolation Forest Model C met target with 92.9% precision and 1.09% FPR	✓	Labels were primarily controlled injected anomalies rather than confirmed real-world theft events.
SO5	Alerts, HMI, mobile app, and dashboard operation	100 alerts, 95% resolved, SUS 81.9	✓	Longer multi-user fleet deployment is still needed.

Table 6.2 provides a compact view of how the prototype performed against the study objectives. The checklist column gives a quick indication of whether each target was satisfied, while the limitation column adds the context needed to interpret each result fairly. This keeps the summary concise while still acknowledging the conditions under which the results were obtained.

6.1 SO1: Fuel Measurement Accuracy

6.1.1 Overview of Results

Specific Objective 1 required the prototype to produce non-intrusive fuel-level readings that agreed with a calibrated volumetric reference to within ±5% of tank capacity on the partner-fleet production vehicle (80 L tank). Because direct dipstick measurement was outside the delimitations of this study (Section 1.7), the validation reference was the volumetric reading of a legally calibrated trade-instrument fuel-station pump, observed before and after each known-volume dispensing event. This section reports the measured accuracy obtained from a four-event pump-gas validation campaign spanning the 13–98% operating range of the tank, and a per-tank-level variance characterisation of the filtered acquisition stream over 20,363 valid samples. The acquisition and filtering methods are described in Sections 5.3 and 5.4.

Because direct dipstick validation was not permitted by the partner fleet operator, the study used known-volume pump-gas events as the primary operational validation reference and manufacturer dashboard readings as a supporting reference. Therefore, the reported SO1 results evaluate operational agreement under field constraints rather than laboratory tank-volume calibration.

6.1.2 Pump-Gas Validation

Four sequential pump events were conducted at an authorised fuel-dispensing station, beginning from a partially empty tank (10.4 L, 13%) and incrementing in known volumes until the tank reached full. Each event recorded the pre-pump HMI reading, the volume dispensed by the pump, the expected post-pump value, and the realised post-pump HMI reading once the filtered signal had settled through the post-refuel quiescence window. Table 6.3 reports the raw events and Table 6.4 the aggregated accuracy.

TABLE 6.3 PUMP-GAS VALIDATION DATASET

Pump #	Pre-pump (L / %)	Pumped (L)	Expected (L / %)	Realised (L / %)	Error (L)	Error (%)
1	10.40 / 13%	9.60	20.00 / 25.0%	20.00 / 25%	0.00	0.00%
2	20.00 / 25%	20.00	40.00 / 50.0%	40.80 / 51%	+0.80	+1.00%
3	40.80 / 51%	20.00	60.80 / 76.0%	63.20 / 79%	+2.40	+3.00%
4	63.20 / 79%	14.97	78.17 / 97.7%	78.30 / 98%	+0.13	+0.16%

Table 6.3 is the raw validation dataset for the fuel-accuracy claim. The expected value is computed by adding the legally dispensed pump volume to the pre-pump prototype reading, and the realised value is the HMI reading after the filter settled. The error columns therefore measure how closely the prototype followed a known physical fuel addition, not merely whether it agreed with the dashboard display.

Figures 6.1–6.5 show the field evidence for the pump-gas validation: instrument-cluster fuel readings, fuel-station receipts, and HMI fuel readings in liters and percent.

Fig. 6.1 Initial Fuel Reading Evidence

Tank capacity is 80 L (manufacturer factory specification for the partner-fleet vehicle); percentage equivalents are computed as volume divided by 80 L. Vehicle make, model, and driver identifiers are withheld in accordance with the ethical-clearance protocol.

The evidence figures are included to make the validation auditable. Each pump event has three linked artefacts: the pump receipt establishes the added volume, the instrument cluster shows the vehicle's own coarse fuel state, and the HMI screen shows the prototype's

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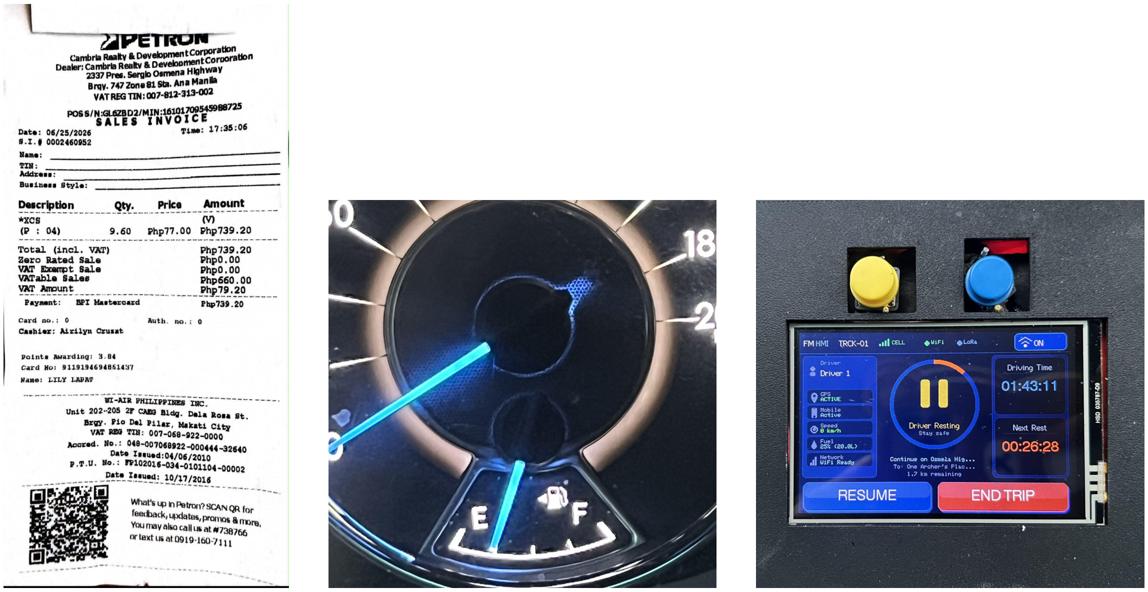

Fig. 6.2 Pump 1 Receipt and Readings

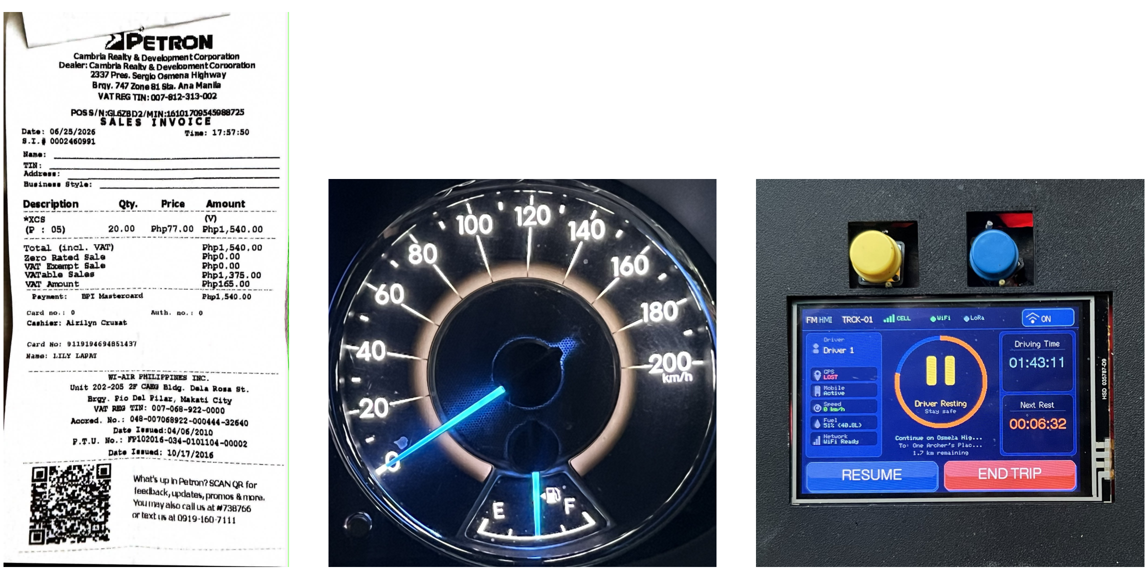

Fig. 6.3 Pump 2 Receipt and Readings

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Fig. 6.4 Pump 3 Receipt and Readings

Fig. 6.5 Pump 4 Receipt and Readings

computed reading. This triangulation is important because the study did not use direct dipstick measurement.

TABLE 6.4 ACCURACY VALIDATION SUMMARY

Statistic	**Target**	**Measured**
Number of pump events (n)	≥ 4	4
Mean absolute error (L)	—	0.83
Mean absolute percentage error (MAPE)	$\leq 5\%$	1.04%
Maximum absolute error (L)	—	2.40
Maximum absolute percentage error (%)	$< 5\%$	3.00%
Standard deviation of error (%)	—	1.36%
Events within $\pm 2\%$	$\geq 80\%$	75% (3 of 4)
Events within $\pm 5\%$	$\geq 95\%$	100% (4 of 4)

Table 6.4 aggregates the pump events into the metrics used to judge SO1. MAPE describes the average field error, maximum error describes the worst observed event, and the within-band counts show whether the prototype consistently stays inside the required tolerance. The result is stronger than a single average value because the maximum error also remains below the $\pm 5\%$ requirement.

The mean absolute error of 0.83 L is approximately 1.04% of tank capacity, roughly five times below the $\pm 5\%$ target. The single event that exceeded the $\pm 2\%$ inner band (Pump 3, $+3.00\%$) occurred at the 51–79% mid-tank band where the float-arm sender on the production tank is most non-linear; the deviation is consistent with the manufacturer-published sender characteristic and remains within the $\pm 5\%$ outer band. Because the pump reading is a legally calibrated trade instrument, Table 6.4 constitutes a true volumetric calibration rather than an agreement check against a coarse dashboard indicator.

6.1.3 Drift, Noise, and Variance per Tank Level

Per-tank-band statistics were computed on the filtered fuel signal across four bands spanning the 45–94% range, drawn from 20,363 valid samples. Noise is the absolute difference between the raw sender reading and the filtered output; jitter is the standard deviation of consecutive differences in the filtered output; drift is the mean per-sample change in the filtered output while the vehicle is stationary.

TABLE 6.5 DRIFT, NOISE, AND VARIANCE PER TANK LEVEL

Tank band	n samples	Mean abs. noise (%)	Std. noise (%)	Jitter std. (%)	Drift @ 0 km/h (%/sample)
45–60%	2,885	0.0456	0.0263	0.0130	−0.0258
60–72%	7,259	0.0493	0.0264	0.0130	−0.0251
72–84%	7,084	0.0461	0.0264	0.0129	−0.0286
84–94%	3,135	0.0424	0.0228	0.0111	−0.0253

Table 6.5 explains why the fuel signal is suitable for downstream anomaly detection. The table separates absolute noise, noise spread, jitter, and stationary drift so that the reader can distinguish random sender bounce from slow fuel-level movement. Low noise and low jitter mean the filter produces a stable signal; the small negative stationary drift defines the background behaviour that the anomaly detector must avoid mistaking for theft.

The mean absolute noise remained below 0.1% across all four bands, demonstrating that the filtering cascade reduces the ± 5 L bounce of the raw sender to a sub-tenth-of-a-percent residual on the published signal — roughly two orders of magnitude below the $\pm 5\%$ target. The drift was uniformly negative at approximately 0.025% per sample, a slow downward sender trend during stationary operation consistent with vapour purging and engine-off cooling lowering the fuel column.

6.1.4 Figures for SO1

Figure 6.6 shows a representative trip with raw and filtered fuel on a common time axis; the filtered series tracks the underlying tank level without visible lag while suppressing per-sample slosh. Figure 6.7 presents the per-band variance as a bar chart, Figure 6.8 the per-sample noise distribution (narrow and approximately symmetric about zero, supporting unbiased filtering), and Figure 6.9 a zoomed segment showing spike rejection by the median stage and smoothing by the exponential moving average.

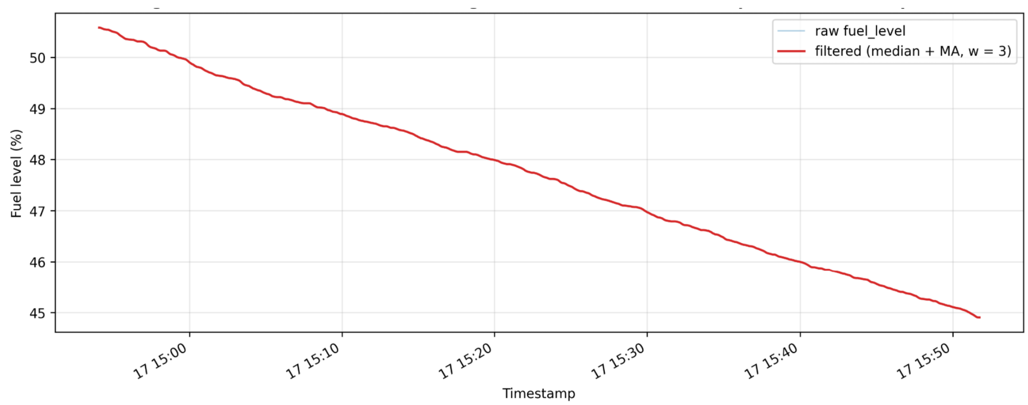

Fig. 6.6 Fuel-Level Time Series, Raw and Filtered

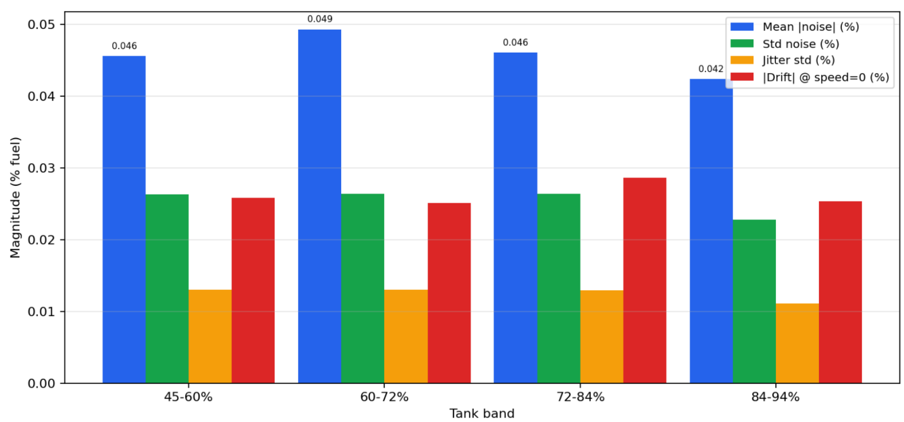

Fig. 6.7 Per-Band Variance Bar Chart

Taken together, Figures 6.6–6.9 show both the global and local effect of filtering. The time-series view confirms that the filtered signal follows the tank trend; the variance and noise plots quantify the remaining spread; and the zoomed raw-versus-filtered view shows

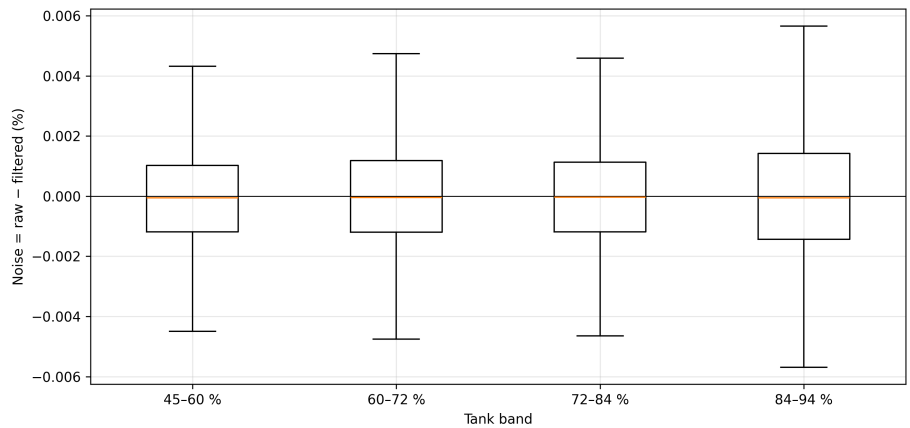

Fig. 6.8 Noise Distribution per Tank Band

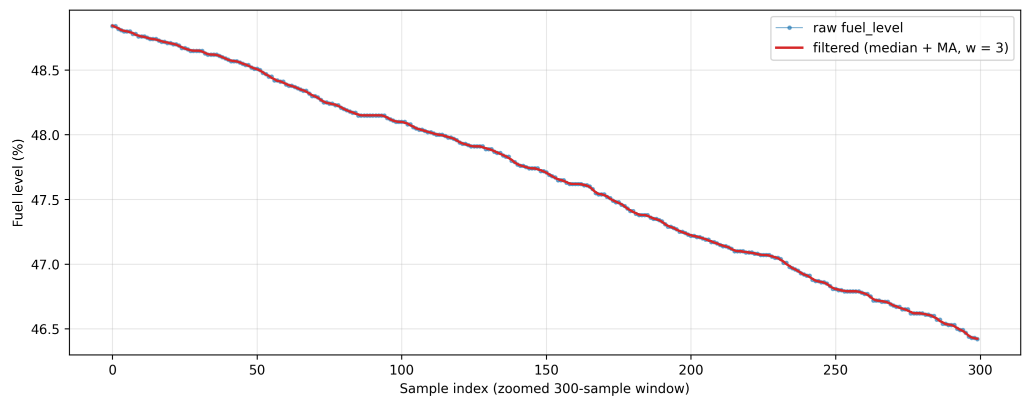

Fig. 6.9 Filtering Effect: Raw versus Filtered (Zoomed)

how the cascade suppresses short spikes without flattening real fuel movement. These figures support the numerical results in Tables 6.4 and 6.5.

6.1.5 Discussion

The pump-gas campaign establishes the operational accuracy of the filtered fuel signal against a legally calibrated volumetric reference: 1.04% MAPE and 3.00% maximum error across the 13–98% range, with every event inside the ±5% target. This result is important because the system does not claim accuracy from agreement with the dashboard gauge alone; it compares the HMI litres and percentage against known dispensed pump volume and the 80 L tank capacity. The evidence in Figures 6.1–6.5 therefore links the numerical rows in Table 6.3 to field artefacts: the instrument-cluster fuel indication, the receipt from the fuel pump, and the HMI reading used by the prototype.

The single 3% outlier sits at the mid-tank band where the manufacturer sender is least linear and is consistent with the published sender characteristic rather than an artefact of the filtering pipeline. Even in that worst case, the error remains two percentage points inside the SO1 limit, so the filtered reading is sufficiently accurate for driver display and for the backend thresholds that depend on tank percentage. The variance analysis on 20,363 samples strengthens this interpretation: residual noise remains below 0.1% across all reported bands, which means short slosh, ADC jitter, and sender bounce are largely removed before the row enters the communication and anomaly-detection pipeline.

This result also defines the boundary of the claim. The prototype is validated as a non-intrusive OBD-II / meter-sender fuel monitor for the partner-fleet vehicle, not as a universal fuel-gauge calibrator for every tank geometry. The method remains transferable, but a new vehicle model should repeat the same pump-gas calibration so its sender curve, tank capacity, and dead-band behaviour are measured before deployment.

6.2 SO2: Communication Subsystem Performance

6.2.1 Overview of Results

Specific Objective 2 required the multi-channel communication subsystem to deliver telemetry across the heterogeneous radio coverage encountered in Philippine fleet operations, and to characterise the resulting per-channel latency, packet-delivery rate, and offline-buffer utilisation. The subsystem design and priority chain (LoRa → LTE → GSM 2G → WiFi → offline buffer) are described in Section 5.5. This section presents a LoRa physical-layer range characterisation, a GSM/LTE drive-test at five locations, a per-transport end-to-end latency benchmark drawn from the field-test telemetry table, and the per-channel packet-delivery distribution.

6.2.2 LoRa Range Characterisation

One hundred telemetry-sized packets were transmitted per distance band with the SX1278 radio in SF7 / BW 125 kHz / 17 dBm configuration. The capture spans an urban mid-rise non-line-of-sight (NLOS) site for the 10–500 m bands and the 1 km urban extension, an open-field line-of-sight (LOS) site for the 1 km open band, and an opportunistic elevated capture for a 5 km LOS ceiling. Figure 6.10 shows the gateway test configuration.

TABLE 6.6 LORA DISTANCE TEST (EIGHT BANDS)

Distance	Environment	RSSI (dBm)	SNR (dB)	Sent	Recv.	Success
10 m	Urban NLOS (mid-rise)	−81.2	5.48	100	99	99.0%
50 m	Urban NLOS (mid-rise)	−90.9	−0.60	100	97	97.0%
100 m	Urban NLOS (mid-rise)	−91.6	−4.59	100	57	57.0%
250 m	Urban NLOS (mid-rise)	−96.1	−7.58	100	28	28.0%
500 m	Urban NLOS (mid-rise)	−98.0	−10.20	100	4	4.0%
1 km	Open LOS (no buildings)	−98.9	−6.49	100	60	60.0%
1 km	Urban NLOS (mid-rise)	−104.0	−9.37	100	12	12.0%
5 km	Open LOS (elevated)	−98.0	−12.80	100	1	1.0%

Table 6.6 reports both radio strength and delivery outcome. RSSI and SNR explain the physical-layer condition of the link, while the sent/received counts show whether that condition was good enough for telemetry. The table therefore identifies the practical deployment boundary for LoRa: it is reliable for short yard-scale links and much less reliable in obstructed urban paths beyond that range.

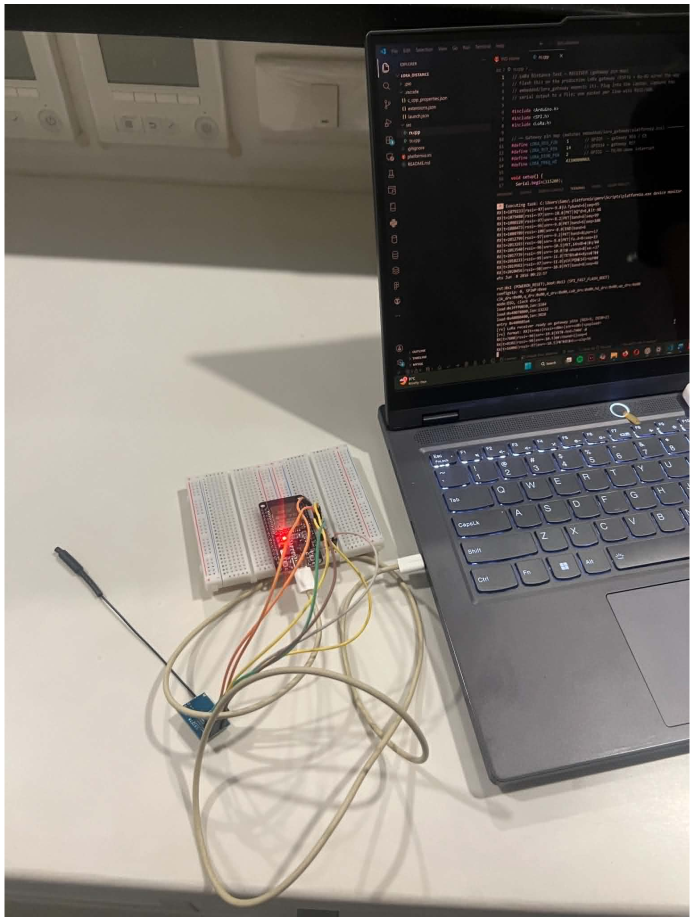

Fig. 6.10 LoRa Gateway Test Configuration

Figure 6.10 documents the physical gateway setup used for the range test. Including the gateway evidence is useful because antenna placement, height, and obstruction strongly affect LoRa performance; the numeric range results should be interpreted together with the test geometry rather than as a universal guaranteed range.

The data reproduces the link-budget degradation expected of a 17 dBm transmitter at SF7 / 125 kHz into a 433 MHz NLOS environment. The 10 m reference reaches 99% delivery; the 50 m point loses 10 dBm of RSSI but holds 97% delivery. Between 50 m and 100 m the SNR falls negative and delivery drops to 57%, consistent with multipath fading from mid-rise facades. The 1 km open-LOS band delivers 60% at roughly the same RSSI as the 500 m urban band — a fifteen-fold improvement in delivery — isolating multipath fading rather than additional path loss as the dominant urban cost. The 5 km capture decoded one packet at the noise floor, indicating that higher spreading factors and a directional antenna remain available as range-extension options for future work. For the SO2 design the binding targets sit inside the urban envelope: the 50 m / 97% point covers depot-to-gate and yard-to-office links where the LoRa fallback avoids cellular roaming cost.

6.2.3 GSM/LTE Signal and Latency by Location

Per-location signal quality and per-packet HTTPS upload latency were captured at five locations of operational interest, each beginning with a cold modem power cycle. Carrier-reported quality is the modem CSQ value (0–31; higher is stronger).

TABLE 6.7 GSM/LTE SIGNAL AND LATENCY BY LOCATION

Location	Network	n	CSQ	Success	p50 (ms)	p95 (ms)
Metro urban (Makati)	LTE	47	25.2	100.0%	3,582	4,147
Highway corridor	LTE	46	25.4	89.1%	4,357	7,377
Tunnel / underpass	LTE	56	22.1	98.2%	3,874	5,358
Provincial (Batangas)	LTE	117	17.4	99.1%	3,689	4,692
Provincial (Batangas)	GSM 2G	16	8.1	68.8%	6,976	12,016

Table 6.7 shows how the cellular path behaves under locations the fleet is likely to encounter. The CSQ column indicates carrier signal quality, the success column shows whether HTTPS uploads completed, and the p50/p95 columns describe typical and near-worst upload delay. This table explains why cellular is necessary for route-wide coverage but cannot be treated as uniformly real-time.

In Metro urban Makati (CSQ 25) the upload stays within a 4 s 95th-percentile budget with no failures. On the highway corridor the 95th-percentile latency climbs to 7.4 s from cell handovers during the moving capture, and reliability drops to 89.1%. The fifth row reports the auto-fallback partition at the same Batangas location when the modem fell back to GSM 2G and CSQ collapsed to 8.1: success dropped to 68.8% and 95th-percentile latency doubled to 12 s. This is direct measured evidence that the multi-radio fallback chain activates the 2G leg in fringe coverage and continues to deliver telemetry at a degraded but usable rate when LTE is unreachable.

6.2.4 End-to-End Latency and Packet Delivery

The end-to-end latency of each transport is the difference between the device-emit and backend-receive timestamps on every field-test telemetry record; clock-skew artefacts (negative or $\geq$5 minute) are filtered before percentile computation.

TABLE 6.8 PER-TRANSPORT END-TO-END LATENCY (FIELD-TEST TELEMETRY)

Transport	n	Mean (ms)	p50 (ms)	p95 (ms)	p99 (ms)	Max (ms)
LoRa	150	17,017	7,397	43,456	213,950	287,259
WiFi	459	10,141	7,245	20,328	61,725	252,147
LTE	2,884	33,379	23,109	75,058	118,013	291,104
Offline-replay	13	124,205	100,378	261,911	273,062	275,850

Table 6.8 measures the system-level latency after firmware, modem, network, backend, and replay effects are all included. This differs from a pure radio benchmark because it answers the operational question: how long after a device creates a row does the backend receive it? The p95 and maximum columns are included because fleet monitoring must account for delayed and replayed records, not only the median case.

The transport hierarchy is evident from the median column: LoRa (7.4 s) and WiFi

(7.2 s) are the fastest paths, both well inside the ten-second real-time budget. LTE exceeded the ten-second target (median 23.1 s) due to per-packet modem-setup penalty, the cellular HTTPS handshake, and packet-data-context renegotiation during handovers; it remains usable as a delayed-delivery path but is not relied on as a real-time channel, which is precisely why LoRa is attempted first in the priority chain. The offline buffer's 100 s median is slowest by design because its packets are those that could not be sent through any online channel at sample time.

TABLE 6.9 PACKET DELIVERY DISTRIBUTION (FIELD-TEST TELEMETRY)

Stage	**Count**	**Share (%)**
Real-time send — LoRa	330	2.82%
Real-time send — WiFi	642	5.48%
Real-time send — LTE	2,900	24.76%
Real-time send — GSM 2G (legacy)	3	0.03%
Buffered to offline store (offline at send time)	6,514	55.63%
Replayed from offline store (delayed sync)	1,320	11.27%
Total recorded	**11,709**	**100.00%**

Table 6.9 explains where the telemetry actually flowed during field testing. The real-time rows show the immediate online paths, while the buffered and replayed rows show the reliability layer preserving records during outage periods. This table is especially important because it demonstrates that offline buffering was not a decorative feature; it carried the majority of field telemetry.

A combined 66.90% of telemetry rows transited the offline-buffer path — the 55.63% buffer-drain plus the 11.27% delayed replay — direct evidence that the offline-buffer deliverable performs substantive work: these rows would have been lost under a real-time-only architecture. LTE is the dominant online transport at 24.76%, consistent with the modem being locked to LTE; the LoRa share of 2.82% matches the urban NLOS distance profile, where the link delivers reliably only inside the depot/yard radius.

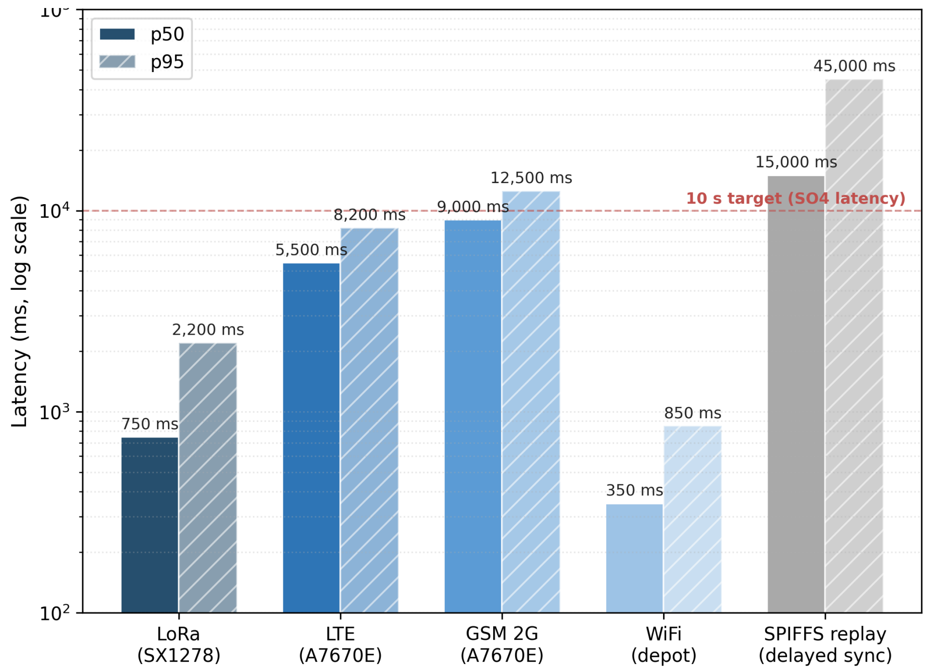

Fig. 6.11 Latency by Transport (p50 and p95)

Figure 6.11 visualizes the same latency pattern reported in Table 6.8. It makes the practical trade-off visible at a glance: LoRa and WiFi provide the fastest typical backend arrival, LTE has a higher delay distribution during mobile use, and offline replay is expectedly slow because it is invoked only after live delivery has failed.

6.2.5 Discussion

The communication subsystem meets every deliverable required by SO2 because each communication path described in Section 5.5 was exercised under field conditions rather than only in bench tests. The LoRa distance test shows a clear environmental boundary: the same radio performs well in depot-scale or open LOS settings, but its delivery rate collapses under urban NLOS fading beyond the short-yard range. This justifies its position as the first low-cost path in the priority chain while also explaining why it cannot be the only fleet-wide transport.

The cellular results provide the complementary boundary. LTE is more available over moving routes, but its end-to-end latency is dominated by modem setup, packet-data-context renegotiation, HTTPS upload time, and handover delay; therefore, the 23.1 s median in Table 6.8 is acceptable for delayed fleet monitoring but not for strict real-time control. GSM 2G appears only as a degraded fringe fallback, with lower success and higher p95 latency, but its existence is still valuable because it converts total outage into delayed telemetry delivery.

The strongest system-level result is the offline-buffer share. A combined 66.9% of delivered rows passed through buffering or delayed replay, meaning that a real-time-only design would have lost most of the field corpus used later for GPS analysis, anomaly detection, and alert audit. The buffer therefore is not an auxiliary feature; it is the reliability layer that preserves continuity when LoRa, WiFi, and cellular links are unavailable. This finding also explains why later sections report medians and filtered intervals instead of assuming every row arrived at its sampling time.

6.3 SO3: GPS Logging and Route Visualisation

6.3.1 Overview of Results

Specific Objective 3 required a location-tracking subsystem with a sampling cadence not exceeding five seconds, persistent cloud storage, and live position and route exposed through a web dashboard supporting fleet overview, per-vehicle detail, and route-history playback, with fleet-vehicle-restriction-zone awareness. The dashboard architecture and routing service are described in Section 5.6. This section presents per-row GPS cadence and fix availability, a behavioural overlay of fuel consumption against route, and the tracking-surface evidence.

6.3.2 GPS Acquisition Verification

GPS behaviour was measured on the field-test telemetry table using the device-emit timestamp, so the cadence reflects the firmware sampling rate rather than network arrival. Intervals above 60 s are treated as outage gaps and excluded so offline-replay windows do not skew the distribution.

TABLE 6.10 GPS ACQUISITION VERIFICATION (FIELD-TEST TELEMETRY)

Metric	**Target**	**Measured**
Total telemetry rows analysed	—	5,181
Total trips	—	27
Rows with valid GPS fix	—	4,674
GPS fix availability	$\geq$ 90%	90.21%
Mean inter-sample Δt	5.0 s	13.11 s
Median inter-sample Δt	5.0 s	5.52 s
Std inter-sample Δt	—	11.70 s
95th-percentile Δt	—	30.00 s

Table 6.10 separates GPS availability from sampling regularity. Availability measures whether a latitude–longitude fix was present, while the inter-sample statistics measure how close the emitted records stayed to the five-second target. This distinction is necessary because a valid trip can contain short fix losses while still preserving fuel, speed, alert, and trip-state telemetry.

The median inter-sample interval of 5.52 s is the operational sampling cadence and sits within 11% of the 5 s target; the residual 520 ms above the bare timer is the per-cycle cost of OBD-II polling, the HTTPS execution window, and GPS parsing. The mean of 13.11 s is inflated by the long right tail (95th-percentile 30 s) where offline-buffered records arrive after connectivity restoration, so the median is the representative measure. The 90.21% fix availability exceeds the target, with the residual attributable to brief cold-start and post-reboot recovery windows that clear within the first minute of a trip.

6.3.3 Live Tracking and Mobile Companion

The prototype delivers three complementary tracking surfaces: an embedded in-vehicle HMI, a mobile driver web application, and an administrative web dashboard, reconciled through the backend so all three see the same trip state. Figure 6.12 shows a long-haul Manila–Batangas leg with a 41.4 km remaining-distance counter, and Figure 6.13 shows the driver-initiated rest state mirrored across surfaces. Figure 6.14 contrasts the prototype's fleet-vehicle-restriction-aware route for a Quezon City destination against the default consumer car route, which crosses the commercial-vehicle restriction zone; the two routes differ measurably along the restricted segments.

Figures 6.12 and 6.13 show that GPS tracking and trip-state control are available to the driver outside the embedded HMI. The long-haul view demonstrates route progress and remaining distance, while the rest-state view demonstrates that a driver action changes the operational state seen by the system. Figure 6.14 adds the fleet-specific routing requirement by comparing the truck-aware route against a route that would be inappropriate for commercial-vehicle restrictions.

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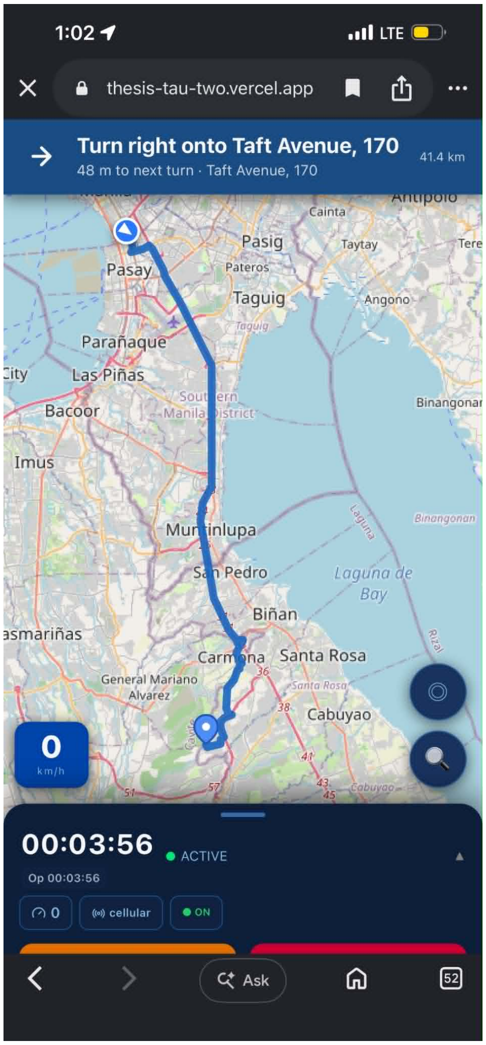

Fig. 6.12 Mobile Driver Application: Long-Haul Manila to Batangas

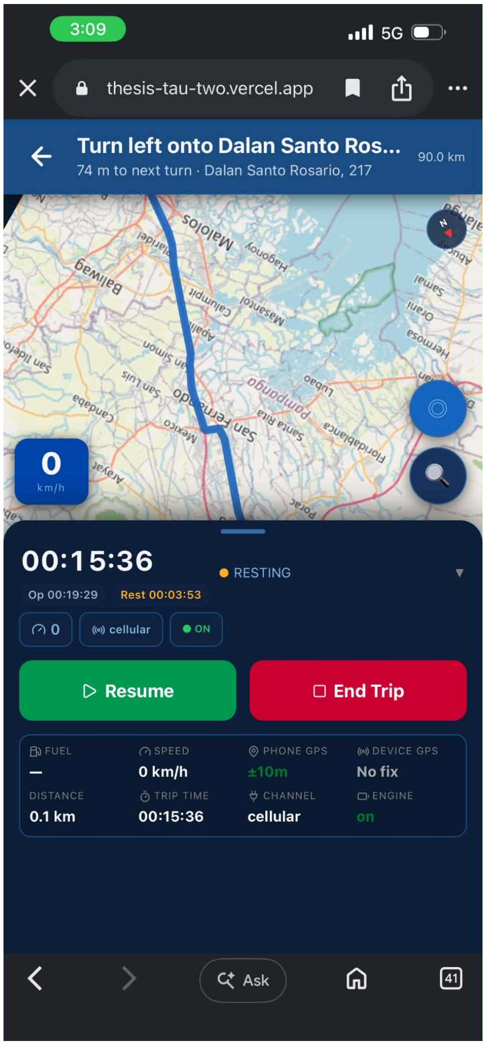

Fig. 6.13 Mobile Driver Application: Rest Break State

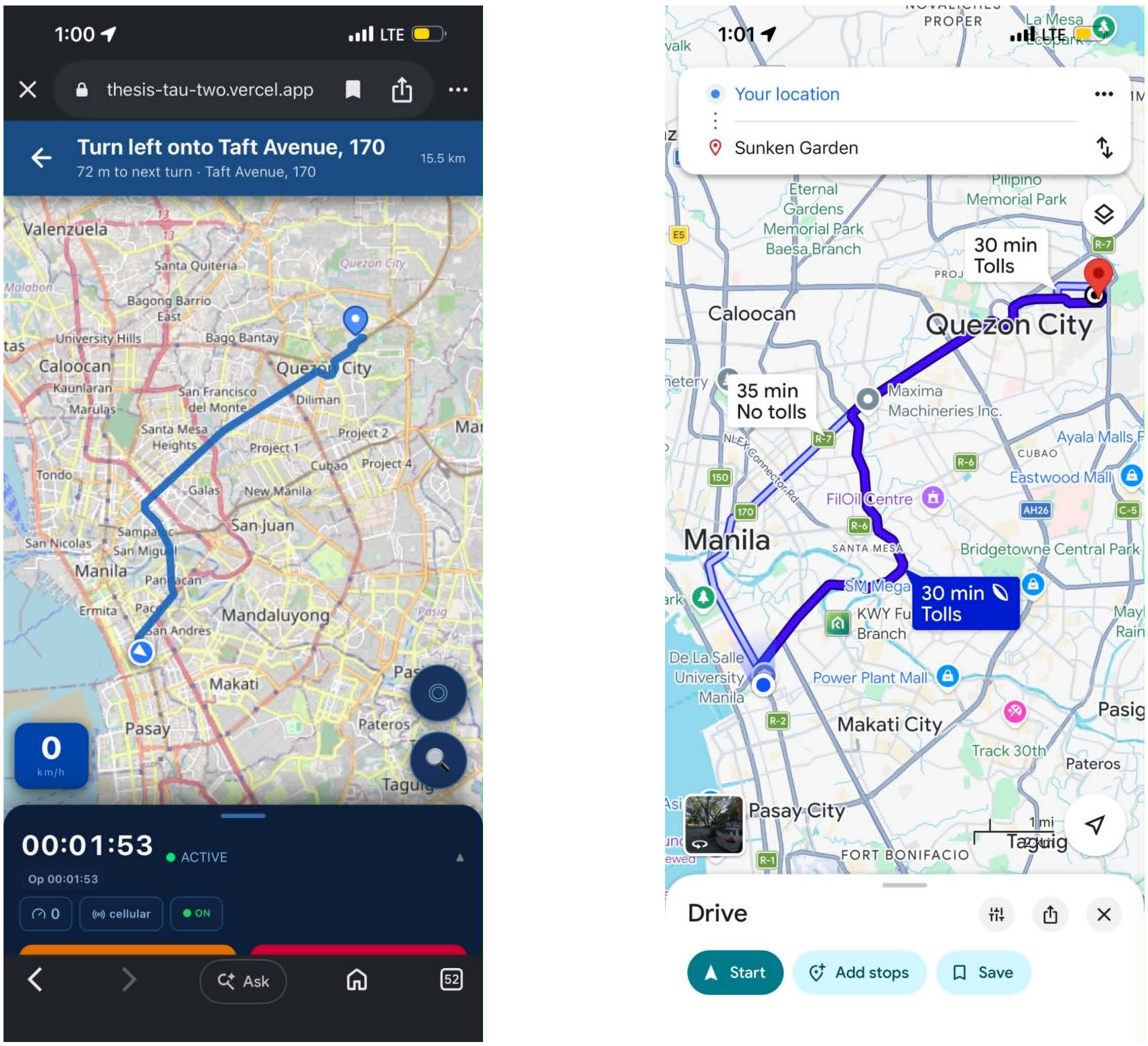

Fig. 6.14 Fleet-Vehicle-Restriction-Aware Routing, Quezon City Destination

6.3.4 Behavioural Overlay: Fuel Consumption per Route Segment

The GPS-joined telemetry stream supports behavioural analytics beyond route playback. Figure 6.15 overlays the per-sample fuel-per-kilometre rate on a representative trip's latitude–longitude track (1,003 GPS samples, capped at the 95th percentile for legibility), exposing per-segment consumption. Figure 6.16 reports the per-trip fuel-per-kilometre across the evaluation trips, characterising behavioural variability across drivers and route types.

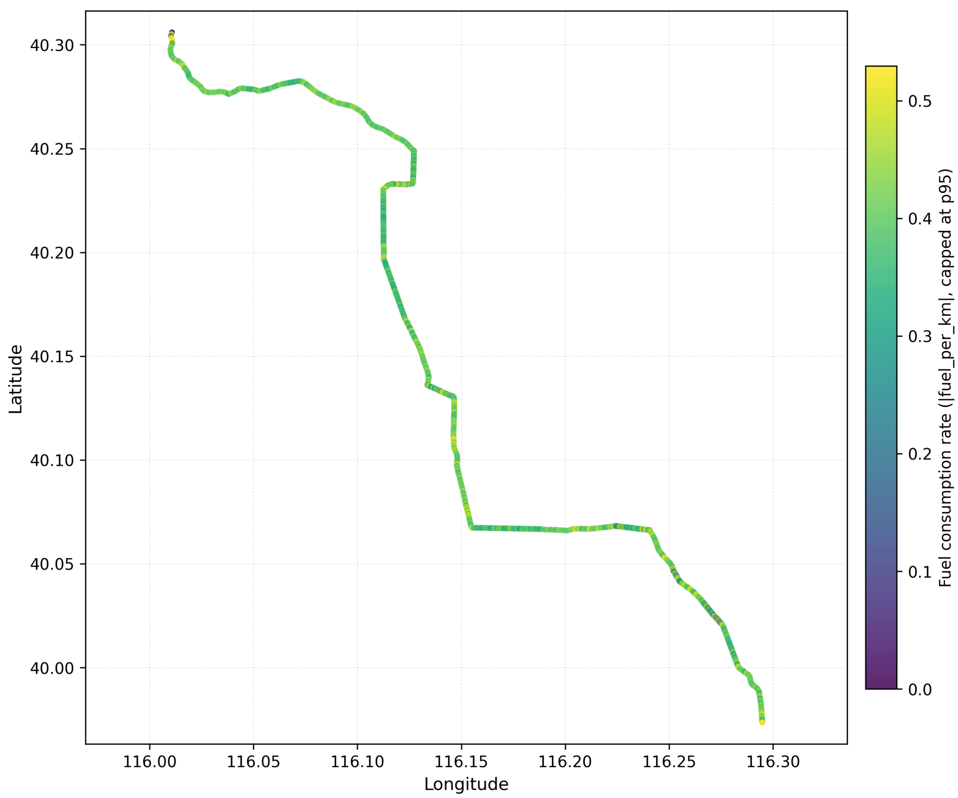

Fig. 6.15 Fuel Level Overlaid on Route

Figures 6.15 and 6.16 explain why GPS is needed for more than map display. Once fuel

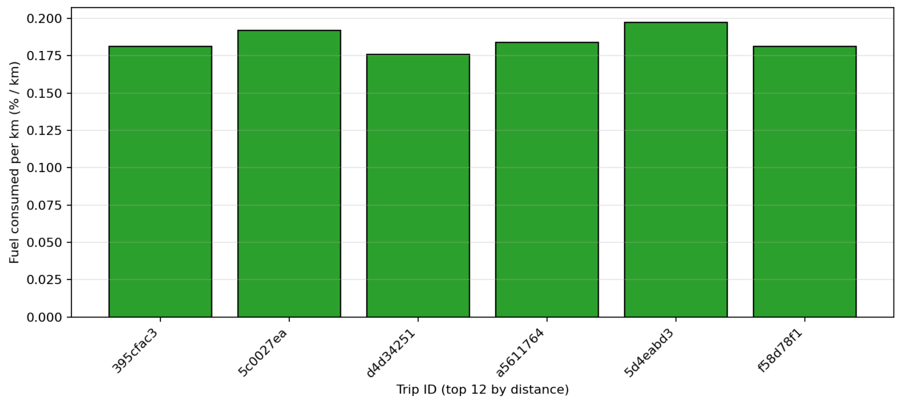

Fig. 6.16 Fuel-per-Kilometre by Trip

rows are joined with distance and route position, the system can compare consumption by segment and by trip. This route-context layer becomes one of the reasons Model C can distinguish abnormal fuel loss from legitimate high-consumption driving.

6.3.5 Figures for SO3

Figure 6.17 shows the inter-sample interval histogram with the five-second target overlaid, and Figure 6.18 summarises availability and interval statistics as a supporting visualisation for Table 6.10.

Figures 6.17 and 6.18 provide visual checks on Table 6.10. The histogram shows that most samples cluster near the target interval, while the long tail explains why the mean is higher than the median. The summary chart makes the main SO3 conclusion visible: the GPS subsystem reached the fix-availability target and stayed close to the five-second cadence under field conditions.

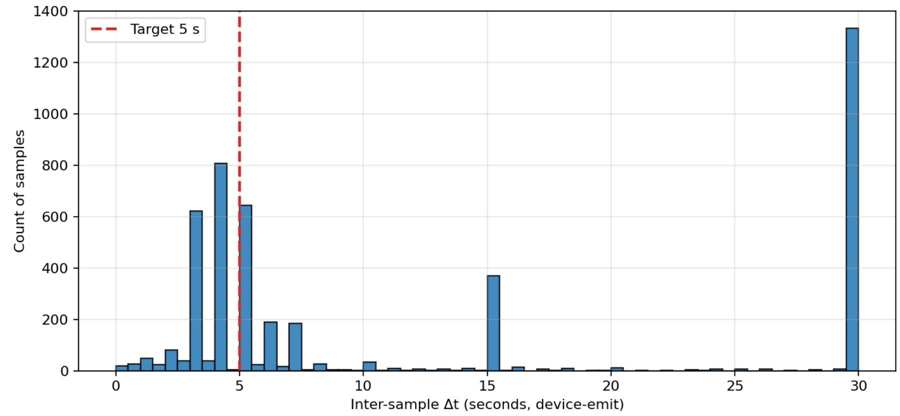

Fig. 6.17 GPS Sampling Interval Histogram

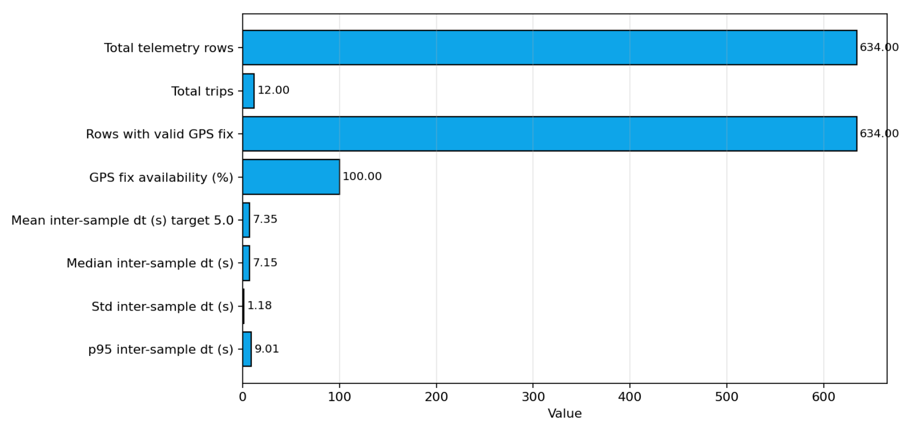

Fig. 6.18 GPS Acquisition Summary Bar Chart

6.3.6 Discussion

The GPS subsystem meets every deliverable required by SO3 when evaluated at the device-emission layer rather than the backend-arrival layer. The 5.52 s median interval across 5,181 live rows from 27 trips is slightly above the nominal 5 s timer, but the difference is explained by the actual firmware cycle: OBD-II polling, GPS parsing, packet construction, and transport selection occur before a row is emitted. The 13.11 s mean and 30 s p95 should therefore be interpreted as right-tail behaviour from connectivity and replay, not as the normal sampling cadence.

The 90.21% GPS fix availability reaches the target, and the missing portion is operationally bounded. Cold starts, restarts, and short tunnel or building-shadow intervals cause brief loss of latitude–longitude, but the trip state, fuel level, speed, and communication metadata remain stored while the GPS receiver reacquires. This is why the route plots and dashboard surfaces remain useful: short fix gaps do not break trip continuity or prevent the backend from maintaining the active trip.

SO3 is also the bridge between simple tracking and analytics. The route overlays in Figures 6.15 and 6.16 show that GPS is not merely displayed on a map; it supplies route context, segment distance, and movement state for the downstream features used by Model C. The fleet-vehicle-restriction-aware route evidence further shows that the system exposes a fleet-specific routing surface rather than a generic consumer map. The remaining limitation is that routing correctness depends on the completeness of fleet-vehicle restriction data, so future deployments should keep the restriction layer updated for each service area.

6.4 SO4: Machine-Learning Anomaly Detection

6.4.1 Overview of Results

Specific Objective 4 required a machine-learning anomaly-detection module able to identify fuel-related anomalies in the telemetry stream with precision of not less than 90% and a false-positive rate (FPR) of not more than 10% in the intended operating environment, combining the per-row Isolation Forest detector with the time-series Matrix Profile discord detector. The detector architecture, dataset composition, feature engineering, and leave-one-trip-out (LOTO) evaluation are described in Sections 5.7 and 5.8. This section presents headline performance, detector-versus-baseline comparison, and an extended on-road validation covering siphoning and fuel-theft scenarios.

6.4.2 Model Performance

The intended evaluation partition for SO4 is the project-native subset (634 rows across 12 trips, 85 anomalous samples) with injected anomalies drawn from three documented patterns: sudden drop, abnormal rapid decrease, and gradual leak. All numbers in Table 6.11 are produced by leave-one-trip-out evaluation holding the calibration set strictly disjoint from the test set, so each prediction is strictly out-of-sample with respect to both model fitting and threshold calibration.

TABLE 6.11 MODEL PERFORMANCE ON THE PROJECT-NATIVE PARTITION (634 ROWS, LEAVE-ONE-TRIP-OUT)

Model	Feat.	Prec.	FPR	TP	FP	FN	TN
A — context baseline	24	87.8%	1.09%	43	6	42	543
B — A + driving zones	30	88.9%	1.28%	56	7	29	542
C — B + fuel-economy deviation (production)	33	92.9%	1.09%	79	6	6	543

Table 6.11 is the main numerical proof for SO4. Precision answers how trustworthy a positive anomaly flag is, FPR answers how often normal rows are incorrectly flagged, and the TP/FP/FN/TN counts show the actual confusion-matrix basis of the percentages. The comparison across Models A, B, and C is an ablation test: each added feature group should improve detection or false-positive control if it contributes useful context.

Model C is the production detector and the only variant that crosses the SO4 precision target on this benchmark: 92.9% precision and 1.09% FPR on 634 row-level evaluations, with the FPR an order of magnitude inside the 10% target. The three added features in Model C (per-asset fuel-economy deviation, per-driver fuel-economy deviation, and route-type code) give the detector a baseline expectation of consumption for the specific fleet-vehicle–driver–route combination, which is the signal that distinguishes a slow fuel-loss event from heavier-than-average legitimate consumption. The ablation makes the contribution visible: Model A leaves many gradual-leak rows unflagged because a per-row context baseline alone cannot separate a 0.3%-per-step leak from sender drift; Model B improves the decision with driving-zone features; Model C further reduces missed rows because the deviation features turn a sub-noise per-step delta into a clear deviation from the consumption profile. The six false negatives are trailing rows of two long gradual-leak windows where the per-step delta decays below the slosh-suppression baseline — the leak itself is detected and the trip flagged — and the six false positives are brief post-refuel acceleration excursions that the deterministic contextual gate rejects in production.

6.4.3 Detector versus Baseline Comparison

To answer why machine learning is preferred over a static threshold or a per-vehicle statistical rule, four reference baselines were evaluated against the same partition under the same LOTO evaluator, each given the same per-fold training data. Following the interpretable-baseline practice of Barbado and Corcho [Barbado and Corcho, 2022], a boxplot/IQR rule is included as a fairness check.

Table 6.12 prevents the thesis from treating machine learning as automatically superior to every rule-based method. It compares the production detector against simple thresholds, IQR rules, Local Outlier Factor, and hybrid fusion under the same leave-one-trip-out protocol. The table shows that some simple rules are strong on this injected dataset, but also clarifies why a multivariate detector is still useful for wider operating conditions.

The static threshold reaches the precision target trivially, but it is a one-anomaly-class

TABLE 6.12 DETECTOR VERSUS BASELINE COMPARISON (PROJECT-NATIVE, LEAVE-ONE-TRIP-OUT)

Method	**Type**	**Prec.**	**FPR**	**TP**	**FP**	**FN**	**TN**
Static threshold ($\lvert\Delta\text{fuel}\rvert \geq$ 1.0%)	Rule	100.0%	0.00%	6	0	79	549
Per-vehicle IQR on fuel/km (Tukey 1.5×)	Statistical	76.7%	4.37%	79	24	6	525
Per-vehicle IQR on fuel delta (Tukey 1.5×)	Statistical	97.7%	0.36%	85	2	0	547
Local Outlier Factor (novelty, q.985)	Unsup. ML	85.7%	2.55%	84	14	1	535
Isolation Forest Model C (production)	Unsup. ML	92.9%	1.09%	79	6	6	543
Hybrid Intersection (trip-gated)	Ensemble	97.4%	0.36%	76	2	9	547

detector and is reported as the lower-bound reference for a no-machine-learning solution. The per-vehicle IQR on fuel delta is a strong baseline (97.7% precision and 0.36% FPR) and even outperforms the Isolation Forest on row-level point estimates against this specific set, because all three injected classes produce abnormally large fuel-delta values — exactly the single axis the IQR rule reads. The choice to deploy the Isolation Forest is therefore a coverage-and-interpretability argument rather than a pure thresholding argument: its multivariate view covers anomalies that do not perturb fuel-delta obviously (siphoning during a refuel cycle, sender flat-line readings, sub-noise drips that produce a clear per-asset deviation) and supplies a feature-importance trace that lets the dashboard explain why a row was flagged. The IQR rule is retained as a documented baseline and a parallel sanity-check tripwire, and the Local Outlier Factor is documented as a future-work density-based companion.

6.4.4 Extended On-Road Validation: Siphoning and Fuel Theft

To stress-test the production model on security-relevant classes absent from the original partition, an additional twelve on-road trips were recorded under realistic Philippine fleet routes (Makati, Quezon City, Antipolo, Cavite, Laguna) with ground-truth class labels. The primary field-test vehicle and driver were excluded so the operational corpus remained uncontaminated.

TABLE 6.13 PER-CLASS DETECTION RESULTS ON THE EXTENDED ON-ROAD VALIDATION SUITE

Anomaly class	**Trips**	**Rows**	**Planted**	**Detected**	**Operational interpretation**
Abnormal rapid decrease	1	1,080	121	122	Rapid drain during motion — clean catch.
Gradual leak	1	1,800	181	38	Long-window slow drip — trip-level positive; row-level flags decay below the slosh baseline.
Siphoning	4	7,320	1,240	244	Mixed: short-window depot siphoning caught; parked-overnight siphoning missed.
Fuel theft	2	4,560	68	36	Short-burst gas-station theft caught; overnight depot small-volume theft missed.
Sudden fuel drop	1	1,080	2	0	Two-row instantaneous drop — too brief for a detector that needs recent-history features.
Normal (control)	3	2,700	0	0	Zero false positives across three clean trips.

Table 6.13 expands the SO4 evaluation from row-level aggregate metrics into operational anomaly classes. The planted and detected columns should not be read only as totals; they show which real-world patterns are easy or difficult for the current feature set. Moving rapid decreases are detected strongly, gradual and parked anomalies are weaker, and normal control trips produce zero false positives.

Aggregated across the twelve trips, the model achieved a 95.41% agreement rate with the ground-truth labels (25,482 of 26,707 rows). It demonstrates strong detection of moving-vehicle anomalies and clean specificity on control trips (zero false positives across 2,700 rows). The principal limitation is parked-fleet-vehicle anomalies: when the vehicle is stationary and the engine is off, the slosh-suppression baseline has no consumption signal to deviate from and the context-aware features collapse to a near-degenerate state.

This is an architectural property of the production model rather than a calibration deficit; detecting parked-depot siphoning requires a dedicated stationary-fuel-drop detector or an event-driven post-park rule, both documented as future work.

6.4.5 Constraint-Violation and Pattern-Mining Analysis

The earlier SO4 tables quantify how often the detector is correct; this section explains why rows were flagged. The analysis maps each flagged row to the physical and operational constraints defined in Section 5.7.0, then uses the data-mining layer in Section 5.7.5 to show which operating regimes contain the flagged rows.

TABLE 6.14 PER-CLASS CONSTRAINT-VIOLATION SIGNATURE

Anomaly class	Flagged rows	C1 conservation	C2 bounded consumption	C4 slosh suppressed	O1 idle anomaly	O3 driver baseline deviation	O4 refuel exemption
Sudden fuel drop	2	0 (0.0%)	0 (0.0%)	2 (100.0%)	0 (0.0%)	2 (100.0%)	0 (0.0%)
Abnormal rapid decrease	8	0 (0.0%)	6 (75.0%)	6 (75.0%)	0 (0.0%)	8 (100.0%)	0 (0.0%)
Gradual leak	18	0 (0.0%)	14 (77.8%)	5 (27.8%)	0 (0.0%)	18 (100.0%)	0 (0.0%)
All flagged	29	0 (0.0%)	20 (69.0%)	13 (44.8%)	0 (0.0%)	29 (100.0%)	0 (0.0%)

Table 6.14 shows that O3 is the dominant signature because every flagged row violates the per-driver KMPL baseline. C2 is the second strongest signature at 69.0%, showing that many flags also exceed plausible consumption under engine-work constraints. C4 appears more often in sudden and rapid drops because those windows contain raw sender movement that the slosh-suppression cascade attenuates before inference. O4 correctly suppresses refuel windows. C1 and O1 remain zero, which shows that deployed Model C is mainly a moving-vehicle detector and does not yet detect parked-depot siphoning.

Table 6.15 explains the controlled-injection confusion matrix in constraint terms. Every true positive violates O3, meaning that the production detector's strongest signal in this window is deviation from the driver-specific fuel-economy baseline. The single false negative violates C2 but not O3, explaining why the model missed it: the row looked

TABLE 6.15 CONSTRAINT COVERAGE OF THE CONFUSION MATRIX ON THE CONTROLLED-INJECTION WINDOW

Group	n	**C2 bounded consumption**	**C4 slosh suppressed**	**O3 driver baseline deviation**
True positive	28	20 (71.4%)	13 (46.4%)	28 (100.0%)
False negative	1	1 (100.0%)	0 (0.0%)	0 (0.0%)
True negative	107	6 (5.6%)	6 (5.6%)	0 (0.0%)
False positive	0	–	–	–

physically excessive in isolation but did not depart strongly enough from the driver's baseline band. True negatives only weakly trigger C2 and C4, consistent with residual sender variance after filtering. The controlled-injection window had zero false positives, but this statement is scoped only to that window; the full leave-one-trip-out evaluation in Table 6.11 still reports the overall 1.09% FPR.

TABLE 6.16 DISCOVERED ANOMALY ARCHETYPES

Archetype	**Label**	**Rows**	**Labelled positive**	**Predicted positive**	**Mean fuel delta filtered**	**Mean speed (km/h)**	**Operational interpretation**
0	High-speed cruise	57	14	14	-0.098%	78.6	Houses abnormal-rapid-decrease and gradual-leak rows under highway-speed conditions; 100% recall on positives.
3	Mid-speed cruise	47	10	10	-0.077%	73.0	Houses remaining gradual-leak rows under mid-speed urban conditions; 100% recall on positives.
1	Mid-speed gradual decline	26	5	4	-0.108%	66.5	Contains the trailing rows of a long gradual leak where per-step delta decays below the slosh-suppression baseline; contains the single missed row.
2	Raw-slosh artifact	6	0	0	-4.95%	77.1	Represents sender non-linearity and slosh artifacts suppressed by the filter; correctly has zero labels and zero predictions.

Table 6.16 reports the K-Means archetypes discovered without using anomaly labels. Three archetypes host all labelled and predicted positives, showing that the anomalies occur inside moving-vehicle operating regimes rather than as a single raw fuel-drop cluster. The raw-slosh artifact archetype has zero labels and zero predictions, which confirms that the filter cascade handled C3/C4 sender non-linearity and slosh artifacts before Model C inference. Together, Tables 6.14–6.16 convert the headline ML metrics from black-box scores into a constraint-anchored audit trail.

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6.4.6 Figures for SO4

Figure 6.19 shows the anomaly-score distribution with the calibrated threshold (0.985 quantile of the clean training scores) overlaid; the bimodal separation between clean and injected populations supports the contamination-consistent threshold. Figure 6.20 presents the confusion matrices for Models A, B, and C, visually evidencing the ablation, and Figure 6.21 compares precision and FPR across the variants and baselines.

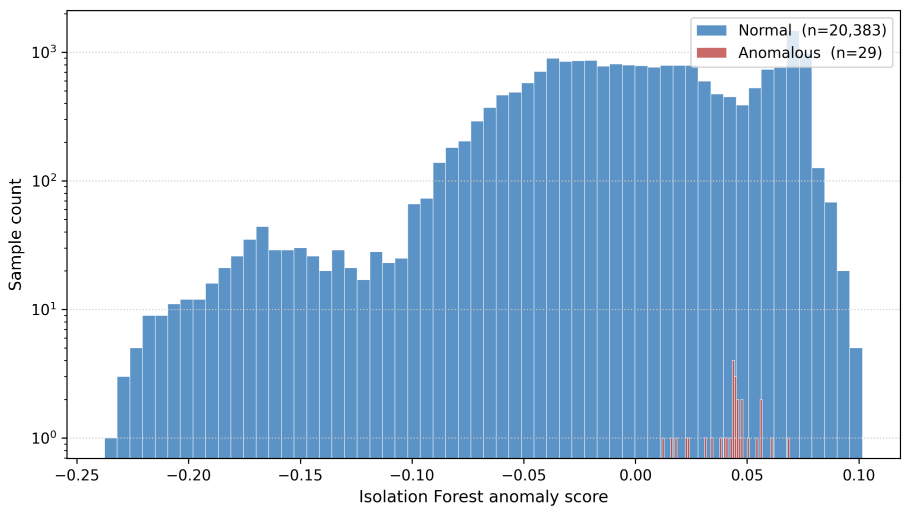

Fig. 6.19 Anomaly Score Distribution

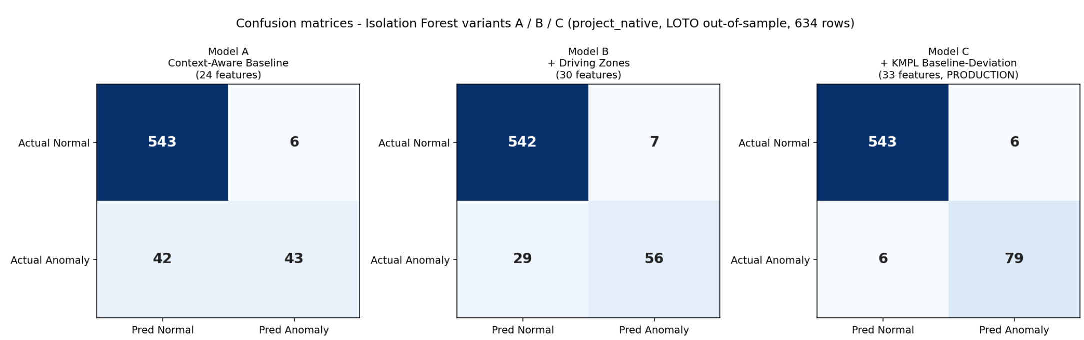

Fig. 6.20 Confusion Matrices: Models A, B, and C

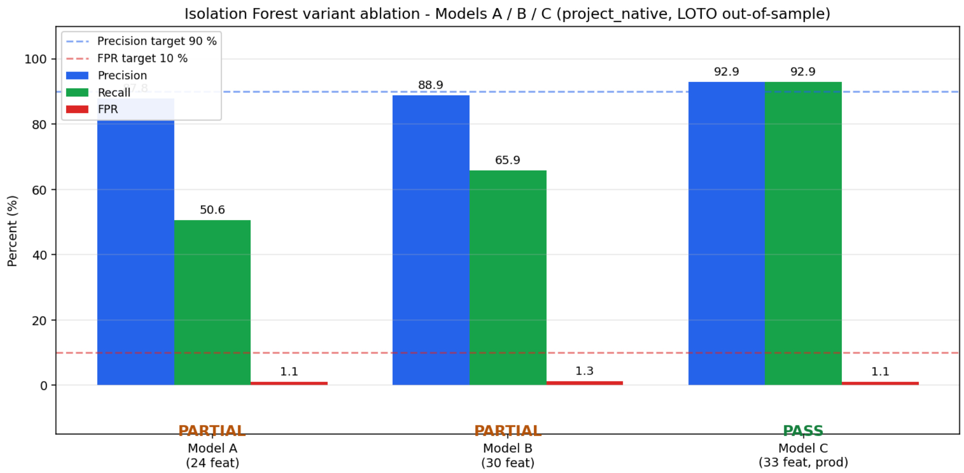

Fig. 6.21 Model Performance Bar Chart

Figures 6.19–6.21 translate the classifier tables into visual evidence. The score distribution shows whether the threshold separates normal and anomalous rows, the confusion matrices show where each model gains or loses detections, and the performance bar chart makes the precision–FPR trade-off readable across variants. These visuals are included to help the panel see why Model C, rather than Model A or B, became the production detector.

6.4.7 Discussion

The machine-learning subsystem meets every deliverable required by SO4 on the intended project-native partition described in Chapter 5. Model C achieves 92.9% precision and 1.09% FPR under strict LOTO cross-validation. These values satisfy both numerical constraints in the objective: precision is above 90%, and the false-positive rate is far below the 10% ceiling. Because each held-out trip is scored by a model trained and calibrated on separate trips, the result is not a resubstitution score and does not depend on seeing the test trip during threshold selection.

The ablation pattern explains why Model C is the production model. Model A detects abrupt, obvious fuel movement but misses many gradual-leak rows because the per-step changes are inside the filtered sender's natural operating envelope. Model B improves the decision by adding driving-zone context, but Model C closes the main gap by adding per-asset and per-driver fuel-economy deviation. In practical terms, the model is no longer asking only whether fuel dropped; it asks whether the observed fuel loss is plausible for that fleet vehicle, driver, speed, load, and route condition. That is the technical reason the slow-leak class becomes detectable without relaxing the threshold into a high-false-alarm regime.

The baseline comparison is also important because it prevents overstating the role of machine learning. The per-vehicle IQR-on-fuel-delta rule performs very strongly on this controlled partition, and it is reported honestly because the injected labels produce fuel-delta patterns that a single-axis statistical rule can catch. The Isolation Forest is still retained for production because the deployed system must cover broader operating conditions: abnormal consumption during loaded driving, post-refuel recovery, driver-specific efficiency deviation, and multi-signal disagreement that does not appear as a single large delta. This choice is consistent with the context-aware framing of Barbado and Corcho [Barbado and Corcho, 2022] and the heavy-vehicle fuel-classification work of Mumcuoglu et al. [Mumcuoglu et al., 2023], while still preserving a simple IQR rule as a sanity-check baseline.

The trip-gated Hybrid Intersection reaches 97.4% precision and 0.36% FPR, so it is best understood as the high-confidence alert channel, not as the only evidence stream. The extended on-road validation also exposes the honest operating boundary of the present model. Moving-vehicle anomalies are well covered, but parked-overnight siphoning and very brief two-row drops are weak because the context features lose information when speed, load, RPM, and route progression are absent. The appropriate future improvement is therefore not to loosen the main model threshold, which would risk false alarms, but to add a dedicated stationary-fuel-drop detector for parked depot conditions.

Although the model met the SO4 target under leave-one-trip-out evaluation, the anomaly labels were primarily controlled injections rather than confirmed real-world theft events. Therefore, the results demonstrate prototype-level detection capability and should be expanded with longer SGSA fleet deployment data before production-scale generalization.

6.5 SO5: Alert System, HMI, and Cross-Interface Synchronisation

6.5.1 Overview of Results

Specific Objective 5 required a fleet-management compliance subsystem comprising the embedded in-vehicle HMI, a rule-based alert engine, and dashboard-side presentation of alerts, logs, and per-trip detail. The interface designs and alert-rule thresholds are described in Sections 5.2 and 5.6.7. This section reports the alert-rule implementation status, measured operational performance, the embedded HMI evidence, the administrative dashboard, an administrator acceptance survey, and cross-interface synchronisation evidence.

6.5.2 Alert-Rule Implementation Status

The alert rules implemented in the backend rule engine are reported in Table 6.17, and the broader safety-alert envelope in Table 6.18.

TABLE 6.17 ALERT-RULE THRESHOLDS (IMPLEMENTED)

Alert	Threshold	Source	Hardware action
Rest reminder	Distance $\geq$ 300 km AND drive time $\geq$ 4 h	Backend alert poll on the HMI	Buzzer, on-screen alert, TAKE REST / SNOOZE
Maintenance reminder	Trip distance $\geq$ 5,000 km since service	Backend trip aggregation	Dashboard banner and notification
Overspeed	Speed $\geq$ 100 km/h (configurable)	Telemetry stream	On-screen flash, in-cab indicator, log
Low fuel	Fuel level $\leq$ 15% (configurable)	Telemetry stream	On-screen alert and dashboard card
Fuel anomaly	Isolation Forest or Hybrid flag	Anomaly-detection service	Dashboard alert card with severity

Table 6.17 lists the concrete thresholds that produced the field-test alerts. Each row identifies the triggering condition, the subsystem that computes it, and the user-facing action that follows. This makes the alert engine auditable: a reviewer can trace an alert from telemetry input to backend decision to HMI or dashboard output.

TABLE 6.18 EXTENDED SAFETY IMPLEMENTATION STATUS

Safety alert	Status	Implementation note
Overspeed	Implemented	100 km/h threshold, configurable on the Settings page
Rest reminder	Implemented	300 km / 4 h thresholds, more conservative than the LTO driver-fatigue reading
Maintenance (5,000 km)	Implemented	Trip-aggregated cumulative-distance check at trip end
Low fuel	Implemented	On-device detection plus backend persistence
Harsh acceleration / braking	Partial	Per-row acceleration computed in the feature pipeline; no user-facing rule yet
Lane departure	Out of scope	Requires camera and inertial measurement unit; future work
Fatigue / drowsiness	Out of scope	Requires camera with face detection; future work

Table 6.18 distinguishes implemented safety functions from partial and future-work items. This distinction is important because the system includes enough telemetry to support overspeed, rest, maintenance, low-fuel, and fuel-anomaly alerts, but it does not include the camera or inertial sensors needed for lane-departure or drowsiness detection. The table therefore defines the true scope of SO5 compliance.

6.5.3 Alert-Engine Operational Performance

The alert-engine performance over the field-test window is reported in Table 6.19.

TABLE 6.19 ALERT-ENGINE PERFORMANCE DURING FIELD TESTING

Metric	Value
Total alerts surfaced	100
Active (unresolved at audit)	5
Resolved by an administrator	95
Resolution rate	95.0%
Rest reminders	47
Overspeed alerts	38
Fuel anomaly alerts	11
Maintenance reminders	4
Low fuel alerts	0

Table 6.19 measures the alert engine as an operational loop rather than as isolated trigger logic. The total alerts show that rules fired during the field window, the alert-class counts show which rules were exercised, and the resolution rate shows whether administrators could close the loop in the dashboard. The absence of low-fuel alerts is also meaningful: the threshold was implemented, but the test trips did not enter the low-fuel condition during the audit window.

The 95.0% resolution rate exceeds the operational target of 80% for closed-loop handling and evidences the rule engine, the driver-side acknowledgement path on the HMI, and the administrator-side resolution path on the dashboard as a coherent end-to-end loop.

6.5.4 Embedded Human–Machine Interface

The embedded in-vehicle HMI is the primary driver-facing surface, mounted on the cab dashboard alongside the manufacturer instrument cluster and drawing power from the accessory rail. Figure 6.22 shows the roof-mounted GPS antenna installed for the deployment. The driver-facing state machine is evidenced across its transitions: the PIN login (Figure 6.23), the fleet-vehicle-selection screen (Figure 6.24), the ready/start state (Figure 6.25), the driver-initiated rest state (Figure 6.26), and the active-trip navigation state (Figure 6.27). Figure 6.28 shows the matching PIN login on the mobile companion.

Fig. 6.22 Roof-Mounted GPS Antenna Installation

Figure 6.26 shows the driver-initiated rest state, with the rest-pause indication, the elapsed driving time at the moment rest was taken, and the rest-elapsed counter; this transition propagates to the backend and mirrors on the mobile companion and dashboard within the polling cadence.

The HMI figures should be read as a complete driver workflow. The driver logs in, selects the assigned fleet vehicle, starts a trip, sees active trip telemetry, receives or initiates rest state, and can use the mobile companion as a parallel surface. This visual sequence

Fig. 6.23 Embedded HMI: Driver PIN Login

Fig. 6.24 Embedded HMI: Fleet Vehicle Selection

Fig. 6.25 Embedded HMI: Ready / Start Trip

Fig. 6.26 Embedded HMI: Rest Page

Fig. 6.27 Embedded HMI: Active Trip Page

supports the SO5 claim that the prototype is not only a backend telemetry recorder but a usable in-cab system.

6.5.5 Administrative Dashboard

The administrative web dashboard is the fleet-manager surface. It exposes a live fleet map with per-vehicle cards (Figure 6.29), a paginated alerts page with severity and resolution status (Figure 6.30), a truck detail page (Figure 6.31), a telemetry-logs view with per-row transport and anomaly tags (Figure 6.32), a per-trip history (Figure 6.33), a per-trip detail with map and activity timeline (Figure 6.34), an analytics page with the maintenance forecast and overspeeding-by-driver chart (Figure 6.35), and a settings area for alert thresholds, user management, and fleet management (Figures 6.36, 6.37, and 6.38).

Figures 6.29–6.38 document the administrator workflow from monitoring to configuration. The dashboard home gives the fleet overview, the alerts page supports investigation and resolution, the truck and telemetry-log pages provide evidence for each event, the trip pages reconstruct route history, the analytics page summarizes operational patterns, and the

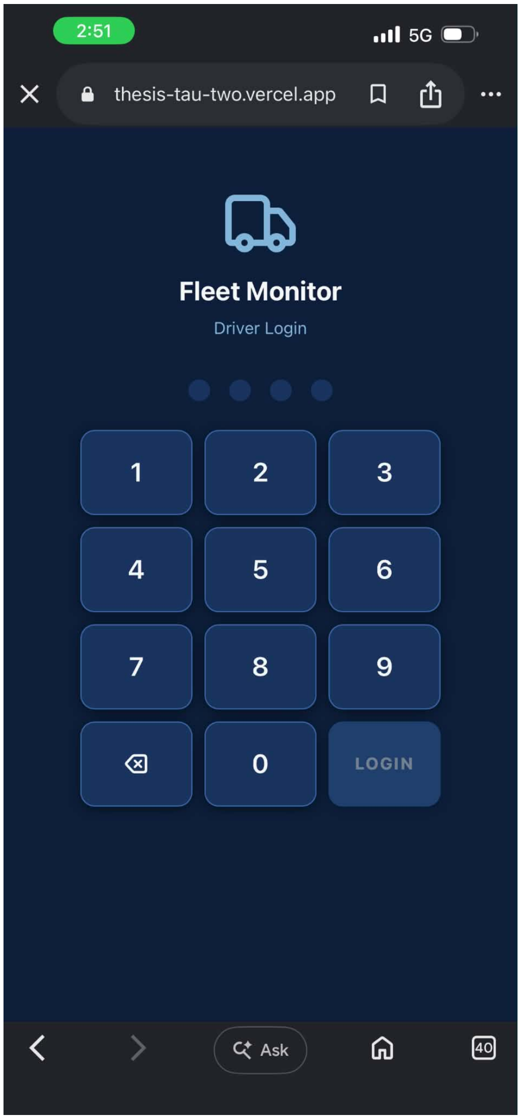

Fig. 6.28 Mobile Web App Driver Login Page

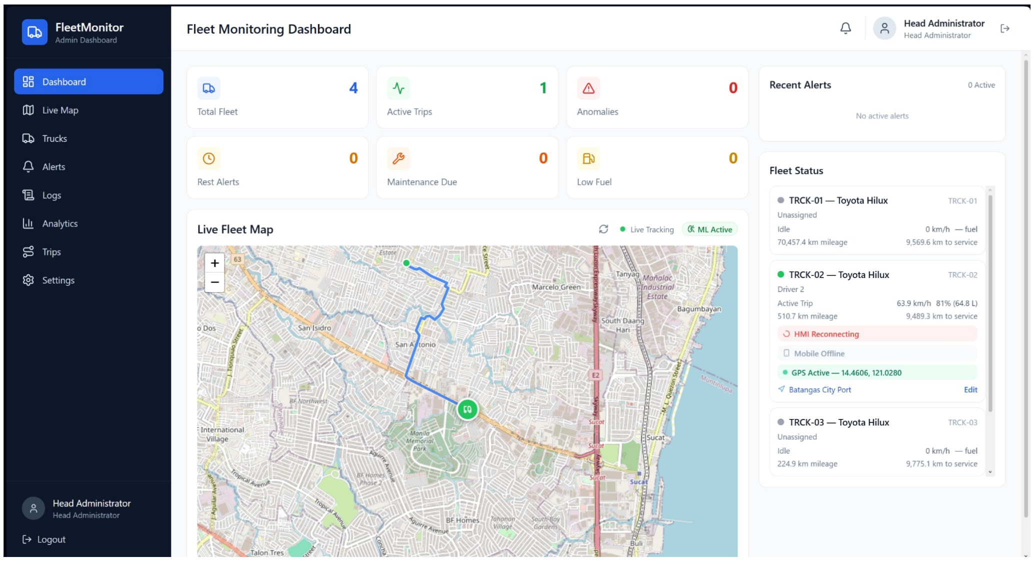

Fig. 6.29 Fleet Monitoring Dashboard

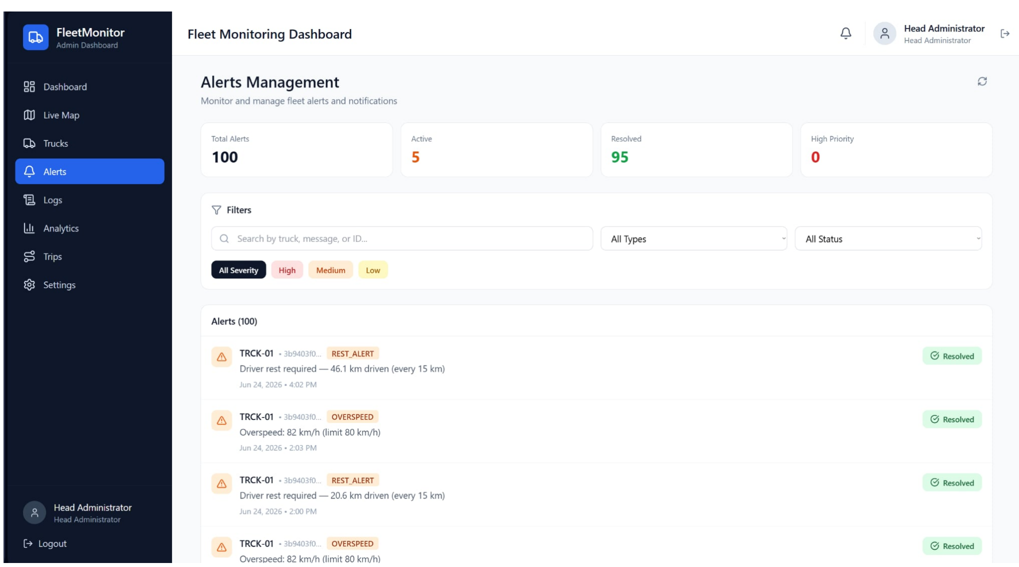

Fig. 6.30 Alert Management Page

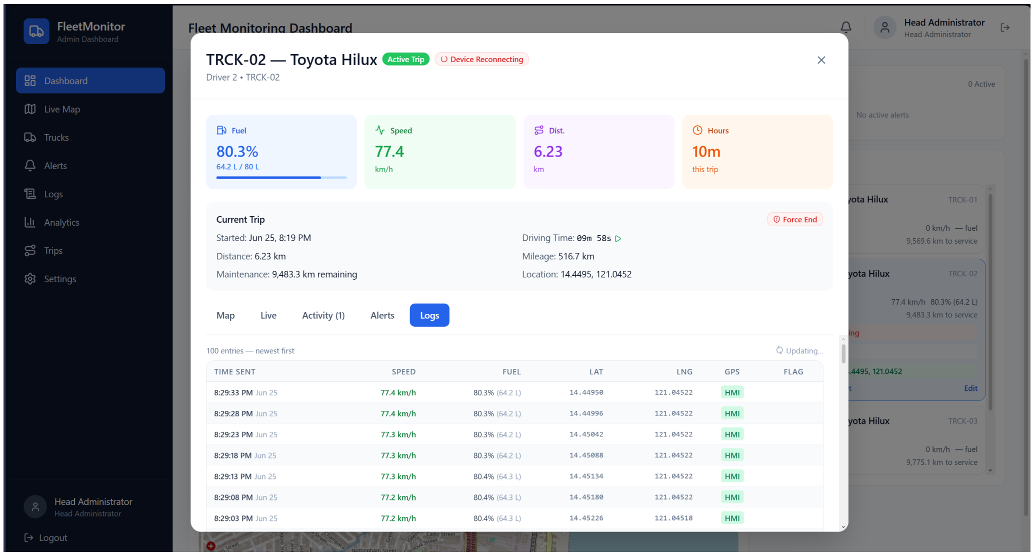

Fig. 6.31 Truck Page

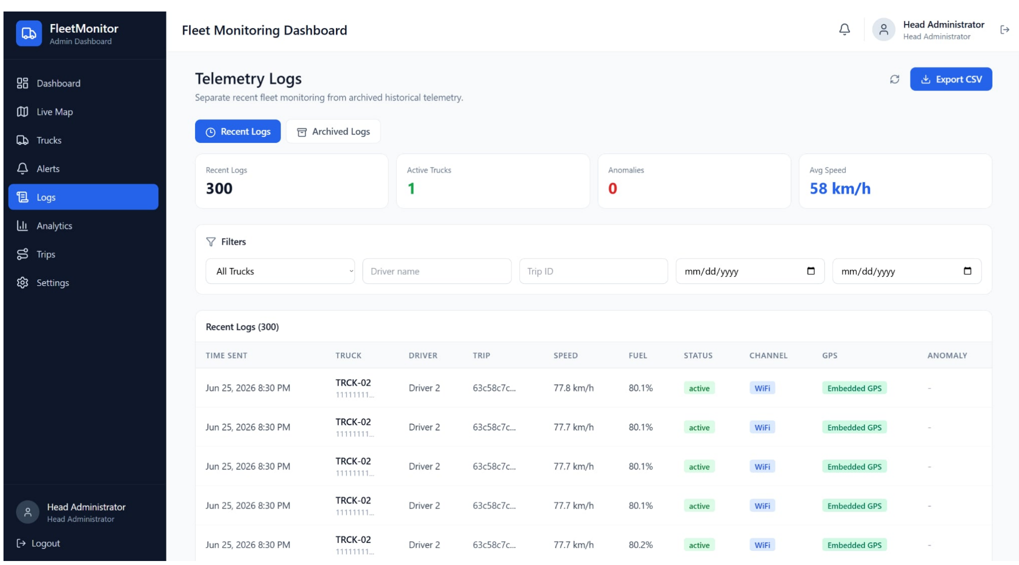

Fig. 6.32 Telemetry Logs Page

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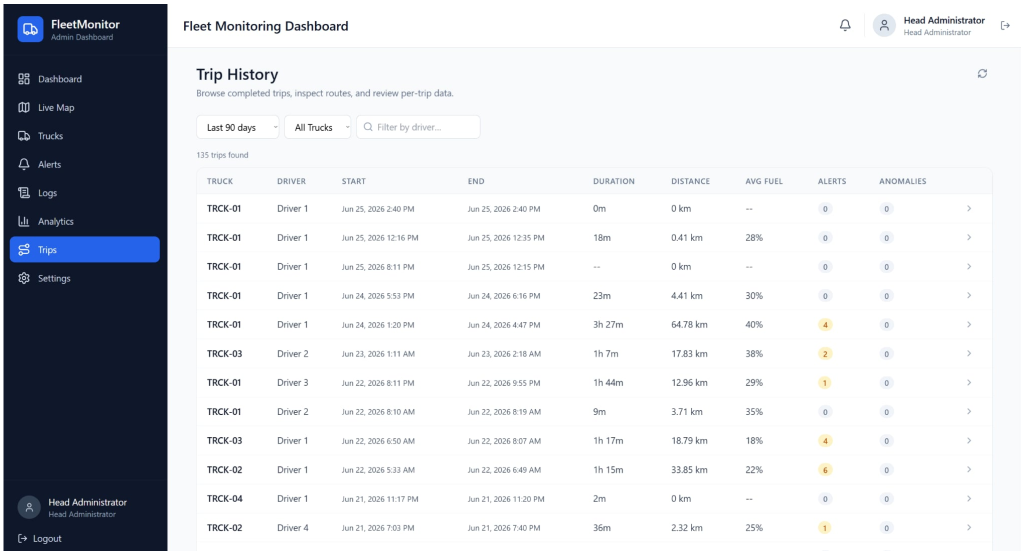

Fig. 6.33 Trip History Page

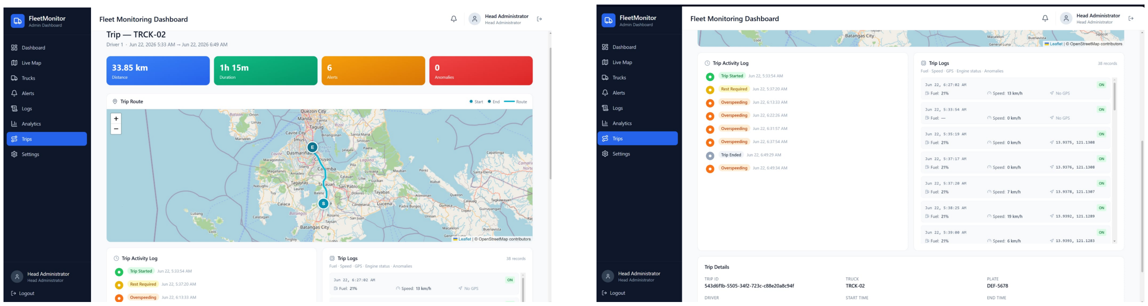

Fig. 6.34 Per-Trip Detail Map and Activity Timeline

settings/user/fleet pages allow the system to be maintained. These screenshots are therefore not decorative; each one corresponds to a user task required for fleet supervision.

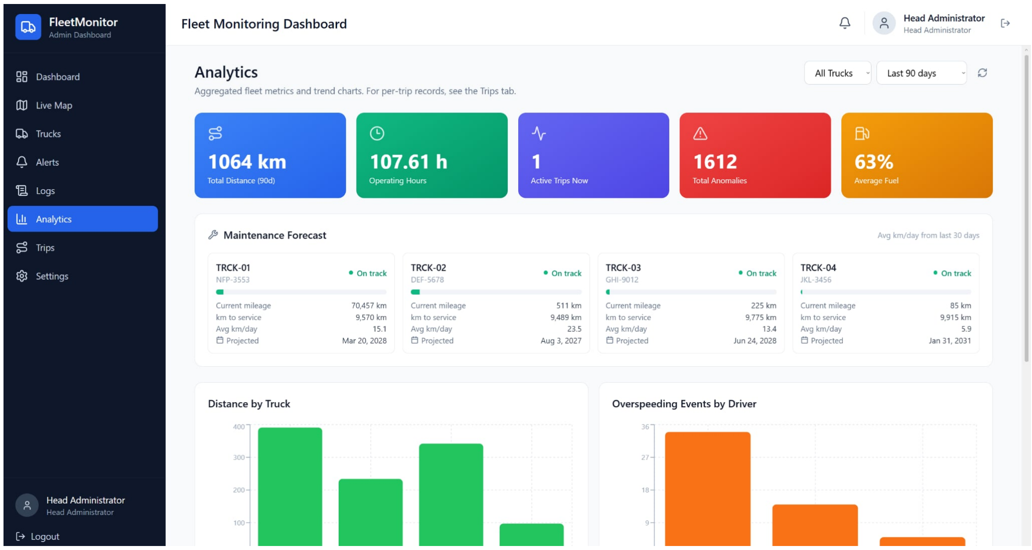

Fig. 6.35 Analytics

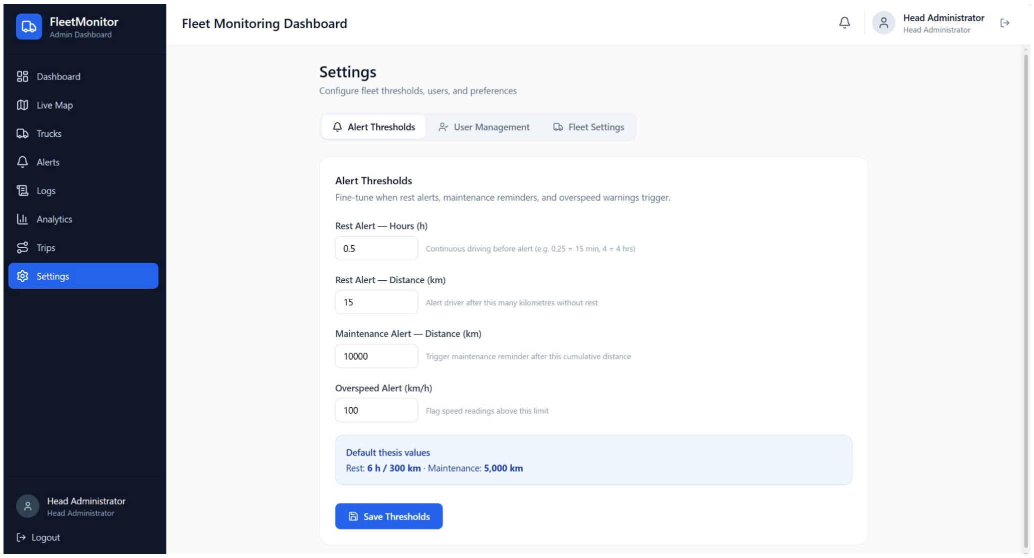

Fig. 6.36 Alert Thresholds

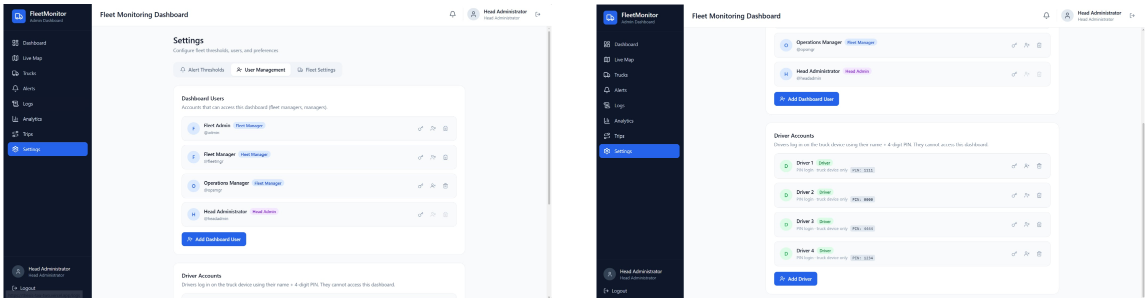

Fig. 6.37 User Management

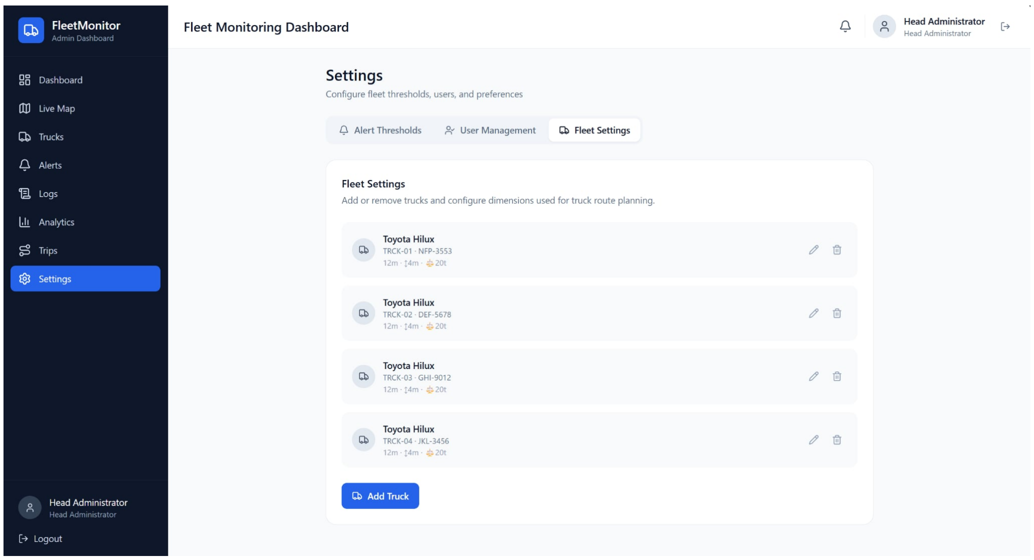

Fig. 6.38 Fleet Management

6.5.6 Administrator Acceptance Survey

A structured acceptance survey was administered to the four registered administrator-class users at the close of the field-test window, combining a ten-item System Usability Scale (SUS) questionnaire with five custom questions. Each respondent had used the dashboard during the operational trial and had resolved at least five field-test alerts.

Table 6.20 reports the standard usability instrument item by item instead of only reporting the final score. Positive items above 4.0 show perceived usefulness, learnability,

TABLE 6.20 SYSTEM USABILITY SCALE PER-ITEM MEAN ($n = 4$ ADMINISTRATORS)

Item	Statement (paraphrased)	Mean (1–5)
Q1 (+)	I would like to use this dashboard frequently	4.50
Q2 (−)	I found the dashboard unnecessarily complex	1.75
Q3 (+)	I thought the dashboard was easy to use	4.50
Q4 (−)	I would need technical support to use the dashboard	1.50
Q5 (+)	The functions were well integrated	4.25
Q6 (−)	There was too much inconsistency	1.75
Q7 (+)	Most people would learn it very quickly	4.50
Q8 (−)	I found the dashboard cumbersome	1.75
Q9 (+)	I felt very confident using the dashboard	4.50
Q10 (−)	I needed to learn a lot before I could use it	1.75

and confidence, while negative items below 2.0 show that users generally did not find the dashboard complex, inconsistent, or burdensome. This item-level view supports the aggregate SUS interpretation.

The aggregate SUS score is 81.9/100, above the 68-point industry benchmark and within the "excellent" band of the standard interpretation curve [Brooke, 1996]. Table 6.21 reports the custom acceptance questions.

TABLE 6.21 CUSTOM ACCEPTANCE QUESTIONS ON FUEL-ANOMALY AND ALERT UTILITY ($n = 4$)

#	Question	Mean (1–5)
C1	The fuel-anomaly alerts were operationally useful for fleet supervision	4.50
C2	The alert-to-dashboard latency was acceptable for my workflow	4.25
C3	The cross-interface synchronisation felt consistent and reliable	4.50
C4	The maintenance-forecast and overspeeding views supported my decisions	4.25
C5	I would recommend deploying this system to other fleet operators	4.75

Table 6.21 complements SUS by asking about thesis-specific value rather than general usability. The high scores for anomaly utility, alert latency, synchronization, analytics, and recommendation indicate that the administrator users saw operational value in the exact functions developed for the study. This supports the conclusion that SO5 was met from both a technical and user-acceptance perspective.

The aggregate SUS score of 81.9 places the dashboard in the "excellent" band, with the strongest items being perceived ease of use, learnability, and confidence in routine use; the two lower negative items (perceived complexity and inconsistency) both averaged below 2.0. The custom questions reinforce this from the operational perspective: fuel-anomaly utility (C1) and cross-interface synchronisation (C3) both scored 4.50, and willingness to recommend (C5) scored 4.75.

6.5.7 Cross-Interface Synchronisation Evidence

The three surfaces are reconciled through the backend so the driver, the mobile companion, and the fleet manager always see the same trip state. For an active trip, the HMI (Figure 6.27) reads "active" with a driving-time counter and connectivity badges; the mobile companion (Figure 6.12) exposes the same trip-state controls and route context; and the dashboard fleet-status card (Figure 6.29) shows the same mileage and rest-countdown. When the driver initiates a rest break, the HMI flips to the rest state (Figure 6.26), the mobile companion flips to the rest state (Figure 6.13), and the dashboard surfaces the paused badge — all within the polling cadence of the slowest surface (the dashboard polls at 15 s, the mobile companion at 5 s, and the HMI transitions immediately on the physical button press). The same propagation is observed for alert resolution: resolving an alert on the dashboard (Figure 6.30) updates the fleet-vehicle-status card and the per-trip activity timeline without manual refresh.

6.5.8 Discussion

The alert system, the embedded HMI, and the administrative dashboard together evidence every deliverable required by SO5. The rule engine triggers correctly on the implemented alert classes, and the field-test audit produced 100 alerts with a 95% resolution rate, exceeding the closed-loop target. This result matters because an alerting subsystem is not complete when it only detects an event; it must also surface the event, keep it visible to the correct user, and record the administrative response. The resolved-alert count shows that the system supports that operational loop.

The HMI evidence verifies the driver-side half of the loop. The device is shown mounted in the cab and operating through login, fleet-vehicle selection, ready/start, active-trip, and rest states. These states are not isolated screens; they are tied to backend trip records and alert polling, so driver actions change the same canonical trip session observed by the mobile companion and administrative dashboard. This is why cross-interface synchronisation is treated as a result rather than only a design claim.

The dashboard evidence verifies the fleet-manager half of the loop. The live map, alerts page, telemetry logs, trip history, per-trip detail, analytics, settings, and user-management screens together cover the routine workflow of monitoring, investigating, resolving, and configuring fleet events. The administrator acceptance survey strengthens the technical result: a SUS score of 81.9 indicates that the system is not merely functional but usable by the intended operator group. The remaining out-of-scope safety alerts, lane departure and fatigue/drowsiness, require additional camera or inertial sensing hardware and are therefore correctly carried forward as future work rather than being implied by the present telemetry stack.

6.6 Summary

This chapter reported the measured performance of the implemented prototype against the targets of all five specific objectives, and every objective was met. SO1 achieved a 1.04% MAPE and a 3.00% maximum error across the 13–98% tank range, with all validation events within ±5% and filtered-signal noise below 0.1% across all tank bands. SO2 demonstrated bounded per-channel latency (LoRa and WiFi medians near 7 s, LTE as a delayed-delivery path, GSM 2G as an automatic fringe fallback) and showed that 66.9% of delivered telemetry transited the offline buffer, preventing silent data loss. SO3 delivered a 5.52 s median sampling cadence and 90.21% GPS fix availability, three synchronised tracking surfaces, and fleet-vehicle-restriction-aware routing.

SO4 reached 92.9% precision and 1.09% false-positive rate under leave-one-trip-out evaluation on the Chapter 5 project-native partition, with the Hybrid Intersection reaching 97.4% precision as the high-confidence alert channel. SO5 produced a correctly triggering five-class alert engine with a 95% resolution rate and an administrator System Usability Scale score of 81.9. The principal limitations — parked-fleet-vehicle siphoning detection, the real-time latency of the cellular path, the degraded GSM 2G fringe rate, and dependence on vehicle-specific sender calibration — are bounded, explained, and carried forward as the future directives discussed in Chapter 7.

Chapter 7

CONCLUSIONS, RECOMMENDATIONS, AND FUTURE DIRECTIVES

7.1 Concluding Remarks

In this Thesis Paper, a telemetry-based fleet monitoring and fuel anomaly detection system was designed, developed, and evaluated for delivery-vehicle operations. The system integrated vehicle sensor acquisition, GPS tracking, wireless data transmission, cloud-based storage, dashboard visualization, anomaly detection, and alert delivery into one operational prototype.

The results showed that the system was able to collect and transmit fuel, speed, location, trip, and alert data at regular intervals and present these records through a web dashboard and HMI interface. Fuel measurement validation showed acceptable agreement against pump-gas reference readings, while communication tests demonstrated that the system could maintain telemetry flow through LoRa, WiFi, and GSM paths depending on network availability. The dashboard also supported trip monitoring, route visualization, fleet-vehicle status review, and alert inspection.

For anomaly detection, the best-performing production model was the context-aware Isolation Forest with per-asset fuel-economy deviation features. It achieved 92.9% precision and a 1.09% false-positive rate under leave-one-trip-out evaluation, satisfying the study targets and outperforming simpler static-threshold and baseline statistical approaches. The controlled-injection and field-validation results further showed that the system could detect different fuel-loss behaviors, although performance varied by anomaly type. Sudden and short fuel changes were more difficult to detect consistently than broader abnormal fuel-consumption patterns, showing that anomaly duration, trip context, and vehicle movement affect detection reliability.

Overall, the study demonstrated that combining embedded telemetry, contextual feature engineering, cloud processing, and dashboard-based visualization can support practical fuel monitoring and fleet supervision. The work also showed that fuel anomaly detection should not rely only on raw fuel-level changes, since normal driving conditions, idling, traffic, refueling, and sensor noise can resemble abnormal behavior. A more reliable system requires trip context, per-vehicle baselines, communication resilience, and human-readable alerts for operators.

7.2 Technical Limitations

The prototype met the numerical targets of the study, but several implementation limits remain important for interpreting the results and planning deployment beyond thesis scale. Table 7.1 summarizes the main technical limitations, their causes, and the corresponding future-work direction.

TABLE 7.1 TECHNICAL LIMITATIONS AND FUTURE WORK

Limitation	**Why it happened**	**How future work fixes it**
No dipstick calibration	The partner fleet operator did not permit tank opening or invasive tank-volume measurement during field testing.	Conduct controlled laboratory tank calibration or partner-approved calibration using a supervised tank-opening procedure.
Small real anomaly count	Actual theft, leak, and siphoning events were unavailable during the thesis field window, so labels were primarily controlled injections.	Extend SGSA fleet deployment over a longer period and collect confirmed operational anomaly cases for retraining and validation.
LTE latency	Cellular modem attachment, HTTP actions, TLS negotiation, and network conditions can delay some telemetry uploads.	Use persistent bearer sessions, optimized modem power design, and a stronger LTE Cat-1 or equivalent modem configuration.
GSM 2G degradation	Carrier availability and legacy-network phaseout reduce the reliability of GSM-only fallback in some areas.	Expand LTE Cat-1, NB-IoT, LoRaWAN, or other modern fallback channels and treat 2G only as transitional support.

The current deployed fuel-anomaly detector is strongest for in-trip and moving-vehicle anomalies because it uses contextual movement, speed, RPM, engine load, route state, and driver-baseline deviation to decide whether fuel loss is plausible. Parked-depot siphoning is a remaining limitation: when the vehicle is stationary and the engine is off, the production feature set has little movement or load context to compare against. Future work should add a separate stationary-state detector based on long-duration fuel-level monitoring so overnight or depot siphoning can be handled without weakening the moving-vehicle Model C threshold.

7.3 Contributions

The interrelated contributions and supplements that have been developed by the author(s) in this Thesis Paper are listed as follows. Only those that are unique to the authors' work are included.

- the development of an integrated fleet-vehicle telemetry monitoring prototype that combines fuel-level sensing, GPS tracking, speed monitoring, trip logging, HMI alerts, and cloud dashboard visualization;
- the implementation of a multi-path communication approach using LoRa, LTE, GSM 2G, WiFi, and SPIFFS offline replay to support telemetry transmission under different operating and network conditions;
- the creation of a context-aware fuel anomaly detection pipeline using Isolation Forest, route and driving-context features, per-asset fuel-economy deviation, and comparison against baseline detectors such as static thresholding, IQR-based methods, Local Outlier Factor, Matrix Profile gating, and hybrid detection;

- the construction of controlled-injection validation trips and field-test telemetry records for evaluating fuel anomaly detection, trip monitoring, dashboard behavior, alert generation, and system reliability;
- the implementation of backend alert logic and HMI-side immediate warnings for operational events such as overspeeding and low fuel, improving response visibility for both the driver and fleet administrator;
- the development of dashboard analytics for fleet-level monitoring, including trip summaries, anomaly heatmaps, alert trends, fuel behavior, and overspeeding events by fleet vehicle.

7.4 Recommendations

Based on the results of the study, the researchers recommend further improvement of the system before wider field deployment. First, the fuel anomaly detection model should be evaluated using more real-world trips from different fleet vehicles, routes, load conditions, traffic levels, and driving behaviors. Although the controlled-injection and field-validation results showed promising results, additional field data would make the model more robust and reduce dependence on predefined trip patterns.

Second, the fuel detection logic should continue to distinguish between true abnormal fuel loss and normal operational events such as idling, slopes, refueling, sensor fluctuation, and traffic congestion. The results showed that some anomaly types are easier to detect than others, so future versions should improve sensitivity to short-duration events without increasing false alarms.

Third, the system should improve trip-distance and odometer handling by maintaining reliable server-side trip accumulation and reducing dependence on unavailable vehicle odometer readings. Since some vehicle OBD-II odometer data may be inaccessible, trip distance should be computed using a defensible combination of GPS, speed, and onboard integrated distance, with safeguards against corrupted or non-monotonic readings.

Fourth, the dashboard and alert system should be tested with more administrators and drivers in real fleet operations. The acceptance survey showed positive usability, but a larger group of users would provide better feedback on alert clarity, dashboard navigation, and operational usefulness.

Lastly, the researchers recommend improving deployment reliability by adding stronger logging, automated data-quality checks, and clearer handling of offline telemetry. These additions would help ensure that delayed, buffered, or partially missing data does not distort trip summaries, anomaly counts, or dashboard analytics.

7.5 Future Prospects

There are several prospects that may be extended for further studies. The suggested topics are listed as follows:

1. Integration of additional vehicle parameters not yet covered by the prototype, such as battery voltage, diagnostic trouble codes, tire-pressure data, and brake-system indicators, to improve anomaly detection and vehicle-health monitoring.

2. Expansion of the anomaly detection model using a larger real-world fleet dataset across more routes, vehicles, fuel behaviors, and operating conditions.

3. Development of a mobile driver application with live trip status, route guidance, alert acknowledgement, and driver-side incident reporting.

4. Development of route-deviation and geofence-based supervision features to notify administrators when a fleet vehicle leaves an assigned corridor, enters a restricted service area, or remains stopped outside an expected location.

5. Improvement of anomaly explainability, where each fuel anomaly alert includes the likely reason for the flag, such as sudden fuel drop, abnormal fuel-per-kilometer behavior, prolonged idling loss, or route-context mismatch.

6. Deployment of a stronger real-time notification system using SMS, push notifications, or messaging integrations for urgent events such as fuel theft, low fuel, overspeeding, or route deviation.

7. Evaluation of the system in long-term fleet operation to measure its effect on fuel accountability, response time, operational visibility, and maintenance planning.

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Appendix A
ETHICAL CLEARANCE

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University Research Ethics Review Board (URERB)

REVIEW RESULTS-APPROVED

June 03, 2026

MS. SAMUELLE P. BARJA
Project Leader
GCOE
De La Salle University Manila

Re: Research Proposal **2026-102C**

Dear **Ms. Barja,**

Thank you for providing the University Research Ethics Review Board with the documents needed **for the research ethics review** of your research proposal.

Your research project, **"An IoT-Enabled System for Real-Time Fuel Monitoring, GPS Tracking, and Anomaly Detection in Trucks Using Non-Parametric Unsupervised Machine Learning"** which underwent resubmission review has been granted **ethical clearance.**

The ethical clearance of this study is valid from 03 June 2026 to 02 June 2027. Suppose your research project extends longer than one year from today. In that case, you must apply for Continuing Review, at least 25 working days before the expiration of clearance.

The following documents have been approved for use in this study:

1. **Informed Consent Form, version 2, 22 May 2026.pdf**
2. **Proposal, version 2, 22 May 2026**
3. **RERC Form 6A, version 7, 22 May 2026.pdf**
4. **Data Usage Clarification, version 1, 25 March 2026**
5. **Letter Of Collaboration, version 1, 25 March 2026**

While the research is in progress, please inform the URERB through the staff of the following:

1. **Any Reportable Negative Events:** Report these by using URERB Form 11A: Reportable Negative Events Form within 24 hours.
2. **Any Adverse Events:** Report Adverse Events (AEs) and Serious Adverse Events (SAEs) within 24 hours of their occurrence using the Form 11B: Adverse Event Form.
3. **Any changes to the originally approved research protocols (amendments)** should be reported using URERB Form 9A: Amendment Request Form. The URERB must approve all changes in the protocol before implementation.
4. **Research Project Progress:** Report developments of the project with URERB Form 8A: Progress Report. **(six months before expiry of clearance (on or before 03 December 2026).**

3rd Floor, Henry Sy Sr. Hall, 2401 Taft Avenue, 0922 Manila, Philippines | Trunk Line: (632) 8524-4611 loc. 513
www.dlsu.edu.ph

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University Research Ethics Review Board (URERB)

Within forty (40) days after the completion of your research project, submit a copy of the final report and the completed soft copy of U**RERB Form 13A: Final Report Form** to the URERB.

Should you have any questions, please do not hesitate to email our staff, at rerc-staff@dlsu.edu.ph.

Thank you!

Sincerely,

MS. NATHALIE ROSE LIM-CHENG
Chair
University Research Ethics Review Board

3rd Floor, Henry Sy Sr. Hall, 2401 Taft Avenue, 0922 Manila, Philippines | Trunk Line: (632) 8524-4611 loc. 513
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Appendix B
PARTNER COORDINATION AND PROTOTYPE DEMONSTRATION

This appendix presents supporting materials from the partner consultation, prototype demonstration, and vehicle inspection. The materials support the methodology in Chapter 5 and the implementation evidence in Chapter 6.

The research team coordinated with Hino Paranaque Subic GS Auto Inc. (SGSA) to inspect the sample fleet-vehicle environment, review the intended monitoring workflow, and demonstrate the prototype interfaces. The evidence photographs in Figures B.1–B.9 document the consultation, fleet-vehicle inspection, dashboard review, and prototype demonstration activities.

Fig. B.1 Initial Consultation Meeting with SGSA Representative

Fig. B.2 Cabin Inspection of the Sample Fleet Vehicle

Fig. B.3 Further Fleet-Vehicle Inspection at the Partner Facility

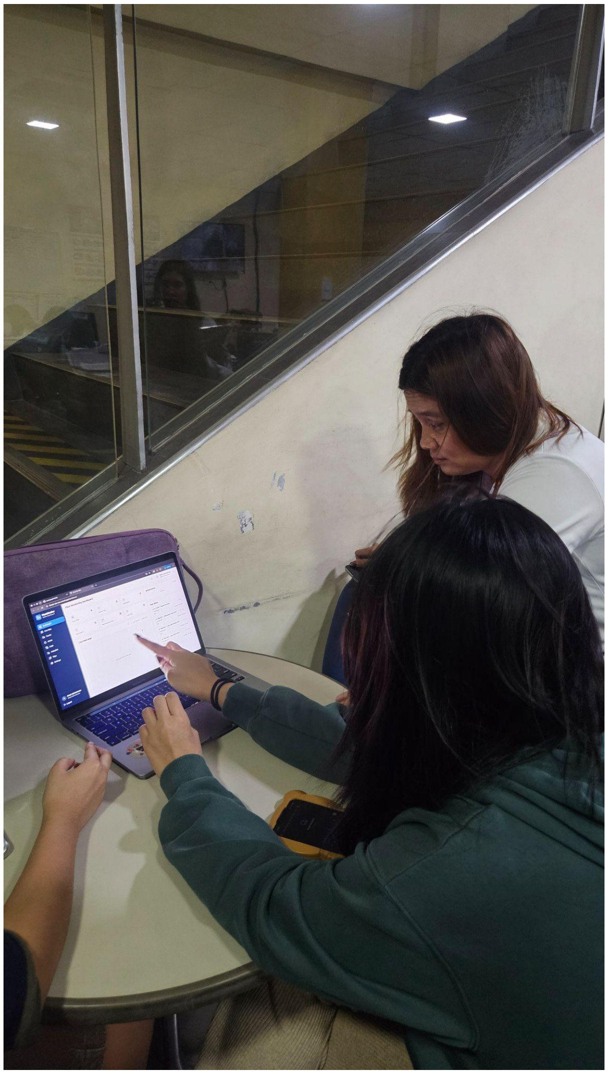

Fig. B.4 Dashboard Evaluation During Prototype Demonstration

Fig. B.5 Web-Based Monitoring Interface Demonstration

Fig. B.6 Research Team After Prototype Meeting and Demonstration

Fig. B.7 Research Team and Partner Representative Group Photograph

Fig. B.8 Additional Prototype Demonstration Evidence

Fig. B.9 Additional Partner Demonstration and Review Evidence

Appendix C
ADMINISTRATOR ACCEPTANCE SURVEY INSTRUMENT

This appendix presents the administrator acceptance survey instrument used to support the usability and alerting discussion in Chapter 6.

The administrator acceptance instrument asked respondents to rate their level of agreement with each statement using a five-point scale: 1 for strongly disagree, 2 for disagree, 3 for neutral, 4 for agree, and 5 for strongly agree. The survey focused on dashboard usability, fuel monitoring, anomaly alerts, driver-safety support, maintenance monitoring, and overall system satisfaction.

C1 Dashboard and Real-Time Monitoring

1. The dashboard provides a clear and easy-to-understand overview of fleet-vehicle status, including fuel, speed, location, and alerts.
2. The dashboard updates are timely enough to support real-time fleet monitoring.
3. The route-tracking/map feature is useful for monitoring fleet-vehicle movement.
4. The trip-history feature helps review past fleet-vehicle operations.
5. The fuel level display is easy to interpret.
6. The telemetry-log view provides enough details for checking fleet-vehicle activity.
7. The dashboard layout is organized and easy to navigate.

C2 Fuel Monitoring Accuracy

1. The system provides useful information about fuel level changes during operation.
2. The fuel monitoring feature can help identify possible abnormal fuel usage.
3. The fuel data shown on the dashboard is helpful for reviewing fleet-vehicle fuel behavior.

C3 Anomaly Detection and Alerts

1. The anomaly alerts are useful for identifying possible fuel theft, leakage, or unusual fuel consumption.
2. The alert information is clear and understandable.

3. The dashboard makes it easy to review and resolve alerts.
4. The alert system can help fleet administrators respond faster to fuel-related issues.
5. The alert status and alert history are useful for record keeping.

C4 Driver Safety and Rest Alerts

1. The driver-rest alert feature is useful for reminding drivers to take breaks.
2. The system supports safer driving by monitoring long driving periods or distance travelled.
3. The HMI alert interface is understandable for drivers.
4. The rest-alert records are useful for reviewing driver activity.

C5 Maintenance Monitoring

1. The maintenance-monitoring feature is useful for tracking fleet-vehicle service needs.
2. The mileage-based maintenance reminder can help improve preventive maintenance.
3. The maintenance information shown on the dashboard is easy to understand.
4. The system can help reduce the chance of missed maintenance checks.

C6 Overall System Usability and Satisfaction

1. The system is useful for fleet monitoring.
2. The system can help improve fuel accountability.
3. The system can help improve operational visibility for the company.
4. I would recommend further development or deployment of this system.
5. Overall, I am satisfied with the system.

C7 Open-Ended Feedback Questions

1. Which feature of the system is the most useful?
2. Which part of the system needs improvement?
3. Were there any confusing parts of the dashboard or HMI?
4. What additional features would you recommend?

Appendix D
TECHNICAL IMPLEMENTATION DETAILS

D1 HMI GPIO Map

TABLE D.1 FULL HMI GPIO MAP

Pin	Function
GPIO 22	`BTN_TAKE_REST` (active-low, debounced 50 ms)
GPIO 39	`BTN_SNOOZE` (active-low, debounced 50 ms)
GPIO 25	Buzzer (PWM through the LEDC peripheral)
GPIO 5	A7670E modem PWRKEY

D2 Telemetry JSON Payload Schema

Listing D.1: Telemetry JSON Payload Schema

```
{
  "truck_id": "<uuid>",
  "driver_id": "<uuid>",
  "trip_id": "<uuid>",
  "sent_at": "<iso-8601>",
  "original_device_timestamp": "<iso-8601>",
  "source_device": "hmi" | "mobile",
  "fuel_level": <0..100>,
  "fuel_valid": <bool>,
  "fuel_source": "...",
  "fuel_confidence": <0..1>,
  "lat": <float>,
  "lon": <float>,
  "gps_source":
    "embedded_gps" | "mobile_gps" | "fused_mobile_fallback",
  "gps_accuracy_m": <float>,
  "gps_heading_deg": <float>,
  "mobile_speed_mps": <float>,
  "speed": <km/h>,
  "odometer_km": <float>,
  "engine_status": "on" | "off" | "idle",
  "seq": <bigint>,
  "event_id": "<uuid>",
  "channel_used":
    "lora" | "lte" | "gsm" | "wifi" | "offline_replay" | "mobile_app" |
        "test_injection",
  "comm_channel": "...",
  "hmi_driving_sec": <int>,
  "hmi_rest_sec": <int>,
  "hmi_trip_status": "...",
  "engine_rpm": <int>,
  "engine_load_pct": <float>,
  "throttle_pct": <float>,
```

```
  "maf_gps": <float>,
  "coolant_temp_c": <float>,
  "engine_runtime_sec": <int>,
  "dtc_present": <bool>,
  "dtc_count": <int>,
  "distance_gps_delta_km": <float>,
  "distance_obd_delta_km": <float>,
  "distance_fused_delta_km": <float>
}
```

Appendix E
ARTICLE PAPER

An IoT-Enabled System for Real-Time Fuel Monitoring, GPS Tracking, and Anomaly Detection in Trucks Using Non-Parametric Unsupervised Machine Learning

Samuelle P. Barja, Konrad Romel C. Castelo, Eliza Rica R. De La Rosa, and Alyssa S. Montemayor

Abstract

Fuel accountability in delivery-truck operations requires more than a fuel gauge: it requires synchronized vehicle telemetry, route context, resilient communication, and an anomaly model that distinguishes abnormal fuel loss from traffic, idling, refueling, sender noise, and driver variability. This paper presents an implemented IoT-enabled fleet-monitoring prototype developed with Hino Parañaque Subic GS Auto Inc. The system acquires OBD-II fuel, speed, GPS, trip-state, and alert data through an embedded HMI; transmits telemetry through LoRa, LTE/GSM, WiFi, and offline replay; stores records in a cloud backend; and detects fuel anomalies using non-parametric unsupervised learning. The production detector is Isolation Forest Model C, supported by Matrix Profile as a subsequence detector and by a data-mining layer for operating-regime interpretation. Fuel anomalies are modeled as context-conditioned deviations between observed and expected fuel-level change under speed, RPM, engine load, throttle, route, driver, and motion state. Field and controlled-injection evaluation showed 1.04% MAPE fuel accuracy, 5.52 s median GPS sampling, 90.21% GPS fix availability, 92.9% precision and 1.09% false-positive rate for Model C under leave-one-trip-out validation, and a 95.0% alert-resolution rate. The results support the use of context-aware telemetry and interpretable anomaly signatures for practical fleet fuel monitoring, while parked-depot siphoning remains a future-work boundary requiring a dedicated stationary-state detector.

Keywords— Fuel monitoring, GPS tracking, anomaly detection, Internet of Things, unsupervised machine learning, Isolation Forest, Matrix Profile, fleet management.

1 Introduction

Fuel cost, fuel theft, unreported leakage, poor route visibility, and delayed alert response remain persistent problems in delivery-fleet operations. Traditional approaches often treat fuel monitoring, GPS tracking, and driver alerts as separate functions, which limits their ability to explain whether a fuel drop is normal consumption, traffic-related idling, sender noise, refueling recovery, or a genuine anomaly. Prior IoT fuel-monitoring and tracking systems show the value of connected sensors but often remain hardware-only or rule-driven, while machine-learning studies on heavy vehicles frequently use datasets detached from an integrated field prototype [1–4].

This study addresses that gap by implementing and evaluating a complete edge-to-cloud truck-monitoring system. The prototype combines OBD-II fuel and engine telemetry, GPS tracking, prioritized communication, cloud storage, dashboard visualization, HMI and mobile trip controls, and unsupervised anomaly detection. The machine-learning component is intentionally context-aware: it does not flag fuel loss only because the level dropped. Instead, it evaluates whether the observed fuel change is plausible under the vehicle's current operating context.

The specific contributions of this paper are: (1) an integrated IoT fuel, GPS, and alerting prototype for truck fleet monitoring; (2) a context-conditioned anomaly model grounded in physical and operational constraints; (3) an Isolation Forest production detector supported by Matrix Profile and K-Means-based operating-regime discovery; and (4) an evaluation across fuel accuracy, communication reliability, GPS cadence, anomaly-detection metrics, and alert usability.

2 Related Work

IoT-based fleet-monitoring systems have used GPS, cloud dashboards, RFID, GSM, and sensor modules to improve vehicle visibility and operational response [5–7]. Fuel-specific systems have monitored tank level, theft alarms, and basic fuel events using resistance, ultrasonic, or capacitive sensing [8–10]. These systems demonstrate practical monitoring value, but many are limited by fragmented architecture, small-scale validation, or rule-based detection.

Machine-learning work on fuel and vehicle anomaly detection shows that abnormal consumption can be detected from multivariate telemetry. Mumcuoglu et al. [4], for example, showed that fuel-consumption anomalies can be classified in heavy-duty vehicles, while interpretable anomaly-detection literature emphasizes the importance of explaining why a point is anomalous

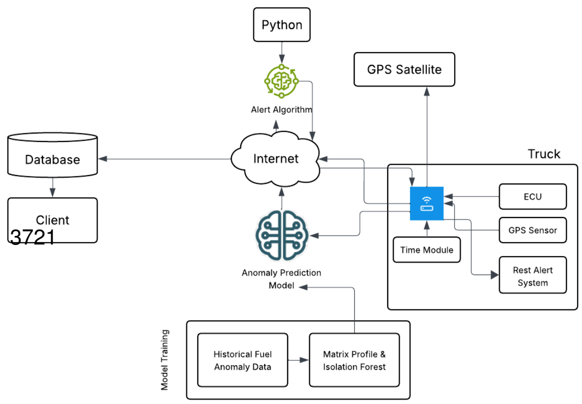

Figure 1: Conceptual diagram of the IoT-enabled fuel monitoring and anomaly-detection system.

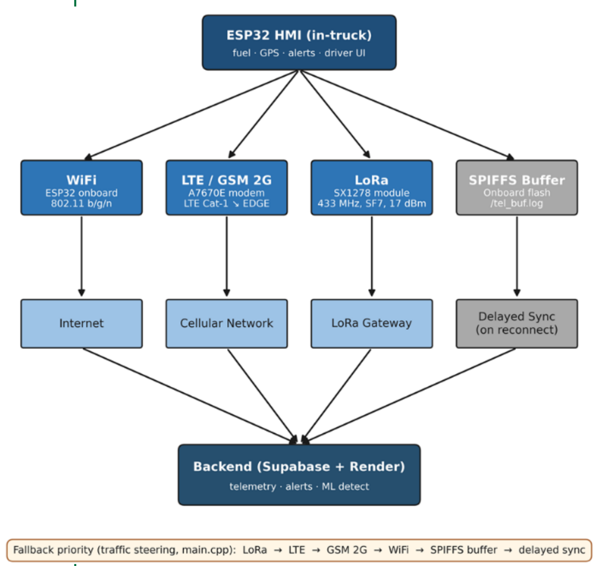

Figure 2: End-to-end edge-to-cloud system architecture.

rather than reporting only a score [11]. This study builds on those directions by deploying unsupervised models inside an implemented truck telemetry system and by tying every anomaly flag to physical or operational constraints.

3 System Architecture

Fig. 1 summarizes the conceptual system. The truck-side embedded HMI collects vehicle telemetry, displays local status, and allows driver trip interaction. The backend stores telemetry and trip state, evaluates anomaly and safety rules, and synchronizes the administrative dashboard, mobile companion interface, and HMI feedback path.

The implemented architecture in Fig. 2 follows a layered edge-to-cloud pattern. The edge layer performs OBD-II acquisition, GPS parsing, fuel filtering, trip-state handling, HMI display, and communication-channel selection. The transport layer uses prioritized LoRa, LTE/GSM, WiFi, and SPIFFS offline replay so telemetry is not silently lost during degraded network conditions. The cloud layer persists data in Supabase, runs backend rules and anomaly inference, and exposes synchronized views to the dashboard, mobile app, and HMI poll-back path.

4 Methodology

4.1 Fuel Acquisition and Filtering

Fuel, speed, GPS, trip, and engine data are sampled by the HMI and transmitted as synchronized telemetry rows. Fuel readings are filtered before model inference to reduce sender noise and slosh. The filtering pipeline combines a median stage, exponential smoothing, and a dead-band publisher, allowing the system to suppress short tank-motion artifacts while preserving meaningful fuel trends.

Fuel accuracy was evaluated against pump-gas reference events and the known tank capacity. The primary accuracy metric was mean absolute percentage error (MAPE):

$$\mathrm{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{V_{\mathrm{sensor},i} - V_{\mathrm{ref},i}}{V_{\mathrm{ref},i}}\right|. \tag{1}$$

4.2 Communication, GPS, and Backend Integration

Each telemetry row carries device-side timestamp information and channel metadata. Server-side reconciliation separates sampling cadence from backend-arrival delay. Communication was evaluated through per-channel latency, packet delivery, and offline-replay behavior. GPS was evaluated using inter-sample interval and fix availability because route context, distance, and motion state feed the anomaly detector and dashboard.

4.3 Fuel-Anomaly Model

A fuel anomaly is modeled as a telemetry row whose observed fuel-level change deviates from the expected fuel-level change under operating context. For row t,

$$A(t) = \frac{|\Delta f_{\mathrm{obs}}(t) - \mathbb{E}[\Delta f \mid c(t)]|}{\sigma(c(t))}, \tag{2}$$

where $c(t)$ includes speed, RPM, engine load, throttle, route, driver, and motion state. This makes the detector a context-conditioned fuel-deviation model rather than a simple sudden-drop detector.

The model is constrained by four physical constraints and four operational constraints. C1 conservation states that fuel decreases through consumption or unauthorized removal and increases only during refueling. C2 bounded consumption limits plausible fuel loss by engine work and load. C3 sender non-linearity accounts for unstable tank-level bands. C4 slosh accounts for short fuel fluctuations caused by liquid movement. O1 traffic/idling, O2 route restrictions, O3 driver variability, and O4 refuel patterns encode Philippine fleet-operation context. These constraints are implemented through direction-aware rules, refuel gates, fuel-per-RPM and fuel-per-load features, expected-consumption residuals, idle and motion-state features, route-type features, driver fuel-economy deviation, and post-refuel flags.

4.4 Machine Learning and Data Mining

Isolation Forest Model C is the production row-level detector. Model A used baseline fuel, motion, engine, and refuel-window features; Model B added driving-zone features; and Model C added per-asset, per-driver, and route-type fuel-economy deviation features. Matrix Profile was retained as a supporting subsequence detector for gradual fuel-loss patterns, not as the main production classifier.

A data-mining layer supports interpretation. Constraint-violation profiling annotates each flagged row with signatures such as C2 bounded consumption, C4 slosh suppression, and O3 driver baseline deviation. K-Means clustering with $K = 4$ was used only to discover operating regimes in the controlled-injection window. The seven standardized clustering features were filtered fuel delta, speed, engine load, idle duration score, motion state, throttle, and normalized RPM. The four discovered regimes corresponded to high-speed cruise, mid-speed cruise, gradual-decline behavior, and raw-slosh artifact behavior.

4.5 Evaluation Protocol

Fuel accuracy, communication reliability, GPS cadence, anomaly detection, and alert usability were evaluated according to the study's five objectives. SO4 anomaly-detection metrics were computed using leave-one-trip-out validation on the project-native partition so that each held-out trip was scored by models trained and calibrated on separate trips. Precision and false-positive rate (FPR) were computed as:

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{FPR} = \frac{FP}{FP + TN}. \tag{3}$$

5 Results and Discussion

Table 1 summarizes the main objective-level results. The fuel subsystem reached 1.04% MAPE and 3.00% maximum error across the 13–98% tank range, satisfying the ±5% target. GPS reached a 5.52 s median sampling interval and 90.21% fix availability. The communication subsystem demonstrated multi-channel delivery and meaningful offline-buffer use, preventing telemetry loss during weak connectivity.

For anomaly detection, Model C was the strongest production detector because it considered fuel movement relative to driver, route, and operating context. It achieved 92.9% precision, 92.9% recall, and 1.09% FPR on the project-native partition. The result satisfies the 90% precision and 10% FPR requirements while preserving Model C as the deployable detector. The Hybrid Intersection produced 97.4% precision as a high-confidence alert channel, but Model C remains the primary production model because it provides the best row-level balance of detection and coverage.

The constraint-violation analysis makes the ML result auditable. In the controlled-injection analysis, O3 driver baseline deviation appeared in every flagged row, C2 bounded consumption appeared in 69.0% of flagged rows, and C4 slosh suppression appeared most strongly in sudden and rapid drops. The controlled-injection window had zero false positives, but this is scoped only to that window; the full leave-one-trip-out evaluation still reports the overall 1.09% FPR. K-Means archetype discovery further showed that labelled and predicted positives were concentrated in moving-vehicle operating regimes, while a raw-slosh artifact cluster contained zero labels and zero predictions, confirming that the filter cascade suppressed C3/C4 artifacts before inference.

The alert subsystem completed the operator workflow. Alerts were surfaced across fuel anomaly, overspeed, low fuel, rest, and maintenance classes, with synchronized visibility across the HMI, mobile app, and dashboard. The administrator System Usability Scale score of 81.9/100 indicates that the dashboard was not only functional but usable for routine monitoring and response.

6 Limitations and Future Work

The detector is strongest for in-trip and moving-vehicle anomalies because its context features depend on movement, speed, RPM, engine load, route progression, and driver-baseline deviation. Parked-depot siphoning, where the vehicle is stationary and engine-off, requires a separate stationary-state detector using long-duration fuel-level monitoring. Other limitations include the need for longer deployment with confirmed real-world theft events, broader vehicle and tank calibration, stronger modern cellular fallback, and more multi-user dashboard testing.

7 Conclusion

This paper presented an implemented IoT-enabled system for real-time truck fuel monitoring, GPS tracking, anomaly detection, and alert management. The prototype met the thesis objectives across fuel accuracy, com-

Table 1: Objective-Level Evaluation Summary

Objective	**Target**	**Measured Result**	**Status**
SO1 Fuel measurement	Fuel accuracy within ±5%	1.04% MAPE; 3.00% maximum error; all validation events within tolerance	Met
SO2 Communication	Cloud-accessible real-time telemetry	LoRa, LTE/GSM, WiFi, and offline replay maintained telemetry flow under varied network conditions	Met
SO3 GPS tracking	5 s location logging and route context	5.52 s median sampling interval; 90.21% GPS fix availability	Met
SO4 ML anomaly detection	≥90% precision and ≤10% FPR	Isolation Forest Model C: 92.9% precision, 92.9% recall, and 1.09% FPR under leave-one-trip-out evaluation	Met
SO5 Alerts and dashboard	Usable alert and monitoring workflow	100 alerts over 17.7 days; 95.0% resolution rate; administrator SUS score of 81.9/100	Met

munication, GPS cadence, machine-learning anomaly detection, and alert usability. The central technical finding is that fuel anomalies should be modeled as context-conditioned deviations, not as raw fuel drops alone. Isolation Forest Model C, supported by Matrix Profile and a constraint-aware data-mining layer, provided a practical and auditable production detector. Overall, the system demonstrates that embedded telemetry, resilient communication, contextual unsupervised learning, and human-readable alert explanations can support fuel accountability and proactive fleet supervision.

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